ON GALOIS REPRESENTATIONS AND p-AIDC HODGE THEORY

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Abstract. In this article, we present a review on the theory of p-adic Galois representations. Precisely, we will state the theory of (φ, Γ)-modules, which establishes a categorical equivalence between representations and linear algebraic objects, offering a framework to classify Galois representations via the category of vector spaces with some additional structures. Focusing on geometric representations, we study the properties about filtrations and Frobenius actions the period rings, and finally prove "admissible implies weakly admissible", which states a categorical equivalence between semi-stable representations and weakly-admissible (φ, N)-modules.

1. Introduction

Since its emergence in the 1970s, p-adic Hodge theory plays an important role in arithmetic geometry and algebraic number theory. One of its main topic is the classification of geometric Galois representations. Around 1990, Jean-Marc Fontaine introduced the theory of (φ, Γ)-modules in [1], which transforms the intricate category of p-adic Galois representations into linear algebraic objects equipped with a Frobenius operator φ and an action of a decomposition group Γ, providing the first systematic approach to classifying p-adic representations. In particular, we are interested in representations come from geometry, i.e. those appear as subquotients of a ℓ-adic étale cohomology of a variety up to a twist. Tsuji's comparison theorem ([6]) suggests that p-adic étale cohomology $H^i(X \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}, \mathbb{Q}_p)$ is potentially semi-stable, and hence any p-adic representation from geometry is potentially semi-stable. Conversely, Fontaine and Mazur make the following conjecture:

Conjecture. *([2]) Let $\rho : G_{\mathbb{Q}} \to GL_n(\mathbb{Q}_\ell)$ be a continuous representation, which is irreducible, unramified away from finitely many primes, and whose restriction to every decomposition group over ℓ is potentially semi-stable. Then ρ is geometric in the sense that, up to a twist, it appears as a subquotient in the étale cohomology of a finite type $\mathbb{Q}$-scheme.*

This motivates us to focus on semi-stable representations. Crystalline representations are semi-stable representations with trivial monodromy action, and furthermore it relates profoundly to the crystalline cohomology. Thus in this article, we will mainly discuss crystalline representations and semi-stable representations of G_K for a p-adic field K. The central objective of this paper is to expound the classification of these special representations using Fontaine's functor $\mathbf{D}_{\mathrm{st}}$. To be more precise, our technical approach revolves around four fundamental period rings:

- The Hodge-Tate ring B_{HT}: Captures Hodge-Tate weights of Galois representations through its graded structure, laying the foundation for p-adic Hodge decompositions.
- The de Rham ring B_{dR}: Carries a complete filtration structure encoding cohomological rigidity of representations.
- The crystalline ring B_{cris} and semi-stable ring B_{st}: Equipped with Frobenius operators and monodromy operators N, enabling algebraic control of geometric representations.

After general discussion about the properties of such period ring, we will pay particular attention to Fontaine's concept of "weak admissibility" on filtered (φ, N)-modules, which described the interaction between the filtration structure and the Frobenius map with some algebraic condition. We will finally see that, these conditions was ultimately shown equivalent to the admissibility of Galois representations through the following theorem.

Theorem A. *(Weakly admissible implies admissible) The functor*

$$\mathbf{D}_{\mathrm{st}} : \boldsymbol{Rep}^{st}_{\mathbb{Q}_p}(G_K) \to \boldsymbol{MF}^{\mathrm{ad}}_K(\varphi, N)$$

is an equivalence of categories, and the functor $\mathbf{V}_{st} : \boldsymbol{MF}^{\mathrm{ad}}_K(\varphi, N) \to \boldsymbol{Rep}^{st}_{\mathbb{Q}_p}(G_K)$ *is a quasi-inverse of* $\mathbf{D}_{\mathrm{st}}$*. Moreover, they are compatible with tensor product, dual, etc.*

The paper is structured as follows: Section 2 (Preliminaries) establishes foundational notations and key structural theorems. Section 3 reviews the basic knowledge about tilting functor and proves Theorem B. Section 4 systematically constructs the Hodge-Tate and de Rham period rings. Section 5 provides a deep analysis of the operator structures on crystalline and semi-stable period rings. Section 6 states the main theorem on weak admissibility. Finally, section 7 completes the proof through reduction to Lubin-Tate cases and applications of Colmez's tools of rigid analysis (Fundamental Lemma). Most of the results can be founded in [3] and [4].

2. Preliminaries

2.1. **Structure of extensions over a p-adic field.** Throughout the paper, unless otherwise specified, let K be a p-adic field. Let k be its residue field, $W = W(k)$ be the Witt ring of k and $K_0 = \mathrm{Frac}W$. Then K_0 is absolutely unramified and K/K_0 is totally unramified. Let $\bar{K}$ be an algebraic closure of K. Let L be a finite extension of K inside $\overline{K}$, then one can define k_L, L_0 as the above. Define

$$L^{cyc} := L(\mu_{p^\infty}) = \bigcup_{n\in\mathbb{N}} L(\mu_{p^n}) = LK_0^{cyc}$$

be the subfield of $\bar{K}$ obtained by adjoining to L all p^n-th roots of unity, and denote

$$H_L := \mathrm{Gal}(\bar{K}/L^{cyc}), \quad \boldsymbol{\Gamma}_L := \mathrm{Gal}(L^{cyc}/L).$$

Then we have the following theorem:

Theorem 2.1. *There exists a constant $n = n(L) \in \mathbb{N}$, such that:*
(1). $L^{cyc}/L(\mu_{p^n})$ is totally ramified and hence $k_L^c := k_{L^{cyc}}$ is a finite extension of k_L.
(2). for any $m \geq n$, $L(\mu_{p^n})$ and $L_0(\mu_{p^m})$ are linearly disjoint over $L_0(\mu_{p^n})$, and hence

$$Gal(L(\mu_{p^n}))/L_0(\mu_{p^n})) = \cdots = Gal(L(\mu_{p^m}))/L_0(\mu_{p^m})) = Gal(L^{cyc}/L_0^{cyc}).$$

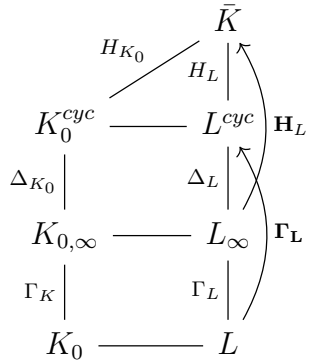
Moreover, if $L = L_0$, then one can take $n = 0$.

The cyclotomic character $\chi : G_L \to \mathbb{Z}_p^\times$ is defined by $\sigma(\zeta) = \zeta^{\chi(\sigma)}$. Clearly $H_L = ker\chi$, thus χ induces an injection $\boldsymbol{\Gamma} \to \mathbb{Z}_p^\times$. By the canonical isomorphism $\mathbb{Z}_p^\times = \mathbb{F}_p^\times \times (1 + p\mathbb{Z}_p)$ when $p \neq 2$ and $\mathbb{Z}_p^\times = \mathbb{F}_2 \times (1 + 4\mathbb{Z}_2)$, one can decompose

$$\boldsymbol{\Gamma}_L = \Delta_L \times \Gamma_L,$$

where Δ_L is a subgroup of $\mathbb{F}_p^\times$ if $p \neq 2$ or $\mathbb{F}_2$ if $p = 2$, and $\Gamma_L \cong \mathbb{Z}_p$. Thus, the study of the extension $\overline{L}/L = \overline{K}/L$ can be reduced to that of an extension L_∞/L with Galois group $\cong \mathbb{Z}_p$

and an extension $\overline{L}/L^{\mathrm{cyc}}$, which serves as a natural base for constructing Fontaine's E-series through its tilts. As a summary, we have the following commutative diagram.



2.2. **Basic theory of B-representations.** Let G be a topological group and B be a topological commutative ring equipped with a continuous action of G compatible with the structure of ring, that is, for all $g \in G$, and $b_1, b_2 \in B$,

$$g(b_1 + b_2) = g(b_1) + g(b_2),\ g(b_1 b_2) = g(b_1)g(b_2).$$

A B-representation X of G is a B-module of finite type equipped with a semi-linear and continuous action of G, where semi-linear means that for all $g \in G, \lambda \in B$, and $x, x_1, x_2 \in X$,

$$g(x_1 + x_2) = g(x_1) + g(x_2), g(\lambda x) = g(\lambda)g(x).$$

A free B-representation X of G is trivial if one of the following two equivalent conditions holds:
1). There exists a basis of X consisting of element of X^G;
2). $X \cong B^d$ which is equipped with natural componentwise action of G.

Check by definition directly, we can get a first classification of free B-representations of G by continuous cohomology:

Theorem 2.2. *Suppose d is a positive integer. The correspondence $X \mapsto [X]$ defines a bijection between the set of equivalence classes of free B-representations of G of rank d and $H^1_{cont}(G, GL_d(B))$. Moreover, X is trivial if and only if $[X]$ is the distinguished point in $H^1_{cont}(G, GL_d(B))$.*

A Galois representation is defined as a B-representation of a Galois group of some field extension. When $B = \mathbb{Q}_p$ (equipped with its natural topology), this defines a p-adic Galois representation. To study those p-adic representations arising from geometry, Fontaine develop a method to construct some big topological rings B and organize those representations into B-admissible categories. The observation starts as the following:

Definition 2.3. Let G be a topological group and B be a topological ring with a continuous G-action. Set $E = B^G$, and F be a closed subfield of E. Then we say that B is (F, G)-regular if the following conditions hold:
(i) B is a domain;
(ii) $B^G = F^G = E \supset F$;
(iii) For $b \in B$, $b \neq 0$, if for any $g \in G$, there exists $\lambda = \lambda(g) \in F$, such that $g(b) = \lambda(b)$, then b is invertible in B.

Definition & Proposition 2.4. *Assume that B is (F,G)-regular, then*
(1) For an F-representation V of G, $\mathbf{D}_B(V) := (B \otimes_E V)^G$ is an E-vector space with dimension $\dim_E \mathbf{D}_B(V) \leq \dim_F V$.
(2) V is called B-admissible if the following equivalent conditions hold:
(i) $B \otimes_F V$ is a trivial B-representation;
(ii) The natural map

$$\alpha_V : B \otimes_E \mathbf{D}_B(V) \to B \otimes_F V$$

is an isomorphism;
(iii) $\dim_F V = \dim_E \mathbf{D}_B(V)$.

Let $\mathbf{Rep}_F^B(G)$ denote the category of B-admissible F-representations of G. Then $\mathbf{Rep}_F^B(G)$ is a Tannakian sub-category of $\mathbf{Rep}_F(G)$, and $\mathbf{D}_B$ is restricted to a faithful tensor functor on $\mathbf{Rep}_F^B(G)$.

Let $\mathbf{Rep}_F(G)$ denote the category of continuous F-representations of G. This is an abelian category with tensor products, dual representations and a unit representation. In other word, $\mathbf{Rep}_F(G)$ is a neutral Tannakian category over F.

2.3. **Galois representations over fields of characteristic p.** Let us consider the Galois representations over a field E of characteristic p as a first example. Let E^s be a separable closure of E, and let $G = \mathrm{Gal}(E^s/E)$. Then by Theorem 2.2 and Hilbert's Theorem 90, all E^s-representations are trivial. Thus $\mathbf{Rep}_{\mathbb{F}_p}(G) = \mathbf{Rep}_{\mathbb{F}_p}^{E^s}(G)$, and the functor $\mathbf{D}_{E^s}$ is faithful on $\mathbf{Rep}_{\mathbb{F}_p}(G)$. This indicates that we can study $\mathbf{Rep}_{\mathbb{F}_p}(G)$ via its image.

Let $\varphi : \lambda \mapsto \lambda^p$ be the absolute Frobenius of E.

Definition 2.5. A φ-module over E is an E-vector space M together with a map $\varphi : M \to M$ which is semi-linear with respect to the absolute Frobenius map, i.e.

$$\varphi(x+y) = \varphi(x) + \varphi(y),\ \varphi(\lambda x) = \varphi(\lambda)\varphi(x) = \lambda^p \varphi(x).$$

Let E_φ be the E-module with E-action $\lambda \cdot \mu := \lambda^p \mu$. Let M be an E-vector space, $M_\varphi := E_\varphi \otimes_E M$ is an E-vector space as well. Given a semi-linear map $\varphi : M \to M$ is equivalent to give an E-linear map $\Phi : M_\varphi \to M, \lambda \otimes x \mapsto \lambda\varphi(x)$.

Definition 2.6. Let M be a φ-module of finite dimension, it is said étale if one of the following equivalent conditions hold:
(1). Φ is an isomorphism;
(2). Φ is injective;
(3). Φ is surjective;
(4). $M = E \cdot \varphi(M)$.

Denote the category of étale φ-module over E by $\mathcal{M}_\varphi^{\text{ét}}(E)$. One can show that it is a Tannakian category.

We define the Frobenius φ on $E^s \otimes_{\mathbb{F}_p} V$ by

$$\varphi(\lambda \otimes v) = \varphi(\lambda) \otimes v = \lambda^p \otimes v,$$

then one can easily check that for all $x \in E^s \otimes_{\mathbb{F}_p} V$,

$$\varphi(g)(x) = g(\varphi(x)),\ \text{for all } g \in G.$$

Thus $\mathbf{D}_{E^s}(V)$ is stable under φ. In fact, it is an étale φ-module.

Theorem 2.7. *If V is a mod p representation of G of dimension d, then $\mathbf{D}_{E^s}(V)$ is an étale φ-module over E and $dim_E\mathbf{D}_{E^s}(V) = d$.*

Proof. Suppose $\{v_1, \cdots, v_d\}$ is a basis of V over $\mathbb{F}_p$, by abuse of notations, write $v_i = 1 \otimes v_i$ in $E^s \otimes_{\mathbb{F}_p} V$. Suppose $\{e_1, \cdots, e_d\}$ is a basis of $\mathbf{D}_{E^s}(V)$ over E. Then

$$e_j = \sum_{i=1}^{d} b_{ij} v_i, \text{ for } B = (b_{ij}) \in \mathrm{GL}_d(E^s).$$

Hence

$$\varphi(e_j) = \sum_{i=1}^{d} b_{ij}^p v_i =: \sum_{i=1}^{d} a_{ij} v_i,$$

where $A = (a_{ij}) = B^{-1}\varphi(B)$, and

$$\det A = (\det B)^{-1}\det(\varphi(B)) = (\det(B))^{p-1} \neq 0.$$

This implies that $\mathbf{D}_{E^s}(V)$ is étale. □

Conversely, let M be an étale φ-module over E. $E^s \otimes_E M$ is viewed as a φ-module via

$$\varphi(\lambda \otimes x) = \lambda^p \otimes \varphi(x),$$

and equipped with a G-action by

$$g(\lambda \otimes x) = g(\lambda) \otimes x, \text{ for } g \in G.$$

We can check that this action commutes with φ as well, and thus

$$\mathbf{V}(M) := \{y \in E^s \otimes_E M \mid \varphi(y) = y\} = (E^s \otimes_E M)_{\varphi = id}$$

is a sub $\mathbb{F}_p$-vector space stable under G.

Theorem 2.8. *The functor*

$$\mathbf{D}_{E^s} : \mathbf{Rep}_{\mathbb{F}_p}(G) \to \mathcal{M}_{\varphi}^{\acute{e}t}(E)$$

is an equivalence of Tannakian categories and the functor

$$\mathbf{V} : \mathcal{M}_{\varphi}^{\acute{e}t}(E) \to \mathbf{Rep}_{\mathbb{F}_p}(G)$$

is a quasi-inverse.

After a "Cohenlization" arguments, we have the theorem for $\mathbb{Z}_p$- and $\mathbb{Q}_p$-representations as well:

Let $\mathcal{O}_{\mathcal{E}} = \mathcal{C}(E)$ be the Cohen ring of E and $\mathcal{E} = \mathrm{Frac}(\mathcal{O}_{\mathcal{E}})$. Let φ denotes the Frobenius map on $\mathcal{O}_{\mathcal{E}}$ and on $\mathcal{E}$.

Definition 2.9. A φ-module over $\mathcal{O}_{\mathcal{E}}$ is an $\mathcal{O}_{\mathcal{E}}$-module M together with a semi-linear map $\varphi : M \to M$ with respect to φ on $\mathcal{O}_{\mathcal{E}}$, and similarly a φ-module over $\mathcal{E}$ is an $\mathcal{E}$-vector space D together with a semi-linear map $\varphi : D \to D$ with respect to φ on $\mathcal{E}$.

Let $\mathcal{O}_{\mathcal{E},\varphi}$ be the $\mathcal{O}_{\mathcal{E}}$-module by $\lambda \cdot x := \varphi(\lambda)x$. For an $\mathcal{O}_{\mathcal{E}}$-module M, $M_\varphi := \mathcal{O}_{\mathcal{E},\varphi} \otimes_{\mathcal{O}_{\mathcal{E}}} M$. Similarly, we define D_φ for an $\mathcal{E}$-vector space. Then, to give a semi-linear map $\varphi : M \to M$ (resp. $D \to D$) is equivalent to give a $\mathcal{O}_{\mathcal{E}}$ (resp. $\mathcal{E}$)-linear map $\Phi : \varphi : M_\varphi \to M$ (resp. $D_\varphi \to D$), $\lambda \otimes x \mapsto \lambda\varphi(x)$.

Definition 2.10. (1). A φ-module over $\mathcal{O}_{\mathcal{E}}$ is called étale if it is finitely generated and $\Phi : M_\varphi \to M$ is an isomorphism.
(2). A φ-module D over $\mathcal{E}$ is étale if $\dim_{\mathcal{E}}(D) < \infty$ and if there exists an $\mathcal{O}_{\mathcal{E}}$-lattice M of D which is stable under φ, such that M is an étale φ-module over $\mathcal{O}_{\mathcal{E}}$.

Theorem 2.11. *With the notations above. Define*

$$\mathbf{D}_{\mathcal{O}_{\widehat{\mathcal{E}^{ur}}}} : \mathbf{Rep}_{\mathbb{Z}_p}(G) \to \mathcal{M}_\varphi^{ét}(\mathcal{O}_{\mathcal{E}}),\ T \mapsto (\mathcal{O}_{\widehat{\mathcal{E}^{ur}}} \otimes_{\mathbb{Z}_p} T)^G,$$

$$\mathbf{T} : \mathcal{M}_\varphi^{ét}(\mathcal{O}_{\mathcal{E}}) \to \mathbf{Rep}_{\mathbb{Z}_p}(G),\ M \mapsto (\mathcal{O}_{\widehat{\mathcal{E}^{ur}}} \otimes_{\mathcal{O}_{\mathcal{E}}} M)_{\varphi=id}.$$

Then the functor

$$\mathbf{D}_{\mathcal{O}_{\widehat{\mathcal{E}^{ur}}}} : \mathbf{Rep}_{\mathbb{Z}_p}(G) \to \mathcal{M}_\varphi^{ét}(\mathcal{O}_{\mathcal{E}})$$

is an equivalence of Tannakian categories and the functor

$$\mathbf{T} : \mathcal{M}_\varphi^{ét}(\mathcal{O}_{\mathcal{E}}) \to \mathbf{Rep}_{\mathbb{Z}_p}(G)$$

is a quasi-inverse. Similarly, define

$$\mathbf{D}_{\widehat{\mathcal{E}^{ur}}} : \mathbf{Rep}_{\mathbb{Q}_p}(G) \to \mathcal{M}_\varphi^{ét}(\mathcal{E}), V \mapsto (\widehat{\mathcal{E}^{ur}} \otimes_{\mathbb{Q}_p} V)^G,$$

$$\mathbf{V} : \mathcal{M}_\varphi^{ét}(\mathcal{E}) \to \mathbf{Rep}_{\mathbb{Q}_p}(G), M \mapsto (\widehat{\mathcal{E}^{ur}} \otimes_{\mathcal{E}} M)_{\varphi=id}.$$

Then the functor

$$\mathbf{D}_{\widehat{\mathcal{E}^{ur}}} : \mathbf{Rep}_{\mathbb{Q}_p}(G) \to \mathcal{M}_\varphi^{ét}(\mathcal{E})$$

is an equivalence of Tannakian categories and the functor

$$\mathbf{V} : \mathcal{M}_\varphi^{ét}(\mathcal{E}) \to \mathbf{Rep}_{\mathbb{Q}_p}(G)$$

is a quasi-inverse.

3. The theory of (φ, Γ)-modules

In this section, we will review the basic theory of (φ, Γ)-modules. Firstly, we introduce the tilting functor, which gives a method to unify the theory over field of positive or zero characteristic. We will then discuss the tilts of the field C and its integer ring, and two distinguished elements in particular. After that, we may have a glance at the tilting equivalence via Fontaine-Wintenberger's Theorem, which provides the most important fact for the theory of (φ, Γ)-module. As a consequence, we find that for every extension over K^{cyc}, we can construct a extension of fields of characteristic p accordingly, which inherits a same Galois group. Finally, we will see that, as an extension of Theorem 2.11 to p-adic fields, we have an equivalence of the category of p-adic representations and étale (φ, Γ)-modules.

3.1. Tilting functors.

Definition 3.1. Assume A is a commutative ring of characteristic p, set

$$A^\flat = \varprojlim_{n \in \mathbb{N}} A_n = \{(x_n) |\ x_n \in A,\ x_{n+1}^p = x_n\},$$

where $A_n = A$ and the transition map is the Frobenius $\varphi : x \mapsto x^p$.

Definition 3.2. Let A be a commutative ring which is separated and complete for the p-adic topology, set

$$A^\flat := \varprojlim_{x \mapsto x^p} A = \{(x^{(n)}) |\ x^{(n)} \in A,\ (x^{(n+1)})^p = x^{(n)}\}.$$

Proposition 3.3. *Assume that A is a commutative ring which is separated and complete for the p-adic topology, then there exists a bijection between $A^\flat$ and $(A/pA)^\flat$.*

By the above proposition, we can identify $A^\flat$ can be considered as a ring by the identification with $(A/pA)^\flat$. For any element $x \in A^\flat$, we can write it as $x = (x^{(n)})$ with $x^{(n)} \in A$ or as $x = (x_n)$ with $x_n \in A/pA$. We have

$$(x+y)^{(n)} = \lim_{m\to\infty}(x^{(n+m)} + y^{(n+m)})^{p^m} \text{ and } (xy)^{(n)} = x^{(n)}y^{(n)}.$$

For an element $x \in A^\flat$, put $x^\sharp := x^{(0)} \in A$.

3.2. **Tilts of $\mathcal{O}_C$ and C.**

Definition 3.4. Let $R = \mathcal{O}_C^\flat$ and $F = \mathrm{Frac}R = C^\flat$.

Theorem 3.5. *The ring R is a complete valuation ring perfect of characteristic p with the valuation $v = v_R$ defined by*

$$v_R(x) = v(x) := v(x^\sharp).$$

The residue field of R is $\bar{k}$ and its fraction field F is a complete non-archimedean perfect field of characteristic p. Furthermore, R is equipped with a natural continuous action of G_{K_0} given by

$$g(x) = (gx^{(n)})_n.$$

Since $\bar{k}$ is perfect and R is complete, by the structure theorem, there exists a section $s : \bar{k} \to R$ of the projection $R \to \bar{k}$, which is a homomorphism of rings. More precisely:

Proposition 3.6. *The section s is given by*

$$a \mapsto ([a^{p^{-n}}])_{n\in\mathbb{N}},$$

where $[a^{p^{-n}}] = (a^{p^{-n}}, 0, \cdots)$ is the Teichmüller representative of $a^{p^{-n}}$.

It is obvious that we have the exact sequence

$$1 \to U_C^+ \to U_C \to \bar{k} \to 1,$$

and the Teichmüller map $\bar{k} \to U_C$ induce an isomorphism $U_C = \bar{k}^\times \times U_C$. Similarly, we have $U_R = \bar{k}^\times \times U_R^+$. In fact, we have more information about the structure of subgroups of R:

Proposition 3.7. *The map*

$$\mathrm{Hom}(\mathbb{Z}[\frac{1}{p}], C^\times) \to F,\ f \mapsto (f(p^{-n}))$$

is a canonical isomorphism of $\mathbb{Z}[G_{K_0}]$-modules. Moreover, identifying these two groups through this isomorphism, then
(1) $U_R = \mathrm{Hom}(\mathbb{Z}[\frac{1}{p}], \mathcal{O}_C^\times) = \bar{k}^\times \times U_R^+$;
(2) $U_R^1 \xrightarrow{\sim} \varprojlim_{n\in\mathbb{N}} U_R^1/(U_R^1)^{p^n}$ is a trosion free $\mathbb{Z}_p$-module and $U_R^+ = \mathrm{Hom}(\mathbb{Z}[\frac{1}{p}], U_C^+) = \mathbb{Q}_p \otimes_{\mathbb{Z}_p} U_R^1$.

Proposition 3.8. *F is an algebraically closed field.*

Proposition 3.9. *Let L be an extension of K_0 contained in $\overline{K}$ and let $H = \mathrm{Gal}(\overline{K}/L)$. Then*

$$R^H = (\mathcal{O}_C^\flat)^H = \mathcal{O}_L^\flat,\ F^H = (C^\flat)^H = \mathrm{Frac}((\mathcal{O}_L/p\mathcal{O}_L)^\flat).$$

The residue field of R^H is $k_L = \bar{k}^H$, the residue field of L.

Proof. Assume $x \in R^H$ (resp. F^H). Write $x \in (x^{(n)})_{n\in\mathbb{N}}$, $x^{(n)} \in \mathcal{O}_C$ (resp. C), then for $h \in H$,

$$h(x) = (h(x^{(n)}))_n.$$

Hence

$$x \in R^H \text{ (resp. } F^H) \Longleftrightarrow x^{(n)} \in \mathcal{O}_C^H \text{ (resp. } C^H) \quad \text{for all } n \in \mathbb{N},$$

then the first assertion follows from the fact $C^H = \widehat{L}$, $\mathcal{O}_C^H = \mathcal{O}_{C^H} = \mathcal{O}_{\widehat{L}} = \varprojlim \mathcal{O}_L/p^n\mathcal{O}_L$ and that $(\mathcal{O}_L/p\mathcal{O}_L)^\flat = (\mathcal{O}_{\widehat{L}}/p\mathcal{O}_{\widehat{L}})^\flat$.

The map $\bar{k} \hookrightarrow R \twoheadrightarrow \bar{k}$ induces the map $k_L \hookrightarrow R^H \twoheadrightarrow k_L$, and the composition map is nothing but the identity map, so the residue field of R^H is k_L. □

Proposition 3.10. *If $v(L^\times)$ is discrete, then*

$$(\mathcal{O}_L)^\flat = R^H = k_L.$$

This is the case if L is a finite extension of K_0.

Proof. From the proof of the previous proposition, we know $k_L \subset R^H = \mathcal{O}_L^\flat$, it remains to show that

$$x = (x^{(n)}) \in R^H,\ v(x) > 0 \Rightarrow x = 0.$$

In fact, we have $v(x^{(n)}) = p^{-n}v(x^{(0)}) = v(x)$. But $v(L^\times) = v(\widehat{L}^\times)$ is discrete, so $v(x) = v(x^{(0)})$, which means that $x = 0$. □

As said in the beginning of this section, the tilting functor establishes a bridge between the categories over fields and those over their tilting. A famous example is the tilting equivalence, stating that the tilting functor gives an equivalence between the categories of extensions over a perfectoid field and those of its tilt, where the perfectoid field is a complete non-discrete semi-perfect valuation field ([5]). However, the above proposition shows that, for those "small" field, which is far from perfectoid, the behavior of tilting functor are quite trivial. Thus, in the following, we will mainly consider the tilts of fields over K^{cyc}.

3.3. **The element ϵ and π.** Let $\epsilon = (1, \epsilon^{(1)}, \cdots)$ be the element in R such that $\epsilon^{(0)} = 1$ and $\epsilon^{(1)} \neq 1$. Then $\epsilon^{(n)}$ is a primitive p^n-th root of unity in $\overline{K}$ satisfying the compatibility condition $(\epsilon^{(n+1)})^p = (\epsilon^{(n)})$. Thus

$$L^{\mathrm{cyc}} = \bigcup_{n\in\mathbb{N}} L(\epsilon^{(n)}).$$

Let $\pi = \epsilon - 1$.

Lemma 3.11. *$\epsilon, \pi \in (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat$, $v(\pi) = \frac{p}{p-1} > 1$ and hence $\epsilon \in U_R^1$.*

Proof. By definition we know that these elements lie in $(\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat$. As for the valuation, note that

$$\pi^{(0)} = \lim_{m\to\infty} (\epsilon^{(m)} - 1)^{p^m}.$$

Let $P(X) = X^{(p-1)p^{m-1}} + X^{(p-2)p^{m-1}} + \cdots + 1 = \frac{X^{p^m}-1}{X^{p^{m-1}}-1}$, and $Q(X) = P(X+1)$. Clearly $\epsilon^{(m)}$ is a root of $P(X)$, whence $\epsilon^{(m)} - 1$ is a root of $Q(X)$. By Eisenstein's criterion, $Q(X)$ is irreducible. Thus $\epsilon^{(m)} - 1$ is uniformizer of the cyclotomic extension $K_0(\epsilon^{(m)})/K_0$. As a result, $v(\epsilon^{(m)} - 1) = \frac{1}{p^{m-1}(p-1)}$, and thus

$$v(\pi) = \lim_{m\to\inf} \frac{p^m}{p^{m-1}(p-1)} = \frac{p}{p-1}.$$

□

Proposition 3.12. *For $g \in G_{K_0}$,*

$$g(\epsilon) = \epsilon^{\chi(g)}, \text{ and } g(\pi) = (1+\pi)^{\chi(g)} - 1.$$

Thus $\epsilon^{\mathbb{Z}_p} \cong \mathbb{Z}_p(1)$ as G_{K_0}-modules.

3.4. **Fontain-Wintenberger's Theorem.** Let $H = H_{K_0} = \mathrm{Gal}(\overline{K}/K^{\mathrm{cyc}})$, then $R^H = (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat$ by Proposition 3.9. Furthermore, its residue field is $k_{K_0^{\mathrm{cyc}}} = k$. Since $\pi \in (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat$, $v(\pi) > 1$, the residue field $k \in R^H = (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat$ and R^H is complete, we have

$$k[[\pi]] \subset R^H = (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat, \text{ and } k((\pi)) \subset F^H.$$

If $x = (x^{(n)}) \in R^H$ and $x = y^p$, then $y = (x^{(n+1)}) \in R^H$ and hence R^H is perfect, and similarly so is F^H. Since R and F are complete, then so are R^H and F^H. As a result,

$$\widehat{k[[\pi]]^{\mathrm{perf}}} \subset R^H, \ \widehat{k((\pi))^{\mathrm{perf}}} \subset F^H.$$

In fact, the converse is true as well:

Theorem 3.13. *$\widehat{k[[\pi]]^{perf}} = R^H$ and $\widehat{k[[\pi]]^{perf}} = F^H$. Furthermore, for $m \in \mathbb{N}$, the projection map*

$$\rho_m : R \to \mathcal{O}_{\overline{K}}/p\mathcal{O}_{\overline{K}}, \ (x_n) \mapsto x_m$$

has image

$$\rho_m(R^H) = \mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}}.$$

Proof. Set $E_0 = k((\pi))$, and $E' = E_0^{\mathrm{perf}}$ $L = K_0^{\mathrm{cyc}}$, it suffices to show that $\mathcal{O}_{E'}$ is dense in R^H. Since R^H is the inverse limit of $\mathcal{O}_L/p\mathcal{O}_L$, the assertion follow from that

$$\rho_m(\mathcal{O}_{E'}) = \mathcal{O}_L/p\mathcal{O}_L \text{ for all } m \in \mathbb{N}.$$

So it suffices to show that $\mathcal{O}_L/p\mathcal{O}_L \subset \rho_m(\mathcal{O}_{E'})$ for all m.

Set $\tilde{\pi}_n = \epsilon^{(n)} - 1$, then

$$\mathcal{O}_{K_0}[\epsilon^n] = W[\tilde{\pi}_n], \ \mathcal{O}_L = \bigcup_{n=0}^{\infty} W[\tilde{\pi}_n].$$

Write $\pi = (\pi_n)_{n\in\mathbb{N}}$. Then $\pi_n = \epsilon_n - 1$ is also the image of $\tilde{\pi}_n$ in $\mathcal{O}_L/p\mathcal{O}_L$, thus $\mathcal{O}_L/p\mathcal{O}_L$ is a k-algebra generated as a k-algebra by π_n's. Since $k \subset \mathcal{O}_{E_0} \subset \rho_m(\mathcal{O}_{E'})$, we are reduced to show that

$$\pi_n \in \rho_m(\mathcal{O}_F) = \rho_m(k[[\pi]]^{\mathrm{perf}})$$

for all $m, n \in \mathbb{N}$. In fact, for all $s \in \mathbb{Z}$, $\pi^{p^{-s}} \in k[[\pi]]^{\mathrm{perf}}$, and

$$\begin{aligned}\pi^{p^{-s}} &= \epsilon^{p^{-s}} - 1 = (\epsilon^{(n+s)}) - 1 \\ &= (\epsilon_{n+s} - 1)\end{aligned}$$

where $\epsilon^{(n)} = 1$ if $n < 0$. Since $\epsilon_{n+s} - 1 = \pi_{n+s}$ for $n + s \geq 0$, let $s = n - m$, we get

$$pi_n = \rho_m(\pi^{p^{m-n}}) \in \rho_m(k[[\pi]]^{\mathrm{perf}}).$$

This completes the proof. □

Theorem 3.14. *Let E_0^s be the separable closure of $E_0 = k((\pi))$ inside F, then E_0^s is dense in F and is stable under G_{K_0}. Moreover, for any $g \in \mathrm{Gal}(\overline{K}/K_0^{\mathrm{cyc}})$, $g|_{E_0^s} \in \mathrm{Gal}(E_0^s/E_0)$, and the map*

$$\mathrm{Gal}(\overline{K}/K_0^{\mathrm{cyc}}) \to \mathrm{Gal}(E_0^s/E_0), g \mapsto g|_{E_0^s}$$

is an isomorphism.

Proof. Let us first show that E_0^s is dense in F. As E_0^s is separably closed, $\widehat{E_0^s}$ is algebraically closed. Let $\overline{E_0}$ be the algebraic closure of E_0 in F, then it is enough to check that $\overline{E_0}$ is dense in F. In other words, we need to prove that $\mathcal{O}_{\overline{E_0}}$ is dense in R. As R is the inverse limit of $\mathcal{O}_{\overline{K}/p\overline{K}}$, let ρ_n be the projection map as in the above theorem, then it suffices to show that

$$\rho_n(\mathcal{O}_{\overline{E_0}}) = \mathcal{O}_{\overline{K}}/p\mathcal{O}_{\overline{K}}$$

for every $m \in \mathbb{N}$. As $\overline{E_0}$ is algebraically closed, it suffice to show that

$$\rho_0(\mathcal{O}_{\overline{E_0}}) = \mathcal{O}_{\overline{K}}/p\mathcal{O}_{\overline{K}}.$$

Since $\mathcal{O}_{\overline{K}} = \varinjlim_{[L:K<\infty],\ L/K_0 \text{ Galois}} \mathcal{O}_L$, it is enough to check that for any finite Galois extension L/K_0,

$$\mathcal{O}_L/p\mathcal{O}_L \subset \rho_0(\mathcal{O}_{\overline{E_0}}).$$

Let $K_{0,n} = K_0(\epsilon^{(n)})$ and $L_n = K_{0,n}L$, then $L_n/K_{0,n}$ is Galois. Suppose J_n is the Galois group of $L_n/K_{0,n}$, then for n sufficiently large, we have $J_n = J_{n+1} = \cdots =: J$ and $K_{0,n} \cap L = K_{0,n+1}L = \cdots =: L'$. Since $\overline{k} \subset \mathcal{O}_{\overline{E_0}}$, we may replace K_0 by a finite unramified extension such that $L_n/K_{0,n}$ is totally ramified for any n. Let ν_n be a generator of the maximal ideal of $\mathcal{O}_{L_n}$, then we can deduce that $\mathcal{O}_{L_n} = \mathcal{O}_{K_{0,n}}[\nu_n]$ since $L_n/K_{0,n}$ is totally ramified. Since $\rho_0(\mathcal{O}_{\overline{E_0}}) \supset \rho_0(R^H) \supset \mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}} \supset \mathcal{O}_{K_{0,n}}/p\mathcal{O}_{K_{0,n}}$, it suffices to show that $\overline{\nu_n} \in \rho_m(\mathcal{O}_{\overline{E_0}})$, where $\overline{\nu_n}$ is the image of ν_n in $\mathcal{O}_{L_n}/p\mathcal{O}_{L_n}$.

Let $P(X) \in K_{0,n}[X]$ be the minimal polynomial of ν_n, which is an Eisenstein polynomial. When n is sufficiently large, P_n is of degree $d = |J|$. Write $P_n(X) = \prod_{g\in J}(X - g(\nu_n))$. We need the following lemma:

Lemma 3.15. *For any $g \in J$, $g \neq 1$, we have $v(g(\nu_n) - \nu_n) \to 0$ as $n \to \infty$.*

Proof. By the ramification theory, we have

$$v_K(\mathfrak{D}_{L_n/K_{0,n}}) = \int_{-1}^{\infty} (|\mathrm{Gal}(K_{0,n}/L')^v|^{-1} - |\mathrm{Gal}(L_n/L')^v|^{-1})dv$$

There exists $h > 0$ such that $\mathrm{Gal}(L/L')^v = 1$ for $v \geq h$, then $\mathrm{Gal}(K_{0,n}/L')^v = \mathrm{Gal}(L_n/L')^v$ for $v \geq h$. Thus

$$v_K(\mathfrak{D}_{L_n/K_{0,n}}) = \int_{-1}^{h} (|\mathrm{Gal}(K_{0,n}/L')^v|^{-1} - |\mathrm{Gal}(L_n/L')^v|^{-1})dv \leq \int_{-1}^{h} |\mathrm{Gal}(K_{0,n}/L')^v|^{-1}dv \to 0$$

as $n \to 0$. On the other hand, $v_K(\mathfrak{D}_{L_n/K_{0,n}}) = \sum_{g\neq 1} i_{L_n/K_{0,n}}(g)$, we conclude that $v(g(\nu_n) - \nu_n) = i_{L_n/K_{0,n}}(g) \to 0$ as $n \to 0$. □

With this lemma, choose n such that $v(g(\nu_n) - \nu_n) < 1/d$ for $g \neq 1$. Let $\overline{P_n}(X) \in \mathcal{O}_{K_{0,n}}/p\mathcal{O}_{K_{0,n}}[X]$ be the polynomial $P_n(X) \pmod p$. We choose $Q(X) \in \mathcal{O}_{\overline{E_0}}[X]$, monic of degree d, a lifting of $\overline{P_n}$, Let x be a root of $Q(X)$. Write $\beta = \rho_0(x)$. Suppose $b \in \mathcal{O}_{\overline{K}}$ is a

lifting of β, then there exists $g_0 \in J$ such that $v(b - g_0(\nu_n)) \geq v(b - g(\nu_n))$ for all $g \in J$. Not that

$$P_n(b) = \prod_{g \in J}(b - g\nu_n), \text{ and } v(P_n(b)) \geq 1,$$

then

$$v(g_0^{-1}b - \nu_n) = v(b_n - g_0\nu_n) \geq 1/d > v(\nu_n - g(\nu_n)), \ \forall g \in J - \{1\}.$$

By Krasner's lemma, $\nu_n \in K_{0,n}(g_0^{-1}b)$, moreover, $\overline{\nu_n} \in \rho_0(\mathcal{O}_{\overline{E_0}})$. This proves the first part of the theorem.

For any $a \in E_0^s$, let $P(X) \in E_0[X]$ be the minimal polynomial of a over E_0. To prove $g(a) \in E^s$ for every $g \in G_{K_0}$, it suffices to show that $g(E_0) = E_0$. In fact, this follows direct from the fact that

$$g(\pi) = (1 + \pi)^{\chi(g)} - 1.$$

Moreover, for any $g \in \mathrm{Gal}(\overline{K}/K_0^{\mathrm{cyc}})$, g fixes E_0, and hence $g(a)$ is a root of $g(P) = P$. Thus we get a homomorphism

$$\mathrm{Gal}(\overline{K}/K_0^{\mathrm{cyc}}) \to \mathrm{Gal}(E_0^s/E_0), g \mapsto g|_{E_0^s}$$

Finally, we prove that this homomorphism is an isomorphism. Injectivity: suppose g is in the kernel of the homomorphism, i.e. $g(a) = a$ for all $a \in E_0^s$. Then $g(a) = a$ for all $a \in F$ by continuity. In particular, $g(a^{(0)}) = a^{(0)}$ for all $a \in F$. Since the projection $F \to C,\ a \mapsto a^{(0)}$ is surjective, this proves that $g = 1$ on C, hence also on $\overline{K}$.

Surjectivity: We identify $H = \mathrm{Gal}(\overline{K}/K_0^{\mathrm{cyc}})$ with a closed subgroup of $\mathrm{Gal}(E_0^s/E_0)$ by injectivity. If the above map is not onto, we have

$$E_0 \subsetneq (E_0^s)^H \subset F^H = \widehat{E_0^{\mathrm{perf}}},$$

that is, $(E_0^s)^H$ is a separable proper extension of E_0 contained in $\widehat{E_0^{\mathrm{perf}}}$. To finish the proof, we just need to prove the following lemma:

Lemma 3.16. *Let E be a complete field of characteristic $p > 0$. There is no nontrivial separable extension F/E contained in E^{perf}.*

Proof. Otherwise, we could find such a nontrivial finite separable extension E' of E contained in E^{perf}. There are $d = [E' : E] > 1$ distinct E-embeddings $\sigma_1, \cdots, \sigma_d$ of E' to E^s. We can extend σ_i to E'^{perf} naturally. The extended map is continuous, hence it can be further extended to $\widehat{E'^{\mathrm{perf}}} = \widehat{E^{\mathrm{perf}}}$. But σ_i acts as the identity map on E^{perf}, so it acts as the identity map on $\widehat{E^{\mathrm{perf}}}$. This is a contradiction. □

□

3.5. **E-series.** From now on, let $E_0 = k((\pi))$ and E_0^s be the separable closure of E_0 inside Frac$R$. Recall that our final goal is to classify p-adic Galois representations of p-adic fields. Since we have known that for fields of characteristic p, this task can be completed by étale ϕ-modules. By Theorem 3.14, we can classify the normal subgroup H_{K_0} of G_{K_0} via this method as well. To make the procedure precise, we use the following notation

Definition 3.17. For any finite extension L of K_0 inside $\overline{K}$, set

$$E = E_0^s, \quad \text{and} \quad E_L = E^{H_L}.$$

And we use the notion $^+$ to specify the ring of integers and $\widetilde{\ }$ means the completion. For example, we have

$$E^+ = \mathcal{O}_E, \quad \widetilde{E} = F, \quad \text{and} \quad \widetilde{E}^+ = R.$$

We can describe E_L and $\widetilde{E}_L$ explicitly:

Proposition 3.18. *For L a finite extension of K_0, let $n(L)$ be given by Theorem 2.1 and k_L^c be the residue field of L^{cyc}. Then*

$$\widetilde{E}_L^+ = (\mathcal{O}_{K_0^{\mathrm{cyc}}}/p\mathcal{O}_{K_0^{\mathrm{cyc}}})^\flat = \{(x_n) \in R \mid x_n \in \mathcal{O}_{L^{\mathrm{cyc}}}/p\mathcal{O}_{L^{\mathrm{cyc}}}, x_{n+1}^p = x^n\},$$

$$E_L^+ = \{\{(x_n) \in R \mid x_n \in \mathcal{O}_{L(\epsilon^{(n)})}/p\mathcal{O}_{L(\epsilon^{(n)})}, x_{n+1}^p = x^n \text{ for } n \geq n(L)\}\}.$$

and

$$E_L = E_L^+[\frac{1}{\pi_L}] = k_L^c((\pi_L)), \quad \widetilde{E}_L = \widetilde{E}_L^+[\frac{1}{\pi_L}] = k_L^c\widehat{(\!(\pi_L)\!)}^{perf},$$

where π_L is any uniformizer of E_L.

Note that $\mathbf{\Gamma}_{K_0} = G_{K_0}/H_{K_0}$ acts on E_0, then G_{K_0} acts on E and hence $\mathbf{\Gamma}_L$ acts on E_L. Set

$$\mathbf{E}_L = E^{\mathbf{H}_L} = E_L^{\Delta_L},$$

and set $\mathbf{E_0} = \mathbf{E_{K_0}}$.

Lemma 3.19. *(1) If $p \neq 2$, set*

$$\bar{\pi}_0 := \sum_{a \in \mathbb{F}_p} \epsilon^{[a]}.$$

Then:
(i) $\bar{\pi}_0 \in \mathbf{E}_0$ and $\bar{\pi}_0 = \pi^{p-1}\lambda$ with $\lambda \equiv 1$ (modπ).
(ii) $\mathbf{E_0} = k((\bar{\pi}_0))$.
(2) If $p = 2$, set $\bar{\pi}_0 := \pi + \pi^{-1}$. Then $\mathbf{E}_0 = k((\bar{\pi}_0))$.

3.6. **Cohen rings of E-series.** Let $A_{\mathrm{inf}} := W(R)$ be the ring of Witt vectors over R. For every $c \in R$, we write $[c]$ for its Teichmüller lift in A_{inf}. Set

$$\boldsymbol{\pi} := [\epsilon] - 1 \in A_{\mathrm{inf}},$$

then $\boldsymbol{\pi} = (\pi, *, \cdots)$ is a lifting of π. Since R is a valuation ring and $k((\pi))^s$ is dense in R, the topology on R is nothing but the π-adic topology. Hence as the Witt ring of R, the topology on A_{inf} is the $(p, \boldsymbol{\pi})$-adic topology, which means the series in p and in $\boldsymbol{\pi}$ converges. In particular, the series

$$\sum_{n=0}^{\infty} \lambda_n \boldsymbol{\pi}^n, \ \lambda_n \in W$$

converges, which gives a continuous embedding

$$W[[\boldsymbol{\pi}]] \hookrightarrow A_{\mathrm{inf}}.$$

We identify $W[[\boldsymbol{\pi}]]$ with a closed subring of A_{inf}. As $\boldsymbol{\pi}$ is invertible in $W(F)$, we have

$$W((\boldsymbol{\pi})) = W[[\boldsymbol{\pi}]][1/\boldsymbol{\pi}] \subset W(F),$$

whose elements are of the form

$$\sum_{n=-\infty}^{\infty} \lambda_n \boldsymbol{\pi}^n, \ \lambda_n \in W, \ \lambda_n = 0 \text{ for } n \ll 0$$

Put

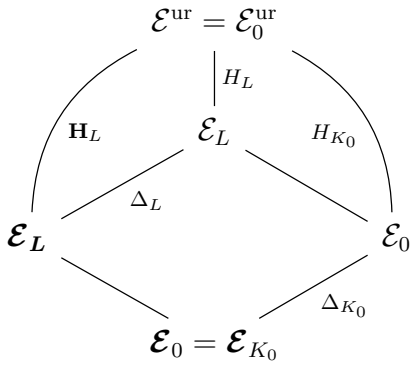
$$\mathcal{O}_{\mathcal{E}_0} := \{ \sum_{n=-\infty}^{\infty} \lambda_n \boldsymbol{\pi}^n,\ \lambda_n \in W,\ \lambda_n \to 0 \text{ as } n \to -\infty\},$$

$\mathcal{O}_{\mathcal{E}_0}$ is the p-adic completion of $W((\boldsymbol{\pi}))$ in $W(F)$, then $\mathcal{O}_{\mathcal{E}_0}$ is a complete DVR whose maximal ideal is generated by p and whose residue field is $E_0 = k((\pi))$. Thus $\mathcal{O}_{\mathcal{E}_0}$ is nothing but the Cohen ring of E_0. Write $\mathcal{E}_0 = \mathcal{O}_{\mathcal{E}_0}[\frac{1}{p}]$ be its fraction field.

Let

$$\pi_0 = -p + \sum_{a \in \mathbb{F}_p} [\epsilon]^{[a]} \quad \text{if } p \neq 2 \text{ and} \quad \pi_0 = [\epsilon] + [\epsilon^{-1}] - 2 \quad \text{if } p = 2,$$

then $\boldsymbol{\mathcal{E}}_0 := \mathcal{E}_0^{\Delta_{K_0}}$ has integer ring $\mathcal{O}_{\boldsymbol{\mathcal{E}}_0} = W((\boldsymbol{\pi}_0))$ which at the same time is the Cohen ring of $\mathbf{E}_0 = k((\bar{\pi}_0))$. We have the following commutative diagram



3.7. **Étale φ, Γ-modules.** Suppose V is a p-adic representation of $H_K = \mathrm{Gal}(\overline{K}/K) = \mathrm{Gal}(E^s/E) = \mathrm{Gal}(\mathcal{E}^{\mathrm{ur}}/\mathcal{E})$, by Theorem 2.11,

$$\mathbf{D}(V) := (\widehat{\mathcal{E}}^{\mathrm{ur}} \otimes)$$

is an étale φ-module over $\mathcal{E}$, and $\mathbf{D}$ defines an equivalence of categories from $\mathbf{Rep}_{\mathbb{Q}_p}(H_K)$ to $\mathcal{M}_{\varphi}^{\text{ét}}(\mathcal{E})$ with a quasi inverse functor given by

$$\mathbf{V} : D \mapsto (D \otimes_{\mathcal{E}} \widehat{\mathcal{E}}^{\mathrm{ur}})^{H_K}.$$

At the same time, there is a similar statement for $\mathbb{Z}_p$-representations of H_K. Now, let V be a $\mathbb{Z}_p$ or p-adic Galois representation of G_K. Set

$$\mathbf{D}(V) := (\mathcal{O}_{\widehat{\mathcal{E}^{ur}}} \otimes_{\mathbb{Z}_p} V)^{H_K} \quad \text{or} \quad (\widehat{\mathcal{E}^{ur}} \otimes_{\mathbb{Q}_p} V)^{H_K}.$$

Since $\boldsymbol{\Gamma}_K = G_K/H_K$, to give a representation of G_K, it is equivalent to give a representation of H_K with a compatible $\boldsymbol{\Gamma}_K$-action. Accordingly, we define the (φ, Γ)-module in a similar way:

Definition 3.20. Let Γ be a topological group. A (φ, Γ)-module D over $(\mathcal{O}_{\mathcal{E}})$ (resp.$\mathcal{E}$) is a φ-module over $(\mathcal{O}_{\mathcal{E}})$ (resp.$\mathcal{E}$) with an action of Γ which is semi-linear, and commutes with φ. D is called étale if it is an étale φ-module and the action of Γ is continuous.

Then we conclude by Theorem 2.11:

Theorem B2. *The functor*

$$\mathbf{D} : \mathbf{Rep}_{\mathbb{Z}_p}(G_K) \to \mathcal{M}_{\varphi, \boldsymbol{\Gamma}_K}^{\acute{e}t}(\mathcal{O}_{\mathcal{E}}), V \mapsto (\hat{\mathcal{O}_{\mathcal{E}^{ur}}} \otimes_{\mathbb{Z}_p} V)^{H_K}$$

is an equivalence of categories and the functor

$$\mathbf{V}:\mathcal{M}^{\acute{e}t}_{\varphi,\mathbf{\Gamma}_K}(\mathcal{O}_{\mathcal{E}})\to\mathbf{Rep}_{\mathbb{Z}_p}(G_K),M\mapsto(\hat{\mathcal{O}_{\mathcal{E}^{ur}}}\otimes_{\mathcal{O}_{\mathcal{E}}}M)_{\varphi=id}.$$

is a quasi-inverse.

Theorem B2. *Then the functor*

$$\mathbf{D}:\mathbf{Rep}_{\mathbb{Q}_p}(G_K)\to\mathcal{M}^{\acute{e}t}_{\varphi,\mathbf{\Gamma}_K}(\mathcal{E}),V\mapsto(\hat{\mathcal{E}^{ur}}\otimes_{\mathbb{Q}_p}V)^{H_K}$$

is an equivalence of categories and the functor

$$\mathbf{V}:\mathcal{M}^{\acute{e}t}_{\varphi,\mathbf{\Gamma}_K}(\mathcal{E})\to\mathbf{Rep}_{\mathbb{Q}_p}(G_K),M\mapsto(\hat{\mathcal{E}^{ur}}\otimes_{\mathcal{E}}M)_{\varphi=id}.$$

is a quasi-inverse.

It turns out that we can replace $\mathcal{M}^{\text{ét}}_{\varphi,\mathbf{\Gamma}_K}(\mathcal{E})$ by the category $\mathcal{M}^{\text{ét}}_{\varphi,\Gamma_K}(\boldsymbol{\mathcal{E}})$, which may be easier to learn since Γ_K is pro-cyclic. Set $\mathbf{D}'(V):=(\widehat{\mathcal{E}^{ur}}\otimes_{\mathbb{Q}_p}V)^{\mathbf{H}_K}$ for every p-adic representation V, then $\mathbf{D}'(V)$ is an étale (φ,Γ_K)-module over $\boldsymbol{\mathcal{E}}$, and

$$\mathbf{D}'(V)=(\mathbf{D}(V))^{\Delta_K}.$$

By Hilbert's Theorem 90, the map

$$\mathcal{E}\otimes_{\boldsymbol{\mathcal{E}}}\mathbf{D}'(V)\xrightarrow{\sim}\mathbf{D}(V)$$

is an isomorphism, and hence both $\mathcal{M}^{\text{ét}}_{\varphi,\mathbf{\Gamma}_K}(\mathcal{E})$ and $\mathcal{M}^{\text{ét}}_{\varphi,\Gamma_K}(\boldsymbol{\mathcal{E}})$ are equivalent categories of $\mathbf{Rep}_{\mathbb{Q}_p}(G_K)$. Accordingly for $\mathbb{Z}_p$-representations, the similar result is true as well.

4. Hodge-Tate Representations and De-Rham Representations

4.1. B_{HT} **and Hodge-Tate Representations.** Let $\mathbb{Z}_p(1)\cong\mathbb{Z}_p\cdot t$ be the multiplicative group free of rank 1, with G_K-action via the cyclotomic character χ:

$$g(t)=\chi(g)t,$$

called the Tate module. For $i\in\mathbb{Z}$, the Tate twist $\mathbb{Z}_p(i)=\mathbb{Z}_p\cdot t^i$ is the free $\mathbb{Z}_p$-module with G_K action through χ^i. Moreover, for a $\mathbb{Z}_p$-module M and $i\in\mathbb{Z}$, the i-th Tate twist of M is $M(i)=M\otimes_{\mathbb{Z}}\mathbb{Z}_p(i)$.

Definition 4.1. The ring of periods of Hodge-Tate, or the Hodge-Tate ring B_{HT} is defined to be

$$B_{\text{HT}}=\bigoplus_{i\in\mathbb{Z}}C(i)=C[t,\frac{1}{t}].$$

Proposition 4.2. *B_{HT} is $(\mathbb{Q}_p,G_K)$-regular, which means that:*
1). B_{HT} is a domain;
2). $\text{Frac}(B_{\text{HT}})^{G_K}=B_{\text{HT}}^{G_K}=K$;
3). For every $b\in B_{\text{HT}}$, $b\neq 0$, if $g(b)\in\mathbb{Q}_pb$ for all $g\in G_K$, then b is invertible.

Definition 4.3. A p-adic representation V of G_K is called Hodge-Tate if it is B_{HT}-admissible.

4.2. B_{dR} **and its properties.** Recall that $F = C^\flat$ is the tilt of C. Fix a valuation ν on C such that $\nu(p) = 1$. Let $\nu^\flat$ be the valuation on F define by $\nu^\flat(c) = \nu(c^\sharp)$ for every $c \in F$.

Definition 4.4. We define the infinitesimal period ring, denoted by A_{inf}, to be the ring of Witt vectors over $\mathcal{O}_F$. For every $c \in \mathcal{O}_F$, we write $[c]$ for its Teichmüller lift in A_{inf}.[1]

Let $a = (a_0, \cdots, a_n, \cdots) \in W(R)$, where $a_n \in R$ for every n. Each a_n can be written as $a_n = (a_{n,0}, a_{n,1}, \cdots)$ with $a_{n,i} \in \mathcal{O}_C/p\mathcal{O}_C$. Then $a \mapsto (a_{0,n}, \cdots, a_{n-1,n})$ gives a map $W(R) \to W_n(\mathcal{O}_C/p\mathcal{O}_C)$. For every $n \in \mathbb{N}$, the following diagram is commutative:

$$\begin{array}{ccc} & & W_{n+1}(\mathcal{O}_C/p\mathcal{O}_C) \\ & \nearrow & \downarrow f_n \\ W(R) & \longrightarrow & W_n(\mathcal{O}_C/p\mathcal{O}_C) \end{array}$$

This induces an isomorphism $W(R) \to \varprojlim_{f_n} W_n(\mathcal{O}_C/p\mathcal{O}_C)$. In fact, one can show that this is an isomorphism. Moreover, It is also a homeomorphism if the right hand side is equipped with the inverse limit topology of the discrete topology.

On the other hand, we have a surjective map

$$W_{n+1}(\mathcal{O}_C) \to W_n(\mathcal{O}_C/p\mathcal{O}_C),\ (a_0, \cdots, a_n) \mapsto (\overline{a_0}, \cdots, \overline{a_{n-1}}).$$

The kernel of the map is $I = \{(pb_0, \cdots, pb_{n-1}, a_n) \mid b_0, \cdots, b_{n-1}, a_n \in \mathcal{O}_C\}$. Recall the map

$$w_n : W_{n+1}(\mathcal{O}_C/p\mathcal{O}_C) \to \mathcal{O}_C, (a_0, \cdots, a_n) \mapsto a_0^{p^n} + pa_1^{p^{n-1}} \cdots + p^n a_n,$$

denote $\overline{w_{n+1}}$ the composite $W_{n+1}(\mathcal{O}_C) \to \mathcal{O}_C \to \mathcal{O}_C/p^n\mathcal{O}_C$. Since

$$w_{n+1}(pb_0, \cdots, pb_{n-1}, a_n) = (pb_0)^{p^n} + \cdots + p^{n-1}(pb_{n-1})^p + p^n a_n \in p^n\mathcal{O}_C,$$

the above map induces homomorphism

$$\theta_n : W_n(\mathcal{O}_C/p\mathcal{O}_C) \to \mathcal{O}_C/p^n\mathcal{O}_C.$$

Furthermore, the diagram

$$\begin{array}{ccc} W_{n+1}(\mathcal{O}_C/p\mathcal{O}_C) & \xrightarrow{\theta_{n+1}} & \mathcal{O}_C/p^{n+1}\mathcal{O}_C \\ \downarrow f_n & & \downarrow \\ W_n(\mathcal{O}_C/p\mathcal{O}_C) & \xrightarrow{\theta_n} & \mathcal{O}_C/p^n\mathcal{O}_C. \end{array}$$

Finally we define the map θ be the composite

$$\theta : A_{\mathrm{inf}} = W(R) \cong \varprojlim_{f_n} W_n(\mathcal{O}_C/p\mathcal{O}_C) \to \mathcal{O}_C$$

Proposition 4.5. *θ is a surjective homomorphism which is continuous and commute with the action of G_{K_0}. Precisely, if $x = (x_0, \cdots, x_n, \cdots) \in A_{\mathrm{inf}} = W(R)$ with $x_n \in R$, write $x_n = (x_n^{(m)})$ for $x_n^{(m)} \in \mathcal{O}_C$, then*

$$\theta(x) = \sum_{n=0}^{\infty} p^n x_n^{(n)}.$$

[1] in Fontaine's notation, $F = R(C) = \mathrm{Fr}(R)$, $A_{\mathrm{inf}} = W(R)$

If we write $x = \sum_n p^n[x_n]$ *for* $x_n \in R$*, then*

$$\theta(x) = \sum_n p^n x_n^{(0)} = \sum_n p^n x_n^\sharp.$$

θ induces $\theta[\frac{1}{p}] : A_{\mathrm{inf}}[\frac{1}{p}] \to C$. It can be computed by $\sum_{n\geq n_0\in\mathbb{Z}} p^n[x_n] \mapsto \sum_{n\geq n_0\in\mathbb{Z}} p^n x_n^\sharp$, which again is surjective and continuous. If there is no confusion, we will denote $\theta[\frac{1}{p}]$ by θ as well.

Let $\varpi := (-p)^\flat$ be an element in R such that $(\varpi)^\sharp = -p$. Let $\xi = [\varpi] + p \in W(R) = A_{\mathrm{inf}}$.

Proposition 4.6. $\ker\theta$ *and* $\ker\theta[\frac{1}{p}]$ *are both principal ideals, generated by* ξ *in* A_{inf} *and* $A_{\mathrm{inf}}[\frac{1}{p}]$ *respectively. Moreover* $\bigcap_{n\geq 0}(\ker\theta)^n = \bigcap_{n\geq 0}(\ker\theta[\frac{1}{p}])^n = 0$.

Definition 4.7. The ring B_{dR}^+ is the $(\ker\theta)$-adic completion of $A_{\mathrm{inf}}[\frac{1}{p}]$, i.e.

$$B_{\mathrm{dR}}^+ = \varprojlim_{n\in\mathbb{N}} A_{\mathrm{inf}}[\frac{1}{p}]/(\ker\theta)^n = \varprojlim_{n\in\mathbb{N}} A_{\mathrm{inf}}[\frac{1}{p}]/(\xi)^n.$$

The field of p-adic periods $B_{\mathrm{dR}} = \mathrm{Frac}(B_{\mathrm{dR}}^+) = B_{\mathrm{dR}}^+[\frac{1}{\xi}]$ is the fractional field of B_{dR}^+.

Remark. B_{dR} is a DVR with residue field $B_{\mathrm{dR}}/\ker\theta = (A_{\mathrm{inf}}/\ker\theta)[\frac{1}{p}] = C$ with the normalized valuation defined by $w(\xi) = 1$. However, we actually do not consider it by this means. Instead of the topology as a DVR, i.e. the ξ-adic topology, we prefer to put on B_{dR} the induced topology by the inverse limit, with the topology on each component $A_{\mathrm{inf}}/(\ker\theta)^n$ being the induced quotient topology of $A_{\mathrm{inf}} = W(R)[\frac{1}{p}]$. In this case, the topology on the residue field C is the natural topology by p-adic valuation.

B_{dR} is not a ring with a proper Frobenius action. This is one of the motivations for the ring of crystalline periods B_{cris}:

Proposition 4.8. *The Frobenius map* $\varphi : A_{\mathrm{inf}} = W(R)[\frac{1}{p}] \to A_{\mathrm{inf}}$ *is not extendable to a continuous map* $B_{\mathrm{dR}}^+ \to B_{\mathrm{dR}}^+$.

Proof. Since $\theta([\varpi]^{\frac{1}{p}} + p) \neq 0$, $[\varpi]^{\frac{1}{p}} + p$ is invertible in B_{dR}^+. But if φ were a natural extension of the Frobenius map, on one hand $\varphi(1/([\varpi]^{\frac{1}{p}} + p))$ should still be invertible in B_{dR}^+, on the other hand one should have $\varphi(1/([\varpi]^{\frac{1}{p}} + p)) = \frac{1}{[\varpi]+p} = \frac{1}{\xi}$, which is not in B_{dR}^+. □

Definition 4.9. For $i \in \mathbb{Z}$, let $\mathrm{Fil}^i B_{\mathrm{dR}}$ be the free B_{dR}^+-module generated by ξ^i. The filtration on B_{dR} is the decreasing exhaustive and separated filtration

$$\cdots \supset \mathrm{Fil}^i B_{\mathrm{dR}} \supset \mathrm{Fil}^{i+1} B_{\mathrm{dR}} \supset \cdots.$$

Lemma 4.10. $\overline{K}$ *is naturally a subfield of* B_{dR}^+*, preserving Galois action and* $\overline{K} \cap \mathrm{Fil}^1 B_{\mathrm{dR}} = 0$.

Proof. Since $A_{\mathrm{inf}} = W(R)$ and $A_{\mathrm{inf}}[\frac{1}{p}]$ are subrings of B_{dR}^+, $K_0 = W(k)[\frac{1}{p}]$ is a subring of B_{dR}^+. For any monic irreducible polynomial $P(X) \in K_0[X]$, under the map $K_0 \hookrightarrow B_{\mathrm{dR}}^+ \xrightarrow{\theta} \mathbb{C} + K$, $P(X) \in C[X]$ has distinct roots in $\mathbb{C} + K$. By Hensel's lemma, $P(X)$ has distinct roots in B_{dR}^+. □

Remark. In fact, B_{dR} depends only on C not on K.

Here we give another description of the filtration of B_{dR}, which indicates the connection between it and B_{HT}. Let $\epsilon \in R$ be the element given by $\epsilon^{(0)} = 1$ and $\epsilon^{(1)} \neq 1$. Since $\theta([\epsilon]-1) = \epsilon^{(0)} - 1 = 0$, $[\epsilon - 1] \subset \ker\theta = \mathrm{Fil}^1 B_{\mathrm{dR}}$, and whence

$$t := \log[\epsilon] = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{([\epsilon]-1)^n}{n} \in B_{\mathrm{dR}}^+$$

converges.

Proposition 4.11. *$t \in \mathrm{Fil}^1 B_{\mathrm{dR}} - \mathrm{Fil}^2 B_{\mathrm{dR}}$, i.e. $\xi B_{\mathrm{dR}}^+ = t B_{\mathrm{dR}}^+$.*

By the definition of t, it is exactly the unit element of the Tate module

Lemma 4.12. *$\epsilon^{\mathbb{Z}_p} \cong \mathbb{Z}_p(1)$ as G_{K_0}-modules. Precisely, for any $g \in G_{K_0}$,*

$$g(\epsilon) = \epsilon^{\chi(g)}, \quad \textit{and} \quad g(t) = \chi(g)t.$$

Thus the Tate module $\mathbb{Z}_p(1)$ can be realized as $\mathbb{Z}_p t \subset B_{\mathrm{dR}}^+$.

Definition 4.13.

$$\begin{aligned} \mathrm{gr} B_{\mathrm{dR}} &= \bigoplus_{i\in\mathbb{Z}} \mathrm{gr}^i B_{\mathrm{dR}} = \bigoplus_{i\in\mathbb{Z}} \mathrm{Fil}^i B_{\mathrm{dR}}/\mathrm{Fil}^{i+1} B_{\mathrm{dR}} \\ &= \bigoplus_{i\in\mathbb{Z}} B_{\mathrm{dR}}^+(i)/tB_{\mathrm{dR}}^+(i) = \bigoplus_{i\in\mathbb{Z}} C(i) \end{aligned}$$

Then by definition:

Proposition 4.14. $\mathrm{gr} B_{\mathrm{dR}} = B_{\mathrm{HT}}$.

Proposition 4.15. *$B_{\mathrm{dR}}^{G_K} = K$. Thus B_{dR} is $(\mathbb{Q}_o, G_K)$-regular since it is a field. Furthermore, $\mathrm{Fil}^i B_{\mathrm{dR}}$ are stable under G_K for every $i \geq 0$.*

4.3. **de-Rham representations.**

Definition 4.16. A p-adic representation is called de-Rham if it is B_{dR}-admissible.

We will establish an equivalence of the category of de-Rham representations and a specific category of linear vector spaces, namely the category of filtered K-vector spaces.

Definition 4.17. Let the category Fil_K of filtered K-vector spaces defined as follows:
i). An object of Fil_K is a finite dimensional K-vector space D equipped with a decreasing filtration indexed by $\mathbb{Z}$ which is exhaustive and separated, i.e.,
1). $\mathrm{Fil}^i D$ are K-subspaces of D;
2). $\mathrm{Fil}^i D = 0$ for $i \gg 0$;
3). $\mathrm{Fil}^i D = D$ for $i \ll 0$.
ii). A morphism $\eta : D_1 \to D_2$ is a morphism between K-vector spaces, such that $\eta(\mathrm{Fil}^i D_1) \subset \mathrm{Fil}^i D_2$ for every $i \in \mathbb{Z}$.

Definition 4.18. 1). Let D_1 and D_2 be filtered K-vector spaces. We define $D_1 \otimes D_2 = D_1 \otimes_K D_2$ to be filtered K-vector space by

$$\mathrm{Fil}^i(D_1 \otimes D_2) = \bigoplus_{i_1+i_2=i} \mathrm{Fil}^{i_1} D_1 \otimes \mathrm{Fil}^{i_2} D_2.$$

2). For every filtered vector space D over K, we define the dual filtration on the dual space $D^\vee = \mathrm{Hom}_K(D, K)$ by

$$\mathrm{Fil}^i(D^\vee) = \{f \in D^\vee \mid \mathrm{Fil}^{1-n}D \subset \ker f\}.$$

3). We define $K[0] = K$ be the trivial filtered K-vector space by

$$\mathrm{Fil}^i K = \begin{cases} K, & \text{if } i \leq 0; \\ 0, & \text{if } i > 0. \end{cases}$$

Definition 4.19. For a filtered K-vector space D, we define its associated graded K-algebra to be

$$\mathrm{gr}D = \bigoplus_{i\in\mathbb{Z}} \mathrm{Fil}^i D/\mathrm{Fil}^{i+1}D.$$

Proposition 4.20. *Let D_1, D_2 be filtered K-vector spaces, then $\mathrm{gr}(D_1 \otimes D_2) \cong \mathrm{gr}D_1 \otimes_K \mathrm{gr}D_2$.*

Definition 4.21. A morphism $\eta : D_1 \to D_2$ is called strict if $\eta(\mathrm{Fil}^i D_1) = \mathrm{Fil}^i D_2 \cap \mathrm{im}\eta$.

Definition 4.22. A short exact sequence in Fil_K is a sequence

$$0 \to D_1 \xrightarrow{\alpha} D_2 \xrightarrow{\beta} D_3 \to 0$$

with α, β strict, α is injective, β is surjective and $\mathrm{im}\alpha = \ker\beta$.

For a p-adic representation V of G_K, put $\mathbf{D}_{\mathrm{dR}}(V) := (B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V)^{G_K}$ with filtration $\mathrm{Fil}^i\mathbf{D}_{\mathrm{dR}}(V) := (\mathrm{Fil}^i B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V)^{G_K}$. Then $\mathbf{D}_{\mathrm{dR}}(V)$ is a filtered K-vector space. Moreover, we have:

Theorem 4.23. *The functor $\mathbf{D}_{\mathrm{dR}} : \mathbf{Rep}^{dR}_{\mathbb{Q}_p}(G_K) \to \mathrm{Fil}_K$ is an exact, faithful and tensor functor.*

Similarly, we put $\mathbf{D}_{\mathrm{HT}}(V) := (B_{\mathrm{HT}} \otimes_{\mathbb{Q}_p} V)^{G_K}$, we have $\mathrm{gr}\mathbf{D}_{\mathrm{dR}} = \mathbf{D}_{\mathrm{HT}}$ and $\mathrm{gr}^i\mathbf{D}_{\mathrm{dR}}(V) = (C(i) \otimes_{\mathbb{Q}_p} V)^{G_K}$. As a result:

Proposition 4.24. *If a p-adic representation is de-Rham, then it is Hodge-Tate.*

But the converse is not true:

Proposition 4.25. *There exists a p-adic representation V of G_K which is a nontrivial extension of $\mathbb{Q}_p(1)$ by $\mathbb{Q}_p$, i.e. there exists a non-split exact sequence of p-adic representations*

$$0 \to \mathbb{Q}_p \to V \to \mathbb{Q}_p(1) \to 0.$$

Such a representation V is Hodge-Tate but not de Rham.

5. Iwasawa Logarithm

Before step further, let us generalize the p-adic logarithm when defining the element t. We aim to define the logarithm map on $C = \mathbb{C}_K^\times$ and $F^\times = (\mathrm{Frac}R)^\times$, satisfying

$$\log(xy) = \log x + \log y.$$

We begin with the lemma from functional equations.

Lemma 5.1. *For any positive integer N, let c_N be the least common multiple of integer from 1 to N, then*

$$\sum_{n=1}^{N}(-1)^{n-1}\frac{X^n}{n}+\sum_{n=1}^{N}(-1)^{n-1}\frac{Y^n}{n}=\sum_{n=1}^{N}(-1)^{n-1}\frac{(X+Y+XY)^n}{n}+\frac{1}{c_N}P_N(X,Y),$$

where $P_N\in\mathbb{Z}[X,Y]$ is a sum of monomials of degree $\geq N+1$.

Logarithm on $C^\times$. We define the logarithm map on C as follows:

(a). For $x\in U_C^1$, set

$$\log x:=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{(x-1)^n}{n}.$$

This is well defined and satisfies $\log xy=\log x+\log y$ by Lemma 5.1. This function is in fact a bijection from U_C^1 to $p\mathcal{O}_C$, whose inverse is the exponential function

$$\exp: p\mathcal{O}_C\to U_C^1,\ \exp(x):=\sum_{n=0}^{\infty}\frac{x^n}{n!}.$$

(b). For $x\in U_C^+=1+\mathfrak{m}_C$, we define log by the formal power series

$$\log x:=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{(x-1)^n}{n}$$

as well.

(c). For $a\in U_C=\mathcal{O}_C$, by the cannonical decomposition

$$a=[\bar{a}]x,$$

where $\bar{a}\in\bar{k}^\times$, $[\bar{a}]\in W(\bar{k})$ and $x\in U_C^1$, set

$$\log a:=\log x.$$

(d). Finally, for $x\in C^\times$, suppose $v(x)=\frac{r}{s}$ with $r,s\in\mathbb{Z}$ and $s\geq 1$, then $v(x^s)=r=v(p^r)$, and $\frac{x^s}{p^r}=:y\in\mathcal{O}_C^\times$. We can thus have

$$s\log x=\log y+r\log p.$$

To define $\log x$, it suffices to define $\log p$. In particular, if set $\log p:=0$, the corresponding logatithm is called the Iwasawa logarithm, which means

$$\log x=\frac{1}{s}\log y.$$

Lemma 5.2. $\ker(\log: U_C^+\to C)=\boldsymbol{\mu}_{p^\infty}(C)$.

Proof. For $x\in U_C^+$, $x^{p^m}\in U_C^1$ for some $m\in\mathbb{N}$. $\log x=\frac{1}{p^m}\log x^{p^m}=0$ if and only if $x^{p^m}=1$ by the injectivity of $\log: U_C^1\to C$. Thus $\log x=\frac{1}{p^m}\log x^{p^m}=0$ if and only if $x\in\boldsymbol{\mu}_{p^\infty}(C)$. □

Logarithm on $F^\times$. We define the logarithm map

$$\log : F^\times \to B_{\mathrm{dR}},\ x \mapsto \log[x]$$

with the key rule

$$\log[xy] = \log[x] + \log[y]$$

in a similar way.

(a) For $x \in U_R^1$, put

$$\log[x] := \sum_{n=1}^{\infty} (-1)^{n-1} \frac{([x]-1)^n}{n}.$$

We need to show the series converges: in fact, one has

$$\theta([x]-1) = x^\sharp - 1 \Rightarrow \overline{\theta}([x]-1) = 0.$$

By Proposition 6.5, $\gamma_n([x]-1) = \frac{([x]-1)^n}{n!} \in A_{\mathrm{cris}}$. We rewrite the equality as

$$\log[x] = \sum_{n=1}^{\infty} (-1)^{n-1}(n-1)!\gamma_n([x]-1),$$

since $(n-1)! \to 0$ p-adically as $n \to 0$, we concludes that the series converges. Moreover, $x \in U_R^1$ means that $x - 1 = \varpi \cdot u$ for some $u \in R^\times$, and by a similar way as the proof of Proposition 6.6, we have $([x]-1)^n \in pA_{\mathrm{cris}}$. Thus we can define the exponential map $\exp : \log(U_R^1) \to A_{\mathrm{cris}}$, such that $\exp \circ \log[x] = [x]$ for every $x \in U_R^1$. It follows that the logarithm map $\log : U_R^1 \to A_{\mathrm{cris}}$ is injective.

(b). For $x \in U_R^+$, $x^{p^m} \in U_R^1$ for some $m \geq 0$. The logarithm map on U_R^1 extends uniquely to B_{cris}^+ by

$$\log : U_R^+ \to B_{\mathrm{cris}}^+,\ x \mapsto \frac{1}{p^m} \log[x^{p^m}],$$

which is injective as well since 1 does not have nontrivial p^n-th root in R.

(c). By Proposition 3.7, we have the decomposition $R^\times = \bar{k}^\times \times U_R^+$. Thus for $x \in R^\times$, $x = [\bar{x}]a$ for some $a \in U_R^+$. Similarly we put

$$\log[x] = \log[a].$$

(d). For $x \in F$, $v(x) = \frac{r}{s}$ for some relatively prime numbers $r, s \in \mathbb{Z}$ and $s \geq 1$. Then

$$v(x^s) = r = v((p^\flat)^r).$$

It remains to define $\mathbf{u} := \log[p^\flat]$.

Note that $\xi = [\varpi] + p$, $\frac{[\varpi]}{-p} = 1 - \frac{\xi}{p}$. Thus

$$\log(\frac{[\varpi]}{-p}) = \sum_{n=1}^{\infty} -\frac{\xi^n}{np^n},$$

which converges in B_{dR}^+. Similarly by setting $\mathbf{u} = \log[\varpi] := \log \frac{[\varpi]}{-p}$, we finish defining the map

$$\log : F \to B_{\mathrm{dR}}^+.$$

Lemma 5.3. *The logarithm map commutes with G_K-action. Moreover, for $x \in F$, $\varphi(\log[x]) = \log(\varphi[x]) = p\log[x]$. In this way, φ extends to im$(\log : F^\times \to B_{\mathrm{dR}}^+)$.*

Definition 5.4. Set $U = \log(U_R^+) \subset (B_{\mathrm{cris}}^+)^{\varphi=p}$.

In the next two subsection, we will see that the inclusion is in fact equality.

Clearly $t = \log[\epsilon] \in U$.

Lemma 5.5. $\ker(\log : R^\times \to U) = \overline{k}^\times$. *Furthermore, we have the following commutative diagramme:*

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \epsilon^{\mathbb{Q}_p} & \longrightarrow & U_R^+ & \xrightarrow{\log^{(0)}} & C & \longrightarrow & 0 \\
 & & \cong\downarrow\log & & \cong\downarrow\log & & \| & & \\
0 & \longrightarrow & \mathbb{Q}_p t & \longrightarrow & U & \xrightarrow{\theta} & C & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \| & & \\
0 & \longrightarrow & \mathrm{Fil}^1 B_{\mathrm{dR}} & \longrightarrow & B_{\mathrm{dR}}^+ & \xrightarrow{\theta} & C & \xrightarrow{\theta} & 0
\end{array}$$

Proof. It suffices to show

$$\begin{array}{ccc}
U_R^+ & \xrightarrow{\log^{(0)}} & C \\
\downarrow\log & & \| \\
U & \xrightarrow{\theta} & C
\end{array}$$

commutes. In fact, this is trivial because

$$\theta(\log[x]) = \theta(\sum_n (-1)^{n-1}\frac{([x]-1)^n}{n}) = \sum_n (-1)^{n-1}\frac{\theta([x]-1)^n}{n} = \sum_n (-1)^{n-1}\frac{(x^0-1)^n}{n} = \log^{(0)} x.$$

□

Clearly, we have:

Proposition 5.6. *For the map* $\log : (F)^\times \to B_{\mathrm{dR}}^+$, *its kernel is* $\overline{k}^\times$ *and its image is* $U \oplus \mathbb{Q}_p\mathbf{u}$, *where* $\mathbf{u} = \log[\varpi]$.

We need to know more about the element $\mathbf{u}$. For $g \in G_{K_0}$, $\log^{(0)}(g(\varpi)/\varpi) = \log(g(-p)/-p) = 1$, thus by Lemma 5.5, $g(\varpi)/\varpi = \epsilon^{\eta(g)}$ for some $\eta(g) \in \mathbb{Q}_p$. By this way we define a character

$$\eta : G_{K_0} \to \mathbb{Q}_p,$$

such that $g(\varpi) = \varpi\epsilon^{\eta(g)}$, and it follows that

$$g(\mathbf{u}) = \mathbf{u} + \eta(g)t.$$

Lemma 5.7. $\mathbf{u} \notin C_{\mathrm{cris}} := \mathrm{Frac}(B_{\mathrm{cris}})$.

Proof. Let $\beta = \xi/p$, then ξ and β are both inside $\mathrm{Fil}^1 B_{\mathrm{dR}}$ but not $\mathrm{Fil}^2 B_{\mathrm{dR}}$. Let $S = A_{\mathrm{inf}}[[\beta]] \subset B_{\mathrm{dR}}^+$. For every $i \in \mathbb{N}$, let $\mathrm{Fil}^i S = S \cup \mathrm{Fil}^i B_{\mathrm{dR}}$, then $\mathrm{Fil}^i S$ is a principal ideal of S generated by β^i. We denote by

$$\theta^i : \mathrm{Fil}^i B_{\mathrm{dR}} \to \mathcal{O}_C$$

the map $\beta^i\alpha \mapsto \alpha$, then $\theta^i(\mathrm{Fil}^i S) = \mathcal{O}_C$.

By construction, $A_{\mathrm{cris}} \subset S$, and hence $C_{\mathrm{cris}} = \mathrm{Frac}(A_{\mathrm{cris}}) \subset \mathrm{Frac}(S)$. We show that $\alpha\mathbf{u} \notin S$ for all $0 \neq \alpha \in S$, which is sufficient for the lemma.

By Lemma 6.1, S is separated by the p-adic topology, it suffices to show that if $r \in \mathbb{N}$ and $\alpha \in S - pS$, then $p^r \alpha \mathbf{u} \notin S$. Write $\alpha = \sum_n c_n \beta^n$ with $c_n \in A_{\mathrm{inf}}$. If for all n, $\theta(c_n) \in p\mathcal{O}_C$, then $c_n \in (p, \xi)A_{\mathrm{inf}} \subset pS$ and thus $\alpha \in pS$, which contradicts the hypothesis. Thus there exists $i < \infty$ such that $\theta(c_i) \notin p\mathcal{O}_C$, and $\theta(c_n) \in p\mathcal{O}_C$ for $n < i$. In other word, we may write

$$\alpha = p\sum_{n<i} b_n\beta^n + b_i\beta^i + \sum_{n>i} b_n\beta^n =: A_1 + A_2 + A_3$$

with $b_n \in A_{\mathrm{inf}}$ and $\theta(b_i) \notin p\mathcal{O}_C$. Suppose $j \in \mathbb{N}$ such that $j > r$ and $p^j > i$. Recall that

$$\mathbf{u} = \sum_{n\geq 1} -\frac{\xi^n}{np^n} = \sum_{n\geq 1} -\frac{\beta^n}{n},$$

write

$$-p^{j-1}\mathbf{u} = \sum_{n\geq 1} \frac{p^{j-1}\beta^n}{n} = \sum_{n<p^j} \frac{p^{j-1}\beta^n}{n} + \frac{\beta^{p^j}}{p} + \sum_{n>p^j} \frac{p^{j-1}\beta^n}{n} =: B_1 + B_2 + B_3.$$

We need only to show that $-p^{j-1}\alpha\mathbf{u} \notin S$. Note that:
(a). $B_1 \in S$, hence $\alpha B_1 \in S$;
(b). A_3B_3, A_3B_2, and A_2B_3 are all in $\mathrm{Fil}^{i+p^j+1}B_{\mathrm{dR}}$;
(c). For all n such that $p^j < n \leq p^j + i < 2p^j$, $\frac{p^{j-1}\beta^n}{n} A_1 \in S$, hence $A_1B_3 \in S + \mathrm{Fil}^{i+p^j+1}B_{\mathrm{dR}}$;
(d). $A_2B_2 = b_i\beta^{i+p^j}/p \in \mathrm{Fil}^{i+p^j}B_{\mathrm{dR}}$.

Thus if $-p^{j-1}\alpha\mathbf{u} \in S$, then

$b_i\beta^{i+p^j}/p \in (\mathrm{Fil}^{i+p^j}B_{\mathrm{dR}}) \cap (S + \mathrm{Fil}^{i+p^j+1}B_{\mathrm{dR}}) = \mathrm{Fil}^{i+p^j+1}S + \mathrm{Fil}^{i+p^j+1}B_{\mathrm{dR}}$.

Now on one hand, $\theta^{i+p^j}(b_i\beta^{i+p^j}/p) = \theta(b_i)/p \notin \mathcal{O}_C$; on the other hand, $\theta^{i+p^j}(\mathrm{Fil}^{i+p^j+1}S + \mathrm{Fil}^{i+p^j+1}B_{\mathrm{dR}}) = \mathcal{O}_C$, we have a contradiction. □

Proposition 5.8. $\mathbf{u}$ *is transcendental over* C_{cris}.

Proof. If $\mathbf{u}$ is not transcendental, suppose $c_0 + c_1X + \cdots + c_{d-1}X^{d-1} + X^d$ is the minimal polynomial of $\mathbf{u}$ in C_{cris}. Since C_{cris} is stable under G_{K_0},

$$g(c_0) + g(c_1)(\mathbf{u} + \eta(g)t) + \cdots + g(c_{d-1})(\mathbf{u} + \eta(g)t)^{d-1} + (\mathbf{u} + \eta(g)t)^d = 0.$$

By the uniqueness of the minimal polynomial, for every $g \in G_{K_0}$, one has $g(c_{d-1}) + d\eta(g)t = c_{d-1}$. Let $c = c_{d-1} + d\mathbf{u}$, then $g(c) = c$ and $c \in (B_{\mathrm{dR}})^{G_{K_0}} = K_0 \subset B_{\mathrm{cris}}$, thus $\mathbf{u} = d^{-1}(c - c_{d-1}) \in C_{\mathrm{cris}}$, which is a contradiction. □

6. Crystalline Representations and Semi-stable Representations

6.1. B_{cris} **and its properties.** Keep the notation $\xi = [\varpi] + p$. For any $n \geq 0$, put $\gamma_n(X) := \frac{X^n}{n!}$.

Lemma 6.1. *The ring* $A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!]$ *is separated and complete by the p-adic topology, i.e.* $\bigcap p^n A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!] = 0$ *and the natural map*

$$A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!] \to \varprojlim A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!]/pA_{\mathrm{inf}}[\![\frac{\xi}{p}]\!]$$

is an isomorphism.

Definition 6.2. Put

$$A^0_{\mathrm{cris}} = \{\sum_{n=0}^{N} a_n\gamma_n(\xi) \mid N<\infty,\ a_n \in A_{\mathrm{inf}}\} \subset A_{\mathrm{inf}}[\frac{\xi}{p}],$$

and define $A_{\mathrm{cris}} = \varprojlim_{n\in\mathbb{N}} A^0_{\mathrm{cris}}/p^nA^0_{\mathrm{cris}}$, and $B^+_{\mathrm{cris}} = A_{\mathrm{cris}}[\frac{1}{p}]$.

By Lemma 6.1, A^0_{cris} is separated by the p-adic topology, whence $A^0_{\mathrm{cris}} \to A_{\mathrm{cris}}$ is injective. Moreover, $A_{\mathrm{cris}} \subset A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!] \subset B^+_{\mathrm{dR}}$ induces injection

$$A_{\mathrm{cris}} \subset A_{\mathrm{inf}}[\![\frac{\xi}{p}]\!] \subset B^+_{\mathrm{dR}},\ \text{and } B^+_{\mathrm{cris}} \subset B^+_{\mathrm{dR}}.$$

We also find that the completion can be identified with the divided power series algebra:

Proposition 6.3.

$$A_{\mathrm{cris}} = \{\sum_{n=0}^{\infty} a_n\gamma_n(\xi) \mid a_n \to 0 \text{ } p\text{-adically in } A_{\mathrm{inf}}\}$$

$$B_{\mathrm{cris}} = \{\sum_{n=0}^{\infty} a_n\gamma_n(\xi) \mid a_n \to 0 \text{ } p\text{-adically in } A_{\mathrm{inf}}[\frac{1}{p}]\}.$$

Remark. As the p-adic completion of the divided power polynomial algebra $A_{\mathrm{inf}}\langle\xi\rangle$, we usually put p-adic topology on A_{cris} and B_{cris}. In particular, the convergence of the series $\sum_{n=0}^{\infty} a_n\gamma_n(\xi)$ depends only on the p-adic topology. Since the canonical topology of B_{dR} is the (p,ξ)-topology, which is coarser than the p-adic topology, the inclusions $A_{\mathrm{cris}} \subset B^+_{\mathrm{dR}}$ and $B^+_{\mathrm{cris}} \subset B^+_{\mathrm{dR}}$ are continuous.

Proposition 6.4. *The kernel* $\ker(\theta : A_{\mathrm{cris}} \to \mathcal{O}_C)$ *is a divided power ideal, which means that, if* $a \in \ker\theta$*, then* $\frac{a^m}{m!}$ *is again in* A_{cris} *and lies in* $\ker\theta$ *for every* $m \in \mathbb{N}$*.*

Proposition 6.5. *For the map* $\overline{\theta} : A_{\mathrm{cris}} \to \mathcal{O}_C \to \mathcal{O}_C/p$*, its kernel* $\ker\overline{\theta} = (\ker\theta, p)$ *is also a divided power ideal.*

Proposition 6.6. $t \in A_{\mathrm{cris}}$ *and* $t^{p-1} \in pA_{\mathrm{cris}}$*.*

Proof. Since $\theta([\epsilon]-1)=0$, $[\epsilon]-1 = b\xi$ for some $b \in A_{\mathrm{inf}}$, thus we have

$$\frac{([\epsilon]-1)^n}{n} = (n-1)!b^n\gamma_n(\xi).$$

Since $(n-1)! \to 0$ p-adically, we have $t \in A_{\mathrm{cris}}$.

To show that $t^{p-1} \in pA_{\mathrm{cris}}$, we just need to show that $([\epsilon]-1)^{p-1} \in pA_{\mathrm{cris}}$. Notice that $[\epsilon]-1 = (\epsilon-1, *, \cdots)$, and

$$(\epsilon-1)^{(n)} = \lim_{m\to\infty} (\zeta_{p^{n+m}}-1)^{p^m},$$

where $\zeta_{p^n} = \epsilon^{(n)}$ is a primitive p^n-th root of unity. By the proof of Lemma 3.11, $v((\epsilon-1)^{(n)}) = \frac{1}{p^{n-1}(p-1)}$, thus

$$(\epsilon-1)^{p-1} = (p^p, 1, *, \cdots) \times \text{unit} = (p^\flat)^p \times \text{unit}.$$

As a result,

$$([\epsilon]-1)^{p-1} \equiv [(\varpi)^p]\cdot(*) \equiv [\varpi]^p\cdot(*) = (\xi-p)^p\cdot(*) \equiv \xi^p\cdot(*) \bmod pA_{\mathrm{cris}},$$

but $\xi^p = p(p-1)!\gamma_n(\xi) \in pA_{\mathrm{cris}}$, we thus get the result. □

Corollary 6.7. $B^+_{\mathrm{cris}}[\frac{1}{t}] = A_{\mathrm{cris}}[\frac{1}{t}]$.

Definition 6.8. We define the ring of crystalline periods by

$$B_{\mathrm{cris}} = A_{\mathrm{cris}}[1/p, 1/t] = A_{\mathrm{cris}}[1/t].$$

Proposition 6.9. *The rings A_{cris}, B^+_{cris}, B_{cris} are stable under the action of G_K. Moreover:*
1). The map $\iota : K \otimes_{K_0} B_{\mathrm{cris}} \to B_{\mathrm{dR}},\ \lambda \otimes x \mapsto \lambda x$ is injective.
2). $B^{G_K}_{\mathrm{cris}} = K_0$.

Proposition 6.10. *B_{cris} is $(\mathbb{Q}_p, G_K)$-regular.*

Unlike on B_{dR}, we can continuously extend the Frobenius map φ on B_{cris} as follows: recall on $A_{\mathrm{inf}} = W(R)$, we have a Frobenius map

$$\varphi((a_0, \cdots, a_n, \cdots)) = (a_0^p, \cdots, a_n^p, \cdots).$$

For all $b \in W(R)$, $\varphi(b) \equiv b^p \bmod p$, thus

$$\varphi(\xi) = \xi^p + p\eta = p(\eta + (p-1)!\gamma_p(\xi))$$

and

$$\varphi(\xi^m) = p^m(\eta + (p-1)!\gamma_p(\xi))^m$$

for some $\eta \in A_{\mathrm{inf}}$. Therefore we can define

$$\varphi(\gamma_m(\xi)) := \frac{p^m}{m!}(\eta + (p-1)!\gamma_p(\xi))^m.$$

As a consequence, $\varphi(A^0_{\mathrm{cris}}) \subset A^0_{\mathrm{cris}}$. By continuity, we extend φ to A_{cris} and B^+_{cris}. Then

$$\varphi(t) = \log([\epsilon^p]) = p\log([\epsilon]) = pt.$$

Consequently φ is extended to B_{cris} by setting $\varphi(\frac{1}{t}) = \frac{1}{pt}$.

The action of φ commutes with the action of G_K.

Proposition 6.11. *There is a natural isomorphism of graded K-algebra*

$$\mathrm{gr}(K \otimes_{K_0} B_{\mathrm{cris}}) \cong \mathrm{gr}B_{\mathrm{dR}} \cong B_{\mathrm{HT}}.$$

6.2. Interaction of Filtration and Frobenius on A_{inf}.

Definition 6.12. For a subring A of B_{dR} (in particular, $A = A_{\mathrm{inf}}$, $A_{\mathrm{inf}}[\frac{1}{p}]$, $W_K(R) = W(R) \otimes_{K_0} K$, A_{cris}, B^+_{cris}, B_{cris}),
1). For every $r \in \mathbb{Z}$, set $\mathrm{Fil}^i A = A \cap \mathrm{Fil}^i B_{\mathrm{dR}}$; denote $\theta : \mathrm{Fil}^0 A = A \cap B^+_{\mathrm{dR}} \to C$ the restriction of $\theta : B^+_{\mathrm{dR}} \to C$.
2). If A is moreover a subring of B_{cris} stable under φ, set

$$I^{[r]} := \{a \in A \mid \varphi^n(A) \subset \mathrm{Fil}^r A \text{ for every } n \in \mathbb{N}\}.$$

We will discuss about $I^{[r]}A_{\mathrm{inf}}$ in this subsection and $I^{[r]}A_{\mathrm{cris}}$ in the next subsection.

For $x \in R$, let $x' = \varphi^{-1}(x) = x^{p^{-1}}$; similarly, for $x \in A_{\mathrm{inf}}$, let $x' = \varphi^{-1}(x)$. Recall that $\boldsymbol{\pi} = [\epsilon] - 1$, $\boldsymbol{\pi}' = [\epsilon'] - 1$. Let

$$\tau := \frac{\boldsymbol{\pi}}{\boldsymbol{\pi}'} = \frac{[\epsilon] - 1}{[\epsilon]' - 1} = 1 + [\epsilon'] + \cdots + [\epsilon']^{p-1}.$$

The image of τ in R is

$$\bar{\tau} = 1 + \epsilon' + \cdots + \epsilon'^{p-1} = \frac{\epsilon - 1}{\epsilon' - 1},$$

hence the valuation of $\bar{\tau}$ is

$$v(\bar{\tau}) = v(\pi) - v(\pi') = \frac{p}{p-1} - \frac{1}{p-1} = 1.$$

We thus conclude that τ is a generator of $\ker\theta$ in $A_{\inf}$: In fact, on one hand,

$$\theta(\tau) = 1 + \epsilon'^{(0)} + (\epsilon'^{(0)})^p + \cdots + (\epsilon'^{(0)})^{p-1} = 1 + \epsilon^{(1)} + (\epsilon^{(1)})^p + (\epsilon^{(1)})^{p-1} = 0,$$

we have $\tau \in \ker\theta$; on the other hand, if $\tau \in (\xi^2)$, then $\tau = b\xi^2$ for some $b \in A_{\inf}$, and $\bar{\tau} = \bar{b} \cdot \varpi^2$, contradicting that $v(\bar{\tau}) = 1$.

Proposition 6.13. *Suppose $r \in \mathbb{N}$. Then*
1). The ideal $I^{[r]}A_{\inf}$ is the principal ideal generated by $\boldsymbol{\pi}$. In particular, $I^{[r]}A_{\inf}$ is the r-th power of $IA_{\inf}$.
2). For every element $a \in I^{[r]}A_{\inf}$, a generates this idea if and only if $v(\bar{a}) = \frac{rp}{p-1}$.

Proof. We first prove the following lemma:

Lemma 6.14. *(1). The ideal $IA_{\inf}$ is principal, generated by $\boldsymbol{\pi}$.*
(2). For every element $a = (a_0, a_1, \cdots) \in IA_{\inf}$, a generates the ideal $IA_{\inf}$ if and only if $v(a_0) = \frac{p}{p-1}$. In this case, $v(a_n) = \frac{p}{p-1}$ for every $n \in \mathbb{N}$.

Proof. For $a = (a_0, a_1, \cdots) \in IA_{\inf}$, let $\alpha_n = a_n^{(n)}$. Then for every $m \in \mathbb{N}$,

$$\theta(\varphi^m a) = \sum_{n\geq 0} p^n (a_n^{p^m})^{(n)} = \sum_{n\geq 0} p^n \alpha_n^{p^n} = 0.$$

We claim that for every pair $(r, m) \in \mathbb{N} \times \mathbb{N}$, one has $v(\alpha_m) \geq p^{-m}(1 + p^{-1} + \cdots + p^{-r})$, as a result, $v(\alpha_m) \geq p^{-m}\frac{p}{p-1}$ and hence $v(a_m) \geq \frac{p}{p-1}$. We prove the claim by induction.
(a). If $r = m = 0$, $\theta(a) = \alpha_0 = 0 \ (\mathrm{mod} p)$, hence $v(\alpha_0) \geq 1$.
(b). If $r = 0$ but $m \neq 0$, then

$$0 = \theta(\varphi^m a) = \sum_{n\geq 0} p^n \alpha_n^{p^n} = \sum_{n=0}^{m-1} p^n \alpha_n^{p^n} + p^m \alpha_n^{p^m}.$$

By induction hypothesis, $v(\alpha_m) \geq p^{-m}$, hence $v(p^n \alpha_n^{p^m}) \geq n + p^{m-n} \geq m + 1$. Thus

$$v(p^m \alpha_m^{p^m}) = m + p^m v(\alpha_m) \geq m + 1,$$

which gives $v(\alpha_m) \geq p^{-m}$.
(c). If $r \neq 0$, one has

$$0 = \theta(\varphi^m a) = \sum_{n\geq 0} p^n \alpha_n^{p^n} = \sum_{n=0}^{m-1} p^n \alpha_n^{p^n} + p^m \alpha_n^{p^m}. + \sum_{m+1}^{\infty} p^n \alpha_n^{p^n}.$$

By induction hypothesis: for $0 \leq n \leq m-1$, $v(\alpha_n) \geq p^{-n}(1 + p^{-1} + \cdots + p^{-r})$, thus

$$v(p^n \alpha_n^{p^m}) \geq n + p^{m-n}(1 + p^{-1} + \cdots + p^{-r}) \geq m + (1 + p^{-1} + \cdots + p^{-r+1});$$

for $n \geq m-1$, $v(\alpha_n) \geq p^{-n}(1 + p^{-1} + \cdots + p^{-r+1})$, thus

$$v(p^n \alpha_n^{p^m}) \geq n + p^{m-n}(1 + p^{-1} + \cdots + p^{-r+1}) \geq m + (1 + p^{-1} + \cdots + p^{-r});$$

one thus has $v(\alpha_m) \geq p^{-m}(1 + p^{-1} + \cdots + p^{-r})$. Notice that for any $n \in \mathbb{N}$, $\theta(\varphi^n \boldsymbol{\pi}) = \theta([\epsilon]^{p^n} - 1) = 0$, thus $\boldsymbol{\pi} \in IA_{\inf}$. On the other hand, as $v(\epsilon - 1) = \frac{p}{p-1}$, the claim implies that

for every $x \in IA_{\inf}$, $\bar{x} = u(\epsilon - 1)$ for some $u \in R$. Since $\boldsymbol{\pi}$ is a lift of $\epsilon - 1$, $x \equiv \boldsymbol{\pi}[u] \ (\mathrm{mod} p)$, where $[u] \in A_{\inf}$. Thus $IA_{\inf} \subset (\boldsymbol{\pi}, p)$. But $\mathcal{O}_C$ is p-torsion free, and notice that

$$\begin{aligned}\theta(\varphi^m(px)) = \theta(\varphi^{m+1}(Vx)) &= \sum_{n=0}^{\infty} p^n((Vx)_n^{(n)})^{p^{m+1}} \\ &= \sum_{n=1}^{\infty} p^n((x)_{n-1}^{(n)})^{p^{m+1}} = p\sum_{n=1}^{\infty} p^{n-1}((x)_{n-1}^{(n-1)})^{p^m} \\ &= p\sum_{n=0}^{\infty} p^n((x)_n^{(n)})^{p^m} = p\theta(\varphi^m(x)),\end{aligned}$$

thus if $x \in IA_{\inf}$, $px \in IA_{\inf}$. Now for any $x \in IA_{\inf}$, $x = a\boldsymbol{\pi} + bp$ for some $a, b \in A_{\inf}$; since $a\boldsymbol{\pi} \in IA_{\inf}$, $bp \in IA_{\inf}$, and by the above argument $b \in IA_{\inf}$. Hence $x = a\boldsymbol{\pi} + pa'\boldsymbol{\pi} + b'p^2$ for some $a', b' \in A_{\inf}$. Repeat this procedure, we conclude that $IA_{\inf} = (\boldsymbol{\pi})$ since $W(R)$ is p-adically complete. □

Let $\mathrm{gr}^i A_{\inf} = \mathrm{Fil}^i A_{\inf}/\mathrm{Fil}^{i+1} A_{\inf}$, and let θ^i be the map $\mathrm{Fil}^i A_{\inf} \to \mathrm{gr}^i A_{\inf},\ a\tau^i \mapsto \theta(a)\bar{\tau}^i$. As $\mathrm{Fil}^i A_{\inf}$ is a principal ideal generated by τ^i, $\mathrm{gr}^i A_{\inf}$ is a free $\mathcal{O}_C$-module of rank 1 generated by $\theta^i(\tau^i) = \theta^1(\tau)^i = \bar{\tau}^i$. Note that $\boldsymbol{\pi} = \boldsymbol{\pi}'\tau$, then

$$\varphi^n(\boldsymbol{\pi}) = \boldsymbol{\pi}'\tau^{1+\varphi+\cdots+\varphi^n}$$

for every $n \in \mathbb{N}$. For $i \geq 1$,

$$\theta(\varphi^i(\tau)) = \theta((1 + [\epsilon'] + \cdots [\epsilon']^{p-1})^{p^i}) = \sum_{i=1}^{p-1} 1^{(p-1)p^{i-1}} = p,$$

hence $\theta^1(\varphi^n(\boldsymbol{\pi})) = p^n(\epsilon^{(1)} - 1)\theta^1(\tau)$.

Proof of (1): The inclusion $\boldsymbol{\pi}^r A_{\inf} \subset I^{[r]}$ is clear. We show $\boldsymbol{\pi}^r A_{\inf} \supset I^{[r]}$ by induction. The case $r = 0$ is trivial. Suppose $r \geq 1$. If $a \in I^{[r]}$, by induction hypothesis, we can write $a = \boldsymbol{\pi}^{r-1}b$ with $b \in A_{\inf}$. Then

$$0 = \theta^{r-1}(\varphi^n(a)) = \theta^{r-1}(\varphi^n(b)\varphi^n(\boldsymbol{\pi}^{r-1})) = \theta(\varphi^n(b))\theta^1(\varphi^n(\boldsymbol{\pi}))^{r-1} = (p^n(\epsilon^{(1)}-1))^{r-1}\cdot\theta(\varphi^n(b))\cdot\theta^1(\tau)^{r-1}.$$

Since $p^n(\epsilon^{(1)} - 1) \neq 0$, one must have $\theta(\varphi^n(b)) = 0$ for every $n \in \mathbb{N}$. Hence $b \in IA_{\inf}$, by the precedent lemma, there exists $c \in A_{\inf}$ such that $b = \boldsymbol{\pi}c$. Thus $a \in \boldsymbol{\pi}^r A_{\inf}$.

Proof of (2): It follows immediately from that $v(\overline{\boldsymbol{\pi}^r}) = \frac{rp}{p-1}$ and taht $x \in A_{\inf}$ is a unit if and only if $\bar{x}$ is a unit in R. □

6.3. **A description of A_{cris}.** For $n \in \mathbb{N}$, write $n = r_n + (p-1)q_n$ with $r_n,\ q_n \in \mathbb{N}$ and $0 \leq r_n < p - 1$. Put

$$t^{\{n\}} := t^{r_n}\gamma_{q_n}(t^{p-1}/p) = (p^{q_n}(q_n!))^{-1}t^n.$$

Note that:
(a). if $p = 2$, $t^{\{n\}} = \gamma_n(t/2)$ and $t^{\{2n\}} = \gamma_{2n}(t/2) = \frac{1}{(2n-1)!!}\gamma_n(t^2/8)$; (b). if $p > 2$, $t^{\{(p-1)n\}} = \gamma_n(t^{p-1}/p)$.

We have shown that $t^{p-1}/p \in A_{\mathrm{cris}}$, therefore $t^{\{n\}} \in A_{\mathrm{cris}}$.

Put

$$\Lambda_\epsilon := \{\sum_{n\in\mathbb{N}} a_n t^{\{n\}} \mid a_n \in W,\ a_n \to 0 \ \ p\text{-adically}\} \subset K_0[[t]] \cap A_{\mathrm{cris}},$$
$$\mathbf{O}_\epsilon := W[[\boldsymbol{\pi}]].$$

Since $\boldsymbol{\pi} = e^t - 1 = \sum_{n\geq 1} \frac{t^n}{n!} = \sum_{n\geq 1} \frac{p^{q_n} q_n!}{n!} t^{\{n\}} \in \Lambda_\epsilon$, one has

$$\mathbf{O}_\epsilon \subset \Lambda_\epsilon \subset A_{\mathrm{cris}}$$

are sub-W-algebras of A_{cris} stable by the action of φ and G_{K_0} which factors through $\Gamma_{K_0} = \mathrm{Gal}(K_0^{\mathrm{cyc}}/K_0)$ since t is stable under H_{K_0}. We also see that

$$t = \log[\epsilon] = \boldsymbol{\pi} \sum_{n\geq 0} (-1)^n \frac{\boldsymbol{\pi}^n}{n+1} = \boldsymbol{\pi} \cdot u,$$

where $u = \sum_{n\geq 0}(-1)^n \frac{\boldsymbol{\pi}^n}{n+1} = \sum_{n\geq 0}(-1)^n \frac{1}{n+1}(\sum_{n\geq 1} \frac{p^{q_n} q_n!}{n!} t^{\{n\}})^n$, clearly the coefficients of $t^{\{n\}}$ go to infinity, and u is a unit in Λ_ϵ.

Recall that Δ_{K_0} is the torsion subgroup of Γ_{K_0}, set

$$\mathbf{O} = \mathbf{O}_\epsilon^{\Delta_{K_0}} \subset \Lambda := \Lambda_\epsilon^{\Delta_{K_0}},$$
$$\pi_0 = -p + \sum_{a\in\mathbb{F}_p} [\epsilon]^{[a]}$$

Lemma 6.15. $\pi_0 = (p-1)\sum_{n\geq 1,\ (p-1)|n} \frac{t^n}{n!}$.

Proof.

$$\pi_0 = \mathrm{Tr}_{K_0((t))/K_0((t^{p-2}))}(e^t - 1) = \mathrm{Tr}_{K_0((t))/K_0((t^{p-1}))}(\sum_{n=1}^{\infty} \frac{t^n}{n!}).$$

Notice that $\mathrm{Tr}_{K_0((t))/K_0((t^{p-1}))}(t^i) \neq 0$ if and only if $(p-1)|i$. Thus the indentity follows. □

Lemma 6.16. *If $p \neq 2$, then*

$$\Lambda = \{\sum_{n\in\mathbb{N}} a_n \gamma_n(t^{p-1}/p) \mid a_n \in W,\ a_n \to 0 \ p\text{-adically}\}$$
$$= \{\sum_{n\in\mathbb{N}} a_n \gamma_n(\pi_0/p) \mid a_n \in W,\ a_n \to 0 \ p\text{-adically}\};$$

if $p = 2$, then replacing t^{p-1}/p by $t^2/8$ and π_0/p by $\pi_0/8$. And

$$\mathbf{O} = W[[\pi_0]], \quad \mathbf{O}_\epsilon \otimes_{\mathbf{O}} \Lambda = \Lambda_\epsilon.$$

Proof. Since $\Delta_{K_0} \cong \mathbb{F}_p^\times$, the subfield of $K_0((t))$ is $K_0((t^{p-1}))$. We thus get the first identity for Λ.

To show the second identity, notice that $\pi_0 = -p + \sum_{a\in\mathbb{F}_p} (\boldsymbol{\pi}+1)^{[a]} = \boldsymbol{\pi}^{p-1} w$ for some unit $w \in \mathbf{O}^\times$. Thus $\mathbf{O} = W[[\boldsymbol{\pi}^{p-1}]] = W[[\pi_0]]$. By the relation $t = \boldsymbol{\pi} u$, $t^{p-1}/p = \boldsymbol{\pi}^{p-1} u/p = \pi_0 w^{-1} u/p$, and the identity follows. The identification $\mathbf{O}_\epsilon \otimes_{\mathbf{O}} \Lambda = \Lambda_\epsilon$ is evident. □

Proposition 6.17. *(1). The element π_0 is a generator of $I^{[p-1]}A_{\mathrm{inf}}$ if $p \neq 2$ and of $I^{[2]}A_{\mathrm{inf}}$ if $p = 2$.*
(2). There exists a unit $u \in \mathbf{O}$ such that $\varphi\pi_0 = u\pi_0(p+\pi_0)^{p-1}$ if $p \neq 2$ and $\varphi\pi_0 = u\pi_0(p+\pi_0)^2$ if $p = 2$.

Proof. (1). is clear. We prove (2) in the case $p \neq 2$, and the case $p = 2$ is analogous.

Let $q = p + \pi_0 = \sum_{a \in \mathbb{F}_p} [\epsilon]^{[a]}$ and $q' = \varphi^{-1}(q)$. We will see that $q = 1 + [\epsilon] + \cdots + [\epsilon]^{p-1}$ and by a similar argument as that about τ, q' is a generator of the $\ker \theta$ in A_{inf}, and it follows that q' and τ generate the kernel of the restriction of θ to $\mathbf{O}'_\epsilon = \varphi^{-1}(\mathbf{O}_\epsilon) = W[[\boldsymbol{\pi}']]$. Thus $\tau = u'_1 q'$ for some unit u'_1 in $\mathbf{O}_\epsilon$,

$$\boldsymbol{\pi} = \varphi(\boldsymbol{\pi}') = \boldsymbol{\pi}'\tau = u'_1 \boldsymbol{\pi}' q',$$

and hence $\varphi(\boldsymbol{\pi}) = u_1 \boldsymbol{\pi} q$ and $\varphi(\boldsymbol{\pi}^{p-1}) = u_1^{p-1} \boldsymbol{\pi}^{p-1} q^{p-1}$. Since π_0 and $\boldsymbol{\pi}^{p-1}$ are two generators of $\mathbf{O}_\epsilon \cap I^{[p-1]} A_{\mathrm{inf}}$, $\varphi(\pi_0) = u \pi_0 q^{p-1}$ with u a unit in $\mathbf{O}_\epsilon$. Now the uniqueness of u and the fact that $\mathbf{O} = \mathbf{O}_\epsilon^{\Delta_{K_0}}$ imply that u and u^{-1} lie in $\mathbf{O}$. □

Theorem 6.18. *One has an isomorphism of A_{inf}-algebras*

$$\alpha : (\widehat{A_{\mathrm{inf}} \otimes_{\mathbf{O}} \Lambda}) \to A_{\mathrm{cris}},$$

which is continuous by p-adic topology, given by

$$\alpha(\sum \alpha_m \otimes \gamma_m(\pi_0/p)) = \sum a_m \gamma_m(\pi_0/p).$$

The isomorphism α thus induces an isomorphism

$$\alpha_\epsilon : \widehat{A_{\mathrm{inf}} \otimes_{\mathbf{O}_\epsilon} \Lambda_\epsilon} \to A_{\mathrm{cris}}.$$

6.4. **Interaction of Filtration and Frobenius on A_{cris}.**

Proposition 6.19. *For every $r \in \mathbb{N}$, write $I^{[r]} = I^{[r]} A_{\mathrm{cris}}$, then if $r \geq 1$, $I^{[r]}$ is a divided power ideal of A_{cris} which is the associated sub-A_{inf}-algebra (and also an ideal) of A_{cris} generated by $t^{\{s\}}$ for $s \geq r$.*

Proof. Let $I(r)$ be the sub-A_{inf}-algebra generated by $t^{\{s\}}$ for $s \geq r$. Notice that $\varphi t^{\{s\}} = p^s t^{\{s\}} \subset I^{[r]} A_{\mathrm{cris}}$, $I(r) \subset I^{[r]}$, and it is clear that $I(r)$ is a divided power ideal. It remains to show $I^{[r]} \subset I(r)$. We show this by induction on r.

The case $r = 0$ is trivial. Suppose $r \geq 1$ and $a \in I^{[r]}$. The induction hypothesis allows us to write $a \in I^{[r]} \subset I^{[r-1]} = I(r-1)$ as the form

$$a = \sum_{s \geq r-1} a_s t^{\{s\}},$$

where $a_s \in A_{\mathrm{inf}}$, and $a_s \to 0$ p-adically. Since $\sum_{s \geq r} a_s t^{\{s\}} \in I(r) \subset I^{[r]}$, $a_{r-1} t^{\{r-1\}} \in I^{[r]}$. But

$$\varphi^n(a_{r-1} t^{\{r-1\}}) = p^{r-1} \varphi^n(a_{r-1}) t^{\{r-1\}} = c \cdot \varphi^n(a_{r-1}) t^{r-1}$$

for a rational number c. Thus $\varphi^n(a_{r-1}) t^{r-1} \in \mathrm{Fil}^r A_{\mathrm{cris}}$, $\varphi^n(a_{r-1}) \in \mathrm{Fil}^1 A_{\mathrm{inf}}$, and hence $a_{r-1} \in I^{[1]} A_{\mathrm{inf}}$. Since $I^{[1]} A_{\mathrm{inf}}$ is generated by $\boldsymbol{\pi}$, $a_{r-1} = \boldsymbol{\pi}\lambda$ for some $\lambda \in A_{\mathrm{inf}}$, and hence $a_{r-1} t^{\{r-1\}} = \lambda \boldsymbol{\pi} t^{\{r-1\}}$. Since $t = \boldsymbol{\pi} u$ for a unit in $\Lambda_\epsilon \subset A_{\mathrm{cris}}$, $a_{r-1} t^{\{r-1\}}$ lies in the ideal generated by $t \cdot t^{\{s-1\}}$, which is contained in the ideal generated by $t^{\{s\}}$. □

Put $A^r_{\mathrm{cris}} := A_{\mathrm{cris}} / I^{[r]} A_{\mathrm{cris}}$ and $A^r_{\mathrm{inf}} := A_{\mathrm{inf}} / I^{[r]} A_{\mathrm{inf}}$.

Proposition 6.20. *A^r_{cris} and A^r_{inf} are torsion free. The natural map*

$$\iota^r : A^r_{\mathrm{inf}} \to A^r_{\mathrm{cris}}$$

are injective, and its cokernel is p-torsion, annihilated by $p^m m!$, where m is the largest integer such that $(p-1)m < r$, i.e. $m = \lfloor \frac{r-1}{p-1} \rfloor$.

Proof. Consider the map

$$A_{\mathrm{cris}} \to (A_{\mathrm{cris}}/\mathrm{Fil}^r A_{\mathrm{cris}})^{\mathbb{N}}, (a \mapsto \varphi^n(a))_{n\in\mathbb{N}},$$

it is clear the kernel is $I^{[r]}A_{\mathrm{cris}}$, thus the map

$$A^r_{\mathrm{cris}} \to (A_{\mathrm{cris}}/\mathrm{Fil}^r A_{\mathrm{cris}})^{\mathbb{N}}$$

is injective. Since $A_{\mathrm{cris}}/\mathrm{Fil}^r A_{\mathrm{cris}} \subset B^+_{\mathrm{dR}}/\mathrm{Fil}^r B_{\mathrm{dR}}$ is torsion free, A^r_{cris} is torsion free. Since the map ι^r is clearly injective, A^r_{inf} is torsion free.

AS A_{inf}-module, A^r_{cris} is generated by the images of $\gamma_s(\pi_0/p)$ for $0 \le (p-1)s < r$. Since $p^s s!\gamma_s(\pi_0/p) \in A_{\mathrm{inf}}$, we have the proposition. □

6.5. **Fundamental exact sequence of p-adic Hodge theory.** For $r \in \mathbb{N}$, let $\mathrm{Fil}^r_p A_{\mathrm{cris}} := \{x \in \mathrm{Fil}^r A_{\mathrm{cris}} \mid \varphi x \in p^r A_{\mathrm{cris}}\}$.

Proposition 6.21. *(1). The sequence*

$$0 \to \mathbb{Z}_p t^{\{r\}} \to \mathrm{Fil}^r_p A_{\mathrm{cris}} \xrightarrow{p^{-r}\varphi - 1} A_{\mathrm{cris}} \to 0$$

is exact.
(2). The ideal $\mathrm{Fil}^r_p A_{\mathrm{cris}}$ is the associated sub-A_{inf}-module of A_{cris} generated by $q'^j\gamma_n(t^{p-1}/p)$, for $j + (p-1)n \ge r$.
(3). If m is the largest integer such that $(q-1)m < r$, then for every $x \in \mathrm{Fil}^r A_{\mathrm{cris}}$, $p^m m!x \in \mathrm{Fil}^r_p A_{\mathrm{cris}}$.

Proof. Proof of (1) and (2): Let $\nu = p^{-r}\varphi - 1$, it is clear that $\mathbb{Z}_p t^{\{r\}} \subset \ker \nu$. Conversely, for $x \in \ker \nu$, in particular, $x \in I^{[r]}A_{\mathrm{cris}}$, and hence we can write

$$x = \sum_{s\ge r} a_s t^{\{s\}}.$$

Since $x = (p^{-r}\varphi)^n(x) \equiv \varphi^n(a_r)t^{\{r\}} \pmod{p^n A_{\mathrm{cris}}}$, $x = bt^{\{r\}}$ for some $b \in A_{\mathrm{inf}}$ and $\varphi(b) = b$, which means that $b \in \mathbb{Z}_p$.

Let N be the sub-A_{inf}-module of A_{cris} generated by $q'^j\gamma_n(t^{p-1}/p)$, for $j + (p-1)n \ge r$. Since $\varphi(q'^j\gamma_n(t^{p-1}/p)) = q^j p^{n(p-1)}\gamma_n(t^{p-1}/p) = p^{j+n(p-1)}(1 + \frac{\pi_0}{p})\gamma_n(t^{p-1}/p)$, $N \subset \mathrm{Fil}^r_p A_{\mathrm{cris}}$.

Since clearly $\mathbb{Z}_p t^{\{r\}} \subset N$, to show (1) and (2), it suffices to show that for every $a \in A_{\mathrm{cris}}$, there exists $x \in N$, such that $\nu(x) = a$. Since A_{cris} and N are separated and complete by the p-adic topology, we only need to show for every $a \in A_{\mathrm{cris}}$, there exists $x \in N$, such that $\nu(x) \equiv a \bmod pA_{\mathrm{cris}}$.

In fact, for $a = \sum_{n\ge \frac{r}{p-1}} a_n\gamma_n(\frac{t^{p-1}}{p})$, we can just take $x = -a$, then

$$\nu(x) = a - \sum_n p^{n(p-1)-r}\varphi(a_n)\gamma_n(\frac{t^{p-1}}{p}) \equiv a \pmod p.$$

For $a = b\gamma_i(t^{p-1}/p)$, where $i \in \mathbb{N}$ such that $(p-1)i \le r$ and $b \in A_{\mathrm{inf}}$, take $x = yq'^{r-(p-1)i}\gamma_i(t^{p-1/p})$, where y is a solution to the equation

$$\varphi(y) - q'^{r-(p-1)i}y = b,$$

then

$$\begin{aligned}\nu(x) &= p^{(p-1)i-r}(b+q'^{r-(p-1)i}y)q^{r-(p-1)i}\gamma_i(t^{p-1}/p)-x\\ &= (b+q'^{r-(p-1)i}y)(1+\frac{\pi_0}{p})^{r-(p-1)i}\gamma_i(t^{p-1}/p)-x\\ &= (1+\frac{\pi_0}{p})^{r-(p-1)i}a+x((1+\frac{\pi_0}{p})^{r-(p-1)i}-1),\end{aligned}$$

thus

$$\nu(x)-a=((1+\frac{\pi_0}{p})^{r-(p-1)i}-1)(a+x)\in(\gamma_i(t^{p-1}/p)\cdot\pi_0/p)A_{\mathrm{cris}}\subset\gamma_{i+1}(t^{p-1}/p)A_{\mathrm{cris}},$$

and by induction, we can return the case discussed above. □

Theorem 6.22. *(1). Suppose*

$$B'_{\mathrm{cris}}:=\{x\in B_{\mathrm{cris}}\mid\varphi^n(x)\in\mathrm{Fil}^0B_{\mathrm{cris}}\text{ for every }n\in\mathbb{N}\}.$$

Then $\varphi(B'_{\mathrm{cris}})\subset B^+_{\mathrm{cris}}\subset B'_{\mathrm{cris}}$ *if* $p\neq 2$ *and* $\varphi^2(B'_{\mathrm{cris}})\subset B^+_{\mathrm{cris}}\subset B'_{\mathrm{cris}}$ *if* $p=2$.
(2). For every $r\in\mathbb{N}$*, the sequence*

$$0\to\mathbb{Q}_p(r)\to\mathrm{Fil}^rB^+_{\mathrm{cris}}\xrightarrow{p^{-r}\varphi-1}B^+_{\mathrm{cris}}\to 0$$

is exact.
(3). For every $r\in\mathbb{Z}$*, the sequence*

$$0\to\mathbb{Q}_p(r)\to\mathrm{Fil}^rB_{\mathrm{cris}}\xrightarrow{p^{-r}\varphi-1}B_{\mathrm{cris}}\to 0$$

is exact.

Proof. (1). $B^+_{\mathrm{cris}}\subset B'_{\mathrm{cris}}$ is trivial. Conversely, suppose $x\in B'_{\mathrm{cris}}$, and suppose that $x=t^{-r}p^{-j}y$ for $r,j\in\mathbb{N}$ and $y\in A_{\mathrm{cris}}$, then

$$\varphi^n(x)=p^{-j-rn}t^{-r}\varphi^n(y)\in\mathrm{Fil}^0B_{\mathrm{cris}},$$

which implies that $\varphi^n(y)\in\mathrm{Fil}^rA_{\mathrm{cris}}$ for every $n\in\mathbb{N}$, i.e. $y\in I^{[r]}A_{\mathrm{cris}}$. By Proposition 6.19, $y=\sum_{s\geq 0}a_st^{\{s+r\}}$ with $a_s\in A_{\mathrm{inf}}$, tending to 0 p-adically. Then

$$\begin{aligned}\varphi(x)=p^{-j-r}t^{-r}\varphi(y)&=p^{-j-r}\sum_{s\geq 0}p^{r+s}\varphi(a_s)t^{\{s+r\}-r}\\ &=p^{-j-r}\sum_{s\geq 0}c_s\varphi(a_s)t^s,\end{aligned}$$

where c_s is a rational number satisfying

$$v(c_s)\geq(s+r)(1-\frac{1}{p-1}-\frac{1}{(p-1)^2})$$

if $p\neq 2$, it is positive integer and $\varphi(x)\in p^{-j-r}W(R)[[t]]\subset B^+_{\mathrm{cris}}$. For $p=2$, the proof is analogous.

(2). follows directly from Proposition 6.21.

(3). By (2), for every integer r and i such that $r+i\geq 0$, one has

$$0\to\mathbb{Q}_p(r+i)\to\mathrm{Fil}^{r+i}B^+_{\mathrm{cris}}\to B^+_{\mathrm{cris}}\to 0,$$

tensoring $\mathbb{Q}_p(-i)$:

$$0\to\mathbb{Q}_p(r)\to t^{-i}\mathrm{Fil}^{r+i}B_{\mathrm{cris}}\to t^{-i}B^+_{\mathrm{cris}}\to 0,$$

the result follows by passing the above exact sequence to the limit. □

Definition 6.23. For $h, d \in \mathbb{Z}$ and $h \geq 1$, set

$$P_{h,d} = \{x \in B_{\mathrm{cris}} \mid \varphi^h(x) = p^d x\}, \quad P^+_{h,d} = P_{h,d} \cap B^+_{\mathrm{cris}}.$$

In particular, set $B_e = P_{1,0} = B_{\mathrm{cris}}^{\varphi=1}$, ${}^hB_e = P_{h,0} = B_{\mathrm{cris}}^{\varphi^h=1}$.

For every $d \in \mathbb{Z}$, $P_{1,d} = B_e t^d$ is a free B_e-module of rank 1. Recall that U is the image of U_R^+ in B_{cris} under the logarithm map.

Theorem 6.24. *(1). $\mathrm{Fil}^0 B_e = \mathbb{Q}_p$, and $P^+_{1,d} = 0$ for every $d < 0$.*
(2). One has $U = P^+_{1,1}$, hence the sequence

$$0 \to \mathbb{Q}_p t \to P^+_{1,1} \xrightarrow{\theta} C \to 0$$

is exact.
(3). Moreover, pick any $u \in U - \mathbb{Q}_p t$, for $d > 0$,

$$P^+_{1.d} = \{x = x_0 t^{d-1} + x_1 u t^{d-2} + \cdots + x_{d-1} u^{d-1} \mid x_0, \cdots, x_{d-1} \in U\}.$$

(4). The sequence

$$0 \to \mathbb{Q}_p \to B_e \oplus B^+_{\mathrm{dR}} \to B_{\mathrm{dR}} \to 0$$

is exact. This sequence is called the fundamental exact sequence of p-adic Hodge theory.

Proof. (1). By Theorem 6.22, the sequence

$$0 \to \mathbb{Q}_p \to \mathrm{Fil}^0 B_{\mathrm{cris}} \xrightarrow{\varphi-1} B_{\mathrm{cris}} \to 0$$

is exact, and the first assertion follows. For the second assertion, if $x \in P^+_{1,d}$, then $x \in B^+_{\mathrm{cris}}$ and $xt^{-d} \in B_e \cap \mathrm{Fil}^{-d} B_{\mathrm{cris}} = 0$ for $d < 0$.

(2). $U \subset P^+_{1,1}$ is clear. Since we have already known the exact sequence

$$0 \to \mathbb{Q}_p \to U \xrightarrow{\theta} C \to 0,$$

for $x \in P^+_{1,1}$, choose $u \in U$ such that $\theta(x) = \theta(u)$, then $u - x = tx_0$ with $x_0 \in \mathbb{Q}_p$. Since $tx_0 \in \mathbb{Q}_p t \subset U$, so is $x = u + x_0 t$.

(3). In general, for $x \in P^+_{1,d}$, suppose $\theta(x) = c$ and $\theta(u) = c_0$, we fine $x_{d-1} \in U$ such that $\theta(x_{d-1}) = \frac{c}{c_0^{d-1}}$, then $\theta(x - x_{d-1}u^{d-1}) = 0$, and we may write $x - x_{d-1}u^{d-1} = ty$ with $y \in B^+_{\mathrm{dR}} \cap P_{1,d-1}$. Moreover, we can easily check that $\varphi^n(y) \in B^+_{\mathrm{dR}}$ for $n \in \mathbb{N}$. By Theorem 6.22, $y \in B^+_{\mathrm{cris}}$ and hence in $P^+_{1,d-1}$. We now proceed by induction.

(4). By the proof of (1), one has $\mathrm{Fil}^i B_e = B_e \cap \mathrm{Fil}^i B_{\mathrm{dR}} = 0$ for $i > 0$, and hence $B_e \cap B^+_{\mathrm{dR}} = \mathrm{Fil}^0 B_e = \mathbb{Q}_p$. Thus it is enough to show that $B_e + B^+_{\mathrm{dR}} = B_{\mathrm{dR}}$. We show by induction on $d > 0$ that $\mathrm{Fil}^{-d} B_{\mathrm{dR}} \subset B_e + B^+_{\mathrm{dR}}$. Suppose $\mathrm{Fil}^{-d+1} B_{\mathrm{dR}} \subset B_e + B^+_{\mathrm{dR}}$, suppose $x = t^{-d}\lambda \in \mathrm{Fil}^{-d} B_{\mathrm{dR}}$ and $\theta(\lambda) = c \neq 0$. By (3), we can find $y \in P^+_{1,d}$, $\theta(y) = c$. Then $t^{-d}y \in B_e$ and $x - t^{-d}y = t^{-d}(\lambda - y) \in \mathrm{Fil}^{-d+1} B_{\mathrm{dR}}$, and the claim is proven. □

6.6. **Lubin-Tate representations and the ring $B_{e,h}$.**

Lemma 6.25. *The map $P_0 \otimes_{\mathbb{Q}_p} B_e \to B_{\mathrm{cris}},\ \lambda \otimes x \mapsto \lambda x$ is injective.*

Proof. Suppose $x = \sum_i \lambda_i \otimes x_i \mapsto 0$, $\lambda_i \in P_0,\ x_i \in B_e$, we need to show $x = 0$. We prove this by induction:

The case $n = 0$ is trivial. Now suppose $n > 1$. $\sum_i \lambda_i x_i = 0$ implies that $\sum_i \varphi(\lambda_i) x_i = 0$. We may assume that $\varphi(\lambda_1) \neq \lambda_1$, otherwise $\lambda_i \in \mathbb{Q}_p$ for every i and we easily conclude that $x = 0$. Then the element

$$\sum_{i=1}^{n} (\varphi(\lambda_1)\lambda_i - \varphi(\lambda_i)\lambda_1) \otimes x_i = \sum_{i=2}^{n} (\varphi(\lambda_1)\lambda_i - \varphi(\lambda_i)\lambda_1) \otimes x_i \mapsto 0,$$

which implies that $\varphi(\lambda_1)\lambda_i - \varphi(\lambda_i) - \lambda_1 = 0$ for every i by the induction hypothesis, i.e. $\lambda_i/\lambda_j \in \mathbb{Q}_p$. Hence $x = 0$. □

Proposition 6.26. *The injection in Lemma 6.25 induces an isomorphism*

$$\iota_h : \mathbb{Q}_{p^h} \otimes_{\mathbb{Q}_p} B_e \to {}^h B_e,\ a \otimes b \mapsto ab.$$

In particular, $\mathrm{Fil}^0 {}^h B_e = \mathbb{Q}_{p^h}$.

Definition 6.27. For $h \in \mathbb{N}$, $h \geq 1$, set

$$V_h = \{x \in B_{\mathrm{cris}}^+ \mid \varphi^h(x) = px, \theta(x) = 0\} = P_{h,1} \cap \mathrm{Fil}^1 B_{\mathrm{dR}}.$$

Lemma 6.28. *If $V_h \neq 0$, then V_h is a 1-dimensional $\mathbb{Q}_{p^h}$-vector space and the map*

$$V_h \to V_1,\ x \mapsto x\varphi(x) \cdots \varphi^{h-1}(x)$$

is onto.

Proof. It is obvious that for every h, V_h is a $\mathbb{Q}_{p^h}$-vector space. Note that $V_1 = P_{1,1} \cap \mathrm{Fil}^1 B_{\mathrm{dR}} = B_e t \cap \mathrm{Fil}^1 B_{\mathrm{dR}} = (B_e \cap \mathrm{Fil}^0 B_{\mathrm{dR}})t = \mathbb{Q}_p t$. For every $x \in V_h$, $x\varphi(x) \cdots \varphi^{h-1}(x)$ is an element in V_1, and thus we can write it as the form $x\varphi(x) \cdots \varphi^{h-1}(x) = at$ for some $a \in \mathbb{Q}_p$. Since the norm map $\mathbb{Q}_{p^h} \to \mathbb{Q}_p$ is onto, we can choose $b \in \mathbb{Q}_{p^h}$ such that $b\varphi(b) \cdots \varphi^{h-1}(b) = a^{-1}$, and hence $t_h := bx$ is an element of V_h with $t_h \varphi(t_h) \cdots \varphi^{h-1}(t_h) = t$. The above argument shows that for every $x \in V_h$, $bx = t_h$ for some $b \in \mathbb{Q}_{p^h}$, and hence $x = b^{-1} t_h$. We conclude that $V_h = \mathbb{Q}_{p^h} t_h$. □

By the Theory of Lubin-Tate formal group laws, one finally sees that $V_h \neq 0$ for every h. Further, we have the followings:

Proposition 6.29. *For every $h \in \mathbb{N}$ and $h \geq 1$, V_h is a 1-dimensional $\mathbb{Q}_{p^h}$-representation of G_K and the sequence*

$$0 \to V_h \to P_{h,1}^+ \xrightarrow{\theta} C \to 0$$

is exact. As a consequence, $\theta(P_{h,d}^+) = C$ for every integer $d \geq 1$.

Definition 6.30. The Lubin-Tate element $\{t_h\}_{h \in \mathbb{N}}$ is a compatible system of elements in B_{cris}^+, such that $V_h = \mathbb{Q}_{p^h} t_h$ and
(1) $\varphi^h(t_h) = pt_h$, $\theta(t_h) = 0$ if $h \neq 0$;
(2) $t_0 = 1$ and $t_1 = t$;
(3) For $r \geq 1$, $t_{hr} \varphi^h(t_{hr}) \cdots \varphi^{h(r-1)}(t_{hr}) = t_h$.

By definition:

Proposition 6.31. *The Lubin-Tate elements $\{t_h\}$ satisfy the following properties:*
(1) t_h is invertible in B_{cris};
(2) For $h \geq 2$, $t_h \in \mathrm{Fil}^1 B_{\mathrm{dR}} - \mathrm{Fil}^2 B_{\mathrm{dR}}$ is a uniformizer of B_{dR}^+ (when considering B_{dR}^+ as a DVR with residue field C);
(3) $\varphi^n(t_h) \in \mathrm{Fil}^0 B_{\mathrm{dR}} - \mathrm{Fil}^1 B_{\mathrm{dR}}$ for $1 \leq n \leq h-1$;
(4) For every d, $P_{h,d} = {}^h B_e t_h^d$ is a free ${}^h B_e$-module of rank 1.

Definition 6.32. For $h \geq 1$, set

$$B_{e,h} := \{x \in B_{\mathrm{cris}}^{\varphi^h=1} \mid xt_h^d \in B_{\mathrm{cris}}^+ \text{ for some } d \in \mathbb{N}\}.$$

The definition of $B_{e,h}$ is independent of the choice of t_h. If $h = 1$, $B_{e,1}$ is nothing but B_e. By computation, we have that:

Lemma 6.33. *${}^h B_e = B_{e,h}[t_h^h/t] = B_{e,h}[t_h/\varphi^n(t_h)]$, and hence $B_{e,h} \subset {}^h B_e \subset \mathrm{Frac} B_{e,h}$.*

By the same method used in the proof of Theorem 7.28, we have:

Proposition 6.34. *Suppose $h \geq 1$ is an integer, then (1) $P_{h,0}^+ = \mathbb{Q}_{p^h}$ and for every $d < 0$, $P_{h,d}^+ = 0$.*
(2) The sequence

$$0 \to \mathbb{Q}_{p^h} t_h \to P_{h,1}^+ \xrightarrow{\theta} C \to 0$$

is exact.
(3) Suppose $u \in P_{h,1}^+ - \mathbb{Q}_{p^h} t_h$, then for $d > 0$,

$$P_{h,d}^+ = \{a_0 t_h^{d-1} + a_1 u t_h^{d-2} + \cdots + a_{d-1} u^{d-1} \mid a_i \in P_{h,1}^+\}.$$

(4) The sequence

$$0 \to \mathbb{Q}_{p^h} \to B_{e,h} \oplus B_{\mathrm{dR}}^+ \to B_{\mathrm{dR}} \to 0$$

is exact.

The exact sequence in (4) is called the fundamental exact sequence of $B_{e,h}$.

6.7. B_{st} and its properties.

Definition 6.35. The ring of semi-stable periods, of log-crystalline periods, B_{st}, is the ring $B_{\mathrm{cris}}[\mathbf{u}]$, the sub-$B_{\mathrm{cris}}$-algebra of B_{dR} generated by $\mathbf{u} = \log[\varpi]$. Let $C_{\mathrm{st}} = \mathrm{Frac} B_{\mathrm{st}}$.

Theorem 6.36. *(1). The map*

$$\iota : K \otimes_{K_0} B_{\mathrm{st}} \to B_{\mathrm{dR}},\ \lambda \otimes b \mapsto \lambda b$$

is injective.
(2). $(C_{\mathrm{st}})^{G_K} = K_0$, hence

$$(B_{\mathrm{cris}}^+)^{G_K} = (B_{\mathrm{cris}})^{G_K} = (B_{\mathrm{st}})^{G_K} = K_0.$$

The ring of semi-stable periods B_{st} is endowed with two operators: the Frobenius φ and the monodromy operator N. By the definition of the logarithm map, we extend $\varphi : B_{\mathrm{cris}} \to B_{\mathrm{cris}}$ to an endomorphism of B_{st} by requiring

$$\varphi(\mathbf{u}) = p\mathbf{u}.$$

Then φ commutes with the action of G_K. One sees that $\varphi : B_{\mathrm{st}} \to B_{\mathrm{st}}$ is injective.

Definition 6.37. The monodromy operator N is defined by

$$N : B_{\mathrm{st}} \to B_{\mathrm{st}},\ \sum_{n\in\mathbb{N}} b_n \mathbf{u}^n \mapsto -\sum_{n\geq 1} n b_n \mathbf{u}^{n-1},$$

which is the unique B_{cris}-derivation such that $N(\mathbf{u}) = 1$.

Definition 6.38. The monodromy operator N is a nilpotent operator satisfying:
(1). the sequence

$$0 \to B_{\mathrm{cris}} \to B_{\mathrm{st}} \xrightarrow{N} B_{\mathrm{st}} \to 0$$

is exact;
(2). $gN = Ng$ for every $g \in G_{K_0}$;
(3). $N\varphi = p\varphi N$.

Proposition 6.39. *The ring B_{cris} and B_{st} are $(\mathbb{Q}_p, G_K)$-regular, which means that:*
(1). B_{cris} and B_{st} are domains;
(2). $B_{\mathrm{cris}}^{G_K} = B_{\mathrm{st}}^{G_K} = C_{\mathrm{st}}^{G_K} = K_0$;
(3). If $b \in B_{\mathrm{cris}}$ (resp. B_{st}), $b \neq 0$, such that $\mathbb{Q}_p b$ is stable under G_K, then b is invertible in B_{cris}.

7. Admissibility and Weak Admissibility

7.1. (φ, N)-modules and filtered (φ, N)-modules.

Definition 7.1. The category of (φ, N)-module over K_0, denoted by $\mathbf{Mod}_{K_0}(\varphi, N)$, is the following category:
(1). An object in $\mathbf{Mod}_{K_0}(\varphi, N)$ is a finite dimensional K_0-vector space D equipped with two maps

$$\varphi, N : D \to D,$$

satisfying the following properties: (a) φ is bijective and semi-linear with respect to the absolute Frobenius σ on K_0;
(b) N is a K_0-linear map;
(c) $N\varphi = p\varphi N$.
(2). A morphism $\eta : D_1 \to D_2$ between two (φ, N)-modules is a K_0-linear map commuting with φ and N.

The category of φ-module over K_0, denoted by $\mathbf{Mod}_{K_0}(\varphi)$, is the full sub-category $\mathbf{Mod}_{K_0}(\varphi, N = 0)$ of $\mathbf{Mod}_{K_0}(\varphi, N)$. An object of it is also called a φ-isocrystal of K_0.

Remark. The forgetful functor from $\mathbf{Mod}_{K_0}(\varphi, N)$ to $\mathbf{Mod}_{K_0}(\varphi)$ is exact.

Proposition 7.2. *The category $\boldsymbol{Mod}_{K_0}(\varphi, N)$ is an abelian category. Moreover, there exist tensor products, unit and dual object in $\boldsymbol{Mod}_{K_0}(\varphi, N)$:*

(1). For D_1 and D_2 in $\boldsymbol{Mod}_{K_0}(\varphi, N)$, the tensor product $D_1 \otimes D_2 = D_1 \otimes_{K_0} D_2$ with

$$\varphi(d_1 \otimes d_2) = \varphi(d_1) \otimes \varphi(d_2), \quad N(d_1 \otimes d_2) = N(d_1) \otimes d_2 + d_1 \otimes N(d_2).$$

(2). K_0 has a structure of (φ, N)-module by $\varphi = \sigma$ and $N = 0$. Moreover,

$$K_0 \otimes D = D \otimes K_0 = D$$

for every (φ, N)-module D.

(3). If D is an object in $\boldsymbol{Mod}_{K_0}(\varphi, N)$, then the dual object $D^* = \mathcal{L}(D, K_0)$ of D is the set of linear maps $\eta : D \to K_0$ with φ and N given by

$$\varphi(\eta) = \sigma \circ \eta \circ \varphi^{-1}, \quad N(\eta) = -\eta \circ N.$$

Proposition 7.3. *For a (φ, N)-module D, the operator N is nilpotent on it.*

Proof. If N is not nilpotent, let h be an integer such that $N^h(D) = N^{h+1}(D) = \cdots = N^m(D)$ for every $m \geq h$, then $D' = N^h(D) \neq 0$ is invariant under N, and by φ since $N^m\varphi = p^m\varphi N^m$ for every integer $m > 0$. Pick a basis of D' and suppose under this basis, the matrices of φ and N are A and B respectively, then A and B must be invertible by the bijectivity of φ and N. By the relation $N\varphi = p\varphi N$, we have $BA = pA\sigma(B)$. Consequently $v_p(\det(B)) = \dim D' + v_p(\det(\sigma(B))) = \dim D' + v_p(\det(B))$, hence $\det(B) = 0$, contradicting that B is invertible. □

Definition 7.4. The category of filtered (φ, N)-modules over K, denoted by $\mathbf{MF}_K(\varphi, N)$, is the following category:
(1). An object of $\mathbf{MF}_K(\varphi, N)$ is a pair (D, D_K), where:
(a). D is a (φ, N)-module over K_0;
(b). $D_K = K \otimes_K D$ is a filtered K-vector space, i.e. D_K is a K-vector space equpped with a decreasing filtration $\cdots \subset \mathrm{Fil}^i D_K \subset \mathrm{Fil}^{i+1} D_K \subset \cdots$ such that $\bigcap_{i\in\mathbb{Z}} \mathrm{Fil}^i D_K = 0$ (aka. separated) and $\bigcup_{i\in\mathbb{Z}} \mathrm{Fil}^i D_K = D_K$(aka. exhausted).
(2). An morphism $\eta : D_1 \to D_2$ between two filtered (φ, N)-modules is a morphism of (φ, N)-modules such that the induced K-linear map $\eta_K = \mathrm{Id}_K \otimes \eta$ is a morphism of filtered K-vector spaces, i.e.

$$\eta_K(\mathrm{Fil}^i D_{1K}) \subset \mathrm{Fil}^i D_{2K}, \quad i \in \mathbb{Z}.$$

Proposition 7.5. *The category $\boldsymbol{MF}_K(\varphi, N)$ is an additive category with kernels and cokernels. More precisely, for a morphism $\eta : D_1 \to D_2$ between two filtered (φ, N)-modules, the kernel and the cokernel are the K_0-vector spaces $\ker\eta$ and cokerη, equipped with a natural structure filtered (φ, N)-module respectively. Moreover, there exist tensor products, unit and dual object in $\boldsymbol{MF}_K(\varphi, N)$.*

7.2. The Newton number t_N.

Definition 7.6. Let D be a φ-crystal over K_0 of dimension 1, set

$$t_N(D) = v_p(\lambda),$$

where $\lambda \in \mathrm{GL}_1(K_0) = K_0^\times$ is a matrix of φ under some basis.

One can easily check that the $t_N(D)$ is well defined, independent to the choice of the basis. Furthermore:

Definition 7.7. Let D be a φ-crystal over K_0 of dimension n, then define

$$t_N(D) := t_N(\bigwedge_n D),$$

where $\bigwedge_n D$ is a 1-dimensional φ-crystal and hence $t_N(\bigwedge_n D)$ is well defined.

Proposition 7.8. *For any φ-crystal over K_0, let A be a matrix of φ under any K_0-basis, then*

$$v_p(detA) = t_N(D).$$

Proposition 7.9. *(1). For any φ-crystals D_1, D_2, D_3 over K_0, suppose there is a exact sequence*

$$0 \to D_1 \to D_2 \to D_3 \to 0,$$

then $t_N(D_2) = t_N(D_1) + t_N(D_3)$;
(2). For any φ-crystals D_1, D_2, $t_N(D_1 \otimes D_2) = t_N(D_1) \dim_{K_0} D_2 + \dim_{K_0} D_1 t_N(D_2)$;
(3). For any φ-crystal D, $t_N(D^*) = -t_N(D)$

Definition 7.10. A φ-crystal is called pure of slope μ if there exists a W-lattice M of D stable under $p^{-d}\varphi^h$, i.e. $p^{-d}\varphi^h(M) = M$, where $\mu = \frac{d}{h}$, $d, h \in \mathbb{Z}$ and $h \geq 1$.

If D is a φ-crystal pure of slope μ, then $\mu \dim_K D = t_N(D)$. The following theorem of Dieudonné and Manin classify φ-crystals:

Theorem 7.11. *(Dieudonné-Manin) For a φ-module D over K_0,*

$$D = \bigoplus_{\mu \in \mathbb{Q}} D_\mu,$$

where D_μ is the part of D pure of slope μ and $D_\mu = 0$ for all but finitely many μ. Thus $t_N(D_\mu) = \mu \dim_{K_0} D_\mu$ and

$$t_N(D) = \sum \mu \dim_{K_0} D_\mu.$$

The proof is just complicated linear algebra, we skip it here. Notice that if the residue field k of K_0 is algebraically closed, then one has:

Proposition 7.12. *If k is algebraically closed, then:*
(1). If D is pure of slope $\mu = \frac{d}{h}$, then $D = K_0 \otimes_{\mathbb{Q}_{p^h}} D_{\varphi^h = p^d}$;
(2). A short exact sequence of φ-crystals over K_0 always splits.

Definition 7.13. For a φ-crystal D over K_0, suppose $D = \bigoplus_{j=1}^m D_{\mu_j}$, where $0 \neq D_{\mu_j}$ is the part of D of slope $\mu_j \in \mathbb{Q}$ and $\mu_1 < \mu_2 < \cdots < \mu_m$. The Newton polygon $P_N(D)$ of D is the convex polygon with break points (0,0) and $(v_1 + \cdots + v_j, \mu_1 v_1 + \cdots \mu_j v_j)$ for $1 \leq j \leq m$, where $v_j = \dim_{K_0} D_{\mu_j}$. Thus the end point of $P_N(D)$ is $(\dim_{K_0} D, t_N(D))$. Set $\mu(D) = \frac{t_N(D)}{\dim_{K_0} D}$, called the slope of D.

7.3. **The Hodge number t_H.**

Definition 7.14. Suppose $\Delta \in \mathbf{Fil}_K$ is a finite dimensional filtered K-vector space.
(1). If $\dim_K \Delta = 1$, define $t_H(\Delta) := \max\{i \in \mathbb{Z} \mid \mathrm{Fil}^i \Delta = \Delta\}$. Thus it is the integer i such that $\mathrm{Fil}^i = \Delta$ and $\mathrm{Fil}^{i+1}\Delta = 0$;
(2). If $\dim_K \Delta = h$, define $t_H(\Delta) := t_H(\bigwedge_h \Delta)$.

Proposition 7.15.

$$t_H(\Delta) = \sum_{i \in \mathbb{Z}} i \cdot \dim_K \mathrm{gr}^i \Delta.$$

Proposition 7.16. *(1). For any filtered K-vector spaces Δ_1, Δ_2, Δ_3 over K_0, suppose there is a exact sequence*

$$0 \to \Delta_1 \to \Delta_2 \to \Delta_3 \to 0,$$

then $t_H(\Delta_2) = t_H(\Delta_1) + t_H(\Delta_3)$;
(2). For any filtered K-vector spaces Δ_1, Δ_2, $t_H(\Delta_1 \otimes \Delta_2) = t_H(\Delta_1) \dim_{K_0} \Delta_2 + \dim_{K_0} \Delta_1 t_H(\Delta_2)$;
(3). For any filtered K-vector space Δ, $t_H(\Delta^*) = -t_H(\Delta)$

Definition 7.17. For $\Delta \in \mathbf{Fil}_K$, suppose $\mathrm{gr}\Delta = \bigoplus_{j=1}^{m} \mathrm{gr}^{i_j}\Delta$ with $i_1 < \cdots < i_m$, let h_j be the K-dimension of $\mathrm{gr}^{i_j}\Delta$. The Hodge polygon $P_H(\Delta)$ of Δ is the convex polygon with break point $(0,0)$ and $(h_1+\cdots+h_j, i_1h_1+\cdots+i_jh_j)$ for $1 \le j \le m$. Thus the end point of $P_H(\Delta)$ is just $(\dim \Delta, t_H(\Delta))$.

7.4. **Admissible and weakly admissible filtered (φ, N)-modules.** For a p-adic representation V of G_K, $\mathbf{D}_{\mathrm{st}}(V) = (B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V)^{G_K}$ is a filtered (φ, N)-module, and the functor

$$\mathbf{D}_{\mathrm{st}} : \mathbf{Rep}_{\mathbb{Q}_p}(G_K) \to \mathbf{MF}_K(\varphi, N)$$

is covariant.

Definition 7.18. A filtered (φ, N)-module D over K is called weakly admissible if:
(1). $t_H(D) = t_N(D)$;
(2). For any sub-object D' of D, $t_H(D') \le t_N(D')$.

It turns out that weakly admissible filtered (φ, N)-modules over K form an abelian category. We denote this category by $\mathbf{MF}_K^{\mathrm{ad}}(\varphi, N)$.

The weak admissibility is described by the Newton number and Hodge number, which reflect the interaction of the Frobenius φ, the monodromy map N and the filtered structure on the representations. As stated in section 1, we will finally establish an equivalence between $\mathbf{MF}_K^{\mathrm{ad}}(\varphi, N)$ and the category of semi-stable representations of G_K:

Definition 7.19. A filtered (φ, N)-module is called admissible if it is an image of $\mathbf{D}_{\mathrm{st}}$.

For a filtered (φ, N)-module D over K, set

$$\mathbf{V}_{\mathrm{st}}(D) := \{v \in B_{\mathrm{st}} \otimes D \mid \varphi v = v,\ Nv = 0,\ v \in \mathrm{Fil}^0(B_{\mathrm{st}} \otimes D)\}.$$

Then $\mathbf{V}_{\mathrm{st}}(D)$ is a sub-$\mathbb{Q}_p$-vector space of $B_{\mathrm{st}} \otimes D$, stable under G_K. We thus reach the final theorem mentioned in the introduction:

Theorem A. *(Weakly admissible implies admissible) The functor*

$$\mathbf{D}_{\mathrm{st}} : \boldsymbol{Rep}_{\mathbb{Q}_p}^{st}(G_K) \to \boldsymbol{MF}_K^{\mathrm{ad}}(\varphi, N)$$

is an equivalence of categories, and the functor $\mathbf{V}_{st} : \boldsymbol{MF}_K^{\mathrm{ad}}(\varphi, N) \to \boldsymbol{Rep}_{\mathbb{Q}_p}^{st}(G_K)$ is a quasi-inverse of $\mathbf{D}_{\mathrm{st}}$. Moreover, they are compatible with tensor product, dual, etc.

And in the next, we will complete our article by giving a detailed proof of this theorem.

8. Weakly admissible implies admissible

8.1. **Unramified representations and modules with trivial filtration.**

Definition 8.1. A filtered K-vector space Δ is said to have trivial filtration if

$$\mathrm{Fil}^0\Delta = \Delta \quad \text{and} \quad \mathrm{Fil}^1\Delta = 0.$$

Lemma 8.2. *A filtered (φ, N)-module D over K with trivial filtration is weakly admissible if and only if D is pure of slope 0. In this case, $N = 0$.*

Proof. Since the filtration of D_K is trivial, $t_H(D') = 0$ for any sub-object D' of D. If D is pure of slope 0, it is clear that $t_N(D') = 0$ for any sub-object D' of D and hence D is weakly admissible. Conversely, if D is weakly admissible, then $t_N(D) = 0$ and $t_N(D') \ge 0$ for any sub-object of D', in particular all slopes of D are ≥ 0. If there is any component of D is pure of positive slope, then the Newton polygon must end up with a point in the first quadrant.

Thus D is pure of slope 0. In this case, since $N\varphi = p\varphi N$, we have $N(D_\mu) \subset D_{\mu-1}$, and hence $N = 0$. □

Let $P_0 = K_0^{\mathrm{ur}}$ be the unramified closure of K_0, a p-adic representation V of G_K is said to be unramified if it is P_0-admissible, i.e. $\mathbf{D}_{\mathrm{ur}}(V) := (P_0 \otimes_{\mathbb{Q}_p} V)^{G_K}$ is a K_0-vector space of dimension $h = \dim V$. Furthermore, since $(P_0 \otimes V)^{I_K}$ is a G_k-representation, and the identity

$$D = \mathbf{D}_{\mathrm{ur}}(V) = ((P_0 \otimes_{\mathbb{Q}_p} V)^{I_K})^{G_k} = (P_0 \otimes_{\mathbb{Q}_p} V)^{G_K}$$

gives that $(P_0 \otimes_{\mathbb{Q}_p} V)^{I_K}$ is of dimension h,[2] and D is an étale φ-module of dimension h, hence a φ-crystal pure of slope 0. The inclusion $P_0 \subset B_{\mathrm{cris}}^+ \subset B_{\mathrm{st}}$ implies that V is crystalline and semi-stable, and $\mathbf{D}_{\mathrm{cris}}(V) = \mathbf{D}_{\mathrm{st}}(V) = \mathbf{D}_{\mathrm{ur}}(V) = D$. Thus D is a filtered (φ, N)-module with $N = 0$. D is also of trivial filtration since $P_0 \subset B_{\mathrm{dR}}^+/\mathrm{Fil}^1 B_{\mathrm{dR}}^+$, and by the above lemma, D is a weakly admissible filtered (φ, N)-module of dimension h with trivial filtration.

Conversely, if D is a weakly admissible filtered (φ, N)-module of dimension h with trivial filtration, then D is a φ-crystal pure of slope 0, i.e. an étale φ-module. Thus

$$V = \mathbf{V}_{\mathrm{ur}}(D) = (P_0 \otimes_{K_0} D)_{\varphi=1}$$

is a p-adic representation of G_k of dimension h, hence a unramified p-adic representation of G_K of dimension h.

Proposition 8.3. *Every unramified p-adic representation of G_K is crystalline and $\mathbf{D}_{\mathrm{st}}$ induces an equivalence of categories between $\boldsymbol{Rep}_{\mathbb{Q}_p}^{ur}(G_K)$, the category of unramified p-adic representations of G_K (equivalently $\boldsymbol{Rep}_{\mathbb{Q}_p}(G_k)$) and the category of weakly admissible filtered (φ, N)-modules with trivial filtration (equivalently, of étale φ-modules over K_0), and $\mathbf{V}_{st}$ is a quasi-inverse of $\mathbf{D}_{\mathrm{st}}$.*

Proof. It remains to show that $\mathbf{V}_{\mathrm{st}}(\mathbf{D}_{\mathrm{st}}(V)) = V$ and $\mathbf{D}_{\mathrm{st}}(\mathbf{V}_{\mathrm{st}}(D)) = D$ for every V and D. □

8.2. **Tate Twist of filtered (φ, N)-modules.**

Definition 8.4. Suppose D is a filtered (φ, N)-module. For $i \in \mathbb{Z}$, the i-th Tate twist $D\langle i\rangle$ of D is the following filtered (φ, N)-module:
(i). $D\langle i\rangle = D$ as K_0-vector space;
(ii). $\mathrm{Fil}^r(D\langle i\rangle)_K = \mathrm{Fil}^{r+i} D_K$;
(iii). $N|_{D\langle i\rangle} = N|_D$, $\varphi|_{D\langle i\rangle} = p^{-i}\varphi|_D$.

Lemma 8.5. *(1). A p-adic representation V of G_K is de-Rham (resp. semi-stable, crystalline) if and only if any Tate twist $V\langle i\rangle$ is de-Rham (resp. semi-stable, crystalline).*
(2). A filtered (φ, N)-module D is weakly admissible if and only if any Tate twist $D\langle i\rangle$ is weakly admissible.
(3). For any $i \in \mathbb{Z}$, $\mathbf{D}_{\mathrm{st}}(V(i)) \xrightarrow{\sim} (\mathbf{D}_{\mathrm{st}}(V))\langle i\rangle$.

8.3. **Change of residue fields.** Recall that I_K is the inertia subgroup of G_K and the sequence

$$0 \to I_K \to G_K \to G_k \to 0$$

is exact.

[2]This also shows that V is unramified if and only if the action of I_K on V is trivial.

Proposition 8.6. *(1). V is de-Rham as a p-adic representation of G_K if and only if V is de-Rham as a p-adic representation of I_K.*
(2). V is semi-stable as a p-adic representation of G_K if and only if V is semi-stable as a p-adic representation of I_K.

Proof. Let $\overline{P}$ be an algebraic closure of $P = P_0K = \widehat{K}^{\rm ur}$ inside C, then $\overline{P} \subset B_{\rm dR}^+$ and $I_K = {\rm Gal}(\overline{P}/P)$. Note that $B_{\rm dR}(\overline{P}/P) = B_{\rm dR}(\overline{K}/K) = B_{\rm dR}$ ($B_{\rm dR}$ only depends on C), thus $B_{\rm dR}^{I_K} = P$. If V is a p-adic representation of G_K,

$$\mathbf{D}_{{\rm dR},P}(V) = (B_{\rm dR} \otimes_{\mathbb{Q}_p} V)^{I_K}$$

is a P-vector space with

$$\dim_P \mathbf{D}_{{\rm dR},P}(V) \leq \dim_{\mathbb{Q}_p} V.$$

V is a de Rham representation of I_K if and only if the equality holds. Note that

$$(\mathbf{D}_{{\rm dR},P}(V))^{G_k} = \mathbf{D}_{\rm dR}(V),$$

and by the theory of representation of Galois group over fields of characteristic p,

$$P \otimes_K (\mathbf{D}_{{\rm dR},P}(V))^{G_k} \to \mathbf{D}_{{\rm dR},P}(V)$$

is an isomorphism, and hence we have (1).

(2). For $\mathbf{D}_{{\rm st},P}(V) = (B_{\rm st} \otimes_{\mathbb{Q}_p} V)^{I_K}$, by a similar way,

$$P_0 \otimes_{K_0} (\mathbf{D}_{{\rm st},P}(V))^{G_k} \to \mathbf{D}_{{\rm st},P}(V)$$

is an isomorphism, and $\mathbf{D}_{\rm st} = (\mathbf{D}_{{\rm st},P}(V))^{G_k}$, the result follows. □

Proposition 8.7. *Let V be a p-adic representation of G_K, associated with*

$$\rho : G_K \to Aut_{\mathbb{Q}_p}(V).$$

Assume $\rho(I_K)$ is finite, then
(1). V is potentially crystalline (potentially semi-stable) and hence de-Rham. (2). The following three conditions are equivalent:
(a). V is semi-stable;
(b). V is crystalline;
(c). $\rho(I_K)$ is trivial, i.e. V is unramified.

Proof. Because of Proposition 8.6, we may assume $k = \bar{k}$ and $K = P$, or $I_K = G_K$. (2) ⇒ (1) is obvious, hence it suffices to prove (2).

The inclusion $P_0 \subset B_{\rm cris}^+ \subset B_{\rm st}$ implies that $(c) \Rightarrow (b) \Rightarrow (a)$. Thus we only need to show $(a) \Rightarrow (c)$.

Let $H = \ker \rho$, H is an open subgroup of G_K, $L = \overline{K}^H$ is a finite Galois extension of K. Write $J = G_K/H$, then

$$\begin{aligned}\mathbf{D}_{\rm st}(V) =& (B_{\rm st} \otimes_{\mathbb{Q}_p} V)^{G_K} = ((B_{\rm st} \otimes_{\mathbb{Q}_p} V)^H)^J \\ =& (B_{\rm st}^H \otimes_{\mathbb{Q}_p} V)^J = (K_0 \otimes_{\mathbb{Q}_p} V)^J = K_0 \otimes_{\mathbb{Q}_p} V^J\end{aligned}$$

Since $B_{\rm st}^H = L_0 = K_0$ by the assumption that $K = P$, therefore

$$V \text{ is semi-stable} \Leftrightarrow \dim_{K_0}(\mathbf{D}_{\rm st}(V)) = \dim_{\mathbb{Q}_p} V^J = \dim_{\mathbb{Q}_p} V \Leftrightarrow V^J = V,$$

which means that $\rho(I_K)$ is trivial. □

8.4. **Weakly admissible implies admissible: 1 Dimensional case.** On one hand, let D be a filtered (φ, N)-module of dimension 1, then we can write $D = K_0 d$. The Frobenius is given by $\varphi(d) = \lambda d$ for some $\lambda \in K_0^\times$, and N must be 0 since N is nilpotent. Thus $t_N(D) = t_p(\lambda)$. Since $D_K = D \otimes_{K_0} K$ is 1-dimensional over K, the filtration is given by

$$\mathrm{Fil}^r D_K = \begin{cases} D_K, \text{ if } r \leq i; \\ 0, \text{ if } r > i. \end{cases}$$

Then the Hodge number of D is $t_H(D) = i$, and therefore D is weakly admissible if and only if $v_p(\lambda) = i$. Conversely, given any $\lambda \in K_0^\times$, it determines a weakly admissible filtered (φ, N)-module D_λ by

$$D_\lambda = K_0, \ \varphi = \lambda\sigma, \ N = 0, \ \mathrm{Fil}^r D_K = \begin{cases} D_K, \text{ if } r \leq v_p(\lambda); \\ 0, \text{ if } r > v_p(\lambda). \end{cases}$$

Thus we have:

Proposition 8.8. *Any admissible (φ, N)-module over K of dimension 1 is of the form D_λ for some $\lambda \in K_0^\times$. Moreover:*
(1). $D_\lambda \cong D_{\lambda'}$ if and only if there exists $u \in W^\times$ such that $\lambda' = \lambda \frac{\sigma(u)}{u}$.
(2). In the special case that $K = K_0 = \mathbb{Q}_p$ and $\sigma = Id$, $D_\lambda \cong D_{\lambda'}$ if and only if $\lambda = \lambda'$.

On the other hand, let V be a p-adic representation of G_K of dimension 1. Write $V = \mathbb{Q}_p v$, $g(v) = \eta(g)v$ for some character

$$\eta : G_K \to \mathbb{Q}_p^\times.$$

Then we can identify the 1-dimensional representation V with the characters η. Moreover, we can make η factors through $\mathbb{Z}_p$ by simply multiplying p^n for some positive integer n, which refers to an isomorphism between representations. For such a character η, it is called B-admissible if the associated representation V is B-admissible.

Proposition 8.9. *Suppose $\eta : G_K \to \mathbb{Z}_p^\times$ is continuous homomorphism, then:*
(1) η is Hodge-Tate if and only if η is of the form $\eta = \eta_0 \chi^i$ for some $i \in \mathbb{Z}$ and character η_0 such that $\eta_0(I_K)$ is finite.
(2) η is de Rham if and only if η is Hodge Tate.
(3) The followingsa are equivalent:
(a) η is semi-stable;
(b) η is crystalline;
(c) There exists an unramified character $\eta_0 : G_K \to \mathbb{Z}_p^\times$ such that $\eta = \eta_0 \chi^i$ for some $i \in \mathbb{Z}$.

Proof. (1) Recall that

$$\mathbf{D}_{\mathrm{HT}}(V) = \bigoplus_{i \in \mathbb{Z}} ((C(i)) \otimes_{\mathbb{Q}_p} V)^{G_K},$$

$V = \mathbb{Q}_p(\eta)$ is Hodge-Tate if and only if there exists a unique $i \in \mathbb{Z}$, such that $(C(-i) \otimes_{\mathbb{Q}_p} V)^{G_K} = (C \otimes V(-i))^{G_K} \neq 0$, if and only if $V(-i)$ is a C-admissible representation. By Sen's theorem, this is equivalent to that $\eta\chi^{-i}(I_K)$ if finite. Let $\eta_0 = \eta\chi^{-i}$, we have (1).

(2) Since de-Rham implies Hodge-Tate, it suffices to show that if η is Hodge Tate, then it is de Rham. In fact, if η is Hodge Tate, let V be the representation associated with η, then $V(-i)$ is C-admissible for some i, and by the inclusion $C \subset B_{\mathrm{dR}}$, $V(-i)$ is de-Rham, and hence V is de-Rham.

(3) follows directly from Proposition 8.7. □

Theorem 8.10. *The functor $\mathbf{D}_{\mathrm{st}}$ gives a bijection of crystalline (equivalently semi-stable) representations of G_K of dimension 1 with admissible filtered (φ, N)-modules over K_0 of dimension 1.*

Proof. If V is a crystalline of dimension 1, then $V = V_0(i)$ with $i \in \mathbb{Z}$ and V_0 unramified by the above proposition, and hence $\mathbf{D}_{\mathrm{st}}(V) = \mathbf{D}_{\mathrm{st}}(V_0)\langle i\rangle$ is an admissible filtered (φ, N)-module over K_0 of dimension 1.

On the other hand, if D is an admissible filtered (φ, N)-module over K_0 of dimension 1, suppose $\mathrm{Fil}^i D_K = D_K$ and $\mathrm{Fil}^{i+1} D_K = 0$, then $D\langle i\rangle$ is with trivial filtration and $V_0 = \mathbf{V}_{\mathrm{st}}(D\langle i\rangle)$ is unramified, and hence $V_0(-i) = \mathbf{V}_{\mathrm{st}}(D)$ is crystalline. □

This gives the following usrful lemma:

Lemma 8.11. *If $b \in B_{\mathrm{cris}}$ satisfies $\varphi b = \lambda b$ with $\lambda \in K_0$ and $v_p(\lambda) = r$, and b is also in $\mathrm{Fil}^{r+1} B_{\mathrm{dR}}$, then $b = 0$.*

Proof. Let $D = K_0 e$ be the one-dimensional filtered (φ, N)-module with $\varphi(e) = \frac{1}{\lambda} e$, $Ne = 0$, and

$$\mathrm{Fil}^i D_K = \begin{cases} D_K, \text{ if } i \leq -r; \\ 0, \text{ if } i > -r. \end{cases}$$

then $t_N(D) = t_H(D) = -r$ and D is admissible. Then $D\langle -r\rangle$ is admissible with trivial filtration. Thus

$$\mathbf{V}_{\mathrm{st}}(D) = \mathbf{V}_{\mathrm{st}}(D\langle -r\rangle)(r)$$

is a crystalline representation of dimension 1. Since for any $b_0 \neq 0$ with $\varphi(b_0) = \lambda b_0$, $V = \mathbb{Q}_p b_0 \otimes e$ is a crystalline representation, and $D = \mathbf{D}_{\mathrm{st}}(V)$, then $\mathbf{V}_{\mathrm{st}}(D) = \mathbf{V}_{\mathrm{st}}(\mathbf{D}_{\mathrm{st}}(V)) = V = \mathbb{Q}_p b_0 \otimes e$, and by the definition of $\mathbf{V}_{\mathrm{st}}$, $b_0 \otimes e \in \mathrm{Fil}^0 D$, and hence $b_0 \in \mathrm{Fil}^r B_{\mathrm{dR}} - \mathrm{Fil}^{r+1} B_{\mathrm{dR}}$. □

8.5. **Reduction of Theorem A.** To prove Theorem A, it is equivalent to show the followings:

Theorem A1. *If V is a semi-stable representation of G_K, then $\mathbf{D}_{\mathrm{st}}(V)$ is weakly admissible and there is a natural isomorphism*

$$V \xrightarrow{\sim} (\mathbf{V}_{\mathrm{st}}(\mathbf{D}_{\mathrm{st}}(V))).$$

Theorem A2. *For any object $D \in \boldsymbol{MF}_K^{\mathrm{ad}}(\varphi, N)$, there exists an object V of $\mathbf{Rep}_{\mathbb{Q}_p}^{st}(G_K)$ such that $\mathbf{D}_{\mathrm{st}}(V) \cong D$.*

Firstly let us construct the isomorphism in Theorem A1. Let $D = \mathbf{D}_{\mathrm{st}}(V)$. Since V is semi-stable, the natural map

$$\alpha_{\mathrm{st}} : B_{\mathrm{st}} \otimes_{K_0} D \to B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V$$

is an isomorphism. We identify them and call them X. Let $\{v_1, \cdots, v_h\}$ be a basis of V over $\mathbb{Q}_p$, and $\{\delta_1, \cdots, \delta_h\}$ be a basis of D over K_0 respectively. Identify v_i with $1 \otimes v_i$ and δ_i with $1 \otimes \delta_i$. Then $\{v_1, \cdots, v_h\}$ and $\{\delta_1, \cdots, \delta_h\}$ are two bases of X over B_{st}.

An element of X can be written as a sum of the form $b \otimes \delta$ where $b \in B_{\mathrm{st}}$ and $\delta \in D$ and also a sum of the form $c \otimes v$ where $c \in B_{\mathrm{st}}$ and $v \in V$. The actions of G_K, φ, adn N on X are listed below:

$$\begin{aligned}
&G_K\text{-action:}\quad g(b\otimes\delta)=g(b)\otimes\delta,\quad g(c\otimes v)=g(c)\otimes g(v)\\
&\varphi\text{-action:}\quad \varphi(b\otimes\delta)=\varphi(b)\otimes\varphi(\delta),\quad \varphi(c\otimes v)=\varphi(c)\otimes v\\
&N\text{-action:}\quad N(b\otimes\delta)=N(b)\otimes\delta+b\otimes N(\delta),\quad N(c\otimes v)=N(c)\otimes v
\end{aligned}$$

We also know that X is endowed with a filtration. By the map $x\mapsto 1\otimes x$, one has the inclusion

$$X\subset X_{\mathrm{dR}}=B_{\mathrm{dR}}\otimes_{B_{\mathrm{st}}}X=B_{\mathrm{dR}}\otimes_K D_K=B_{\mathrm{dR}}\otimes V.$$

Then the filtration of X is induced by

$$\mathrm{Fil}^i B_{\mathrm{dR}}=\mathrm{Fil}^i B_{\mathrm{dR}}\otimes_{\mathbb{Q}_p}V=\bigoplus_{r+s=i}\mathrm{Fil}^r B_{\mathrm{dR}}\otimes_K\mathrm{Fil}^s D_K.$$

Clearly $V\subset\mathbf{V}_{\mathrm{st}}(D)=\{x\in X\mid \varphi(x)=x,\ N(x)=0\, x\in\mathrm{Fil}^0X\}$. We need to check that $\mathbf{V}_{\mathrm{st}}(D)=V$.

For any $x\in\mathbf{V}_{\mathrm{st}}(D)$, write $x=\sum_{n=1}^h b_n\otimes v_n$ where $b_n\in B_{\mathrm{st}}$. Firstly $N(x)=\sum_n N(b_n)\otimes v_n=0$ implies that $N(b_n)=0$, thus $b_n\in B_{\mathrm{cris}}$, and secondly $\varphi(x)=x$ means $\sum_n\varphi(b_n)\otimes v_n=\sum_n b_n\otimes v_n$, then $\varphi(b_n)=b_n$, which means that $b_n\in B_e$. Finally, $\sum_n b_n\otimes v_n\in\mathrm{Fil}^0X$ implies that $b_n\in\mathrm{Fil}^0B_{\mathrm{dR}}$. Since $\mathrm{Fil}^0B_e=\mathbb{Q}_p$ by the fundamental exact sequence, we have that $b_n\in\mathbb{Q}_p$. Therefore $x\in V$, which implies that $V=\mathbf{V}_{\mathrm{st}}(D)$.

Then we complete Theorem A1 by showing the admissibility of D.

Suppose D' is a sub-object of D, i.e. D' be a sub-K_0-vector space of D stable under φ and N.

(1) Assume first that $\dim_{K_0}D'=1$. Let $\{v_1,\cdots,v_h\}$ be a basis of V over $\mathbb{Q}_p$. Write $D'=K_0\delta$, suppose $\varphi(\delta)=\lambda\delta$, $\lambda\in K_0^\times$, then $t_N(D')=v_p(\lambda)=:r$ and $N\delta=0$.

Write $\delta=\sum_{i=1}^h b_i\otimes v_i$, then

$$\varphi(\delta)=\sum_{i=1}^h b_i\otimes v_i\quad\text{and}\quad N\delta=\sum_{i=1}^h Nb_i\otimes v_i,$$

so $\varphi b_i=\lambda b_i$ and $N(b_i)=0$, which implies that $b_i\in B_{\mathrm{cris}}$.

Assume $t_H(D')=s$, which means that $\delta\in\mathrm{Fil}^s(B_{\mathrm{dR}}\otimes_{\mathbb{Q}_p}V)-\mathrm{Fil}^{s+1}(B_{\mathrm{dR}}\otimes_{\mathbb{Q}_p}V)$, and this implies that $b_i\in\mathrm{Fil}^sB_{\mathrm{cris}}$. By Lemma 8.11, $0\neq b\notin\mathrm{Fil}^{r+1}$, and thus $s\leq r$.

Furthermore, in the last subsection we see that is $D=D'$ is of dimension 1, then $t_H(D)=t_N(D)$.

(2) In general, let $\dim_{K_0}D'=m$. Let $V_1=\bigwedge_m V$, then V_1 is semi-stable as well and

$$\mathbf{D}_{\mathrm{st}}(V_1)=\bigwedge_m\mathbf{D}_{\mathrm{st}}(V)=\bigwedge_m D.$$

Now $\bigwedge_m D'\subset\bigwedge_m D$ is a sub-object of dimension 1, and

$$t_H(\bigwedge_m D')=t_H(D'),\quad t_N(\bigwedge_m D')=t_N(D'),$$

and the general case is reduced to the one dimensional case.

Thus now, to prove Theorem A, it suffices to prove Theorem A2.

8.6. **Reduction of Theorem A2.**

Proposition 8.12. *Let D be an admissible filtered (φ, N)-module over K of dimension $h \geq 1$. Let $V = \mathbf{V}_{\mathrm{st}}(D)$, then $\dim_{\mathbb{Q}_p} V \leq h$, V is semi-stable and $\mathbf{D}_{\mathrm{st}} \subset D$ is a sub-object.*

With this proposition, we now only need to prove the following equivalent conditions:
(1) $D \cong \mathbf{D}_{\mathrm{st}}(\mathbf{V}_{\mathrm{st}}(D))$;
(2) $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) \geq h$;
(3) $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) = h$.
We need the following lemma:

Lemma 8.13. *Let F be a filed, J be a subgroup of $Aut(F)$ and $E = F^J$. Let Δ be a finite dimensional E-vector space, and*

$$\Delta_F = F \otimes_E \Delta.$$

Let J acts on Δ_F through $j(\lambda \otimes \delta) = j(\lambda) \otimes \delta$. By the map $\delta \mapsto 1 + \delta$, we identify Δ with $1 \otimes_E \Delta = (\Delta_F)^J$. Let L be a sub-F-vector space of Δ_F. Then there exists Δ', a sub-E-vector space such that $L = F \otimes_E \Delta'$ if and only if L is stable under J, i.e. $j(L) = L$ for any $j \in J$.

Proof. If $L = \Delta' \otimes_E F$, then $j(L) = j(\Delta' \otimes_E F) = L$ is clear. Conversely, if L is stable under J, then we have an exact sequence of F-vector spaces with J-action

$$0 \to L \to \Delta_F \to \Delta_F/L \to 0,$$

taking the J-invariants, we have

$$0 \to L^J \to \Delta \to (\Delta_F/L)^J.$$

Then

$$\dim_E \Delta = \dim_F \Delta_F = \dim_F L + \dim_F(\Delta_F/J) \leq \dim_E L^J + \dim_E(\Delta_F/L)^J.$$

But $\dim_E L^J = \dim_F L$, thus $\Delta' = L^J$ satisfied $L = F \otimes \Delta'$. □

Proof of Theorem 8.12 We may assume that $V \neq 0$. Apply the above lemma to the case

$$\Delta = D,\ F = C_{\mathrm{st}} = \mathrm{Frac}(B_{\mathrm{st}}),\ J = G_K,\ E = C_{\mathrm{st}}^{G_K} = K_0,$$

then

$$\Delta_F = C_{\mathrm{st}} \otimes_{K_0} D \supset B_{\mathrm{st}} \otimes_{K_0} D \supset V.$$

Let L be the sub-C_{st}-vector space generated by V. The actions of φ, N and G_K on B_{st} extend to C_{st}, thus L is stable under φ, N and G_K. By the lemma, there exists a sub-K_0-vector space D' of D, such that

$$L = C_{\mathrm{st}} \otimes_{K_0} D'.$$

Since L is stable under φ, N and G_K, so is D'. Choose a basis $\{v_1, \cdots, v_r\}$ of L over C_{st} consisting of element of V. Choose a basis $\{d_1, \cdots, d_r\}$ of D' over K_0 which is a basis of L over C_{st}. Write

$$v_i = \sum_{j=1}^{r} b_{ij} d_j, \quad b_{ij} \in B_{\mathrm{st}}.$$

By the inclusion $B_{\mathrm{st}} \otimes_{K_0} D' \subset B_{\mathrm{st}} \otimes_{K_0} D$, we have

$$\bigwedge_r (B_{\mathrm{st}} \otimes_{K_0} D') = B_{\mathrm{st}} \otimes_{K_0} \bigwedge_r D' \subset \bigwedge_r (B_{\mathrm{st}} \otimes D) = B_{\mathrm{st}} \otimes_{K_0} \bigwedge_r D.$$

Let $v_0 = v_1 \wedge \cdots \wedge v_r$, $d_0 = d_1 \wedge \cdots \wedge d_r$ and $b = \det(b_{ij})$, then d_0 is a basis of $\bigwedge_r D'$ and so is $v_0 = bd_0$. Hence $b \neq 0$. Since $N = 0$ in $\bigwedge_r D'$, $b \in B_{\mathrm{cris}}$. Suppose $\varphi(d_0) = \lambda d_0$,

then $t_N(D') = t_p(\lambda) =: r$. Since $\varphi(v_0) = v_0$, $\varphi(b) = \lambda^{-1}b$, and by Lemma 8.11, we have that $b \in \mathrm{Fil}^{-s}B_{\mathrm{dR}}$ for some $-s \leq -r$, and hence $d_0 = b^{-1}v_0 \in \mathrm{Fil}^s B_{\mathrm{cris}}$ for some $s \geq r$. Thus $t_H(D') \geq t_N(D')$. On the other hand, the admissibility implies that $t_H(D') \leq t_N(D')$, thus $t_H(\bigwedge_r D') = t_H(D') = t_N(D') = t_N(\bigwedge_r D')$, thus D' and $\bigwedge^r D'$ is an admissible filtered (φ, N)-module of dimension 1, and $\mathbf{V}_{\mathrm{st}}(\bigwedge_r D')$ is a crystalline representation of dimension 1. Since $v_i \in \mathbf{V}_{\mathrm{st}}(D')$ and hence $v_0 \in \mathbf{V}_{\mathrm{st}}(\bigwedge_r D')$, we have

$$\mathbf{V}_{\mathrm{st}}(\bigwedge_r D') = \mathbb{Q}_p v_0.$$

For any $v \in \mathbf{V}_{\mathrm{st}}(D') = V$, write $v = \mathbf{V}_{\mathrm{st}}(D') = V$, write $v = \sum_{i=1}^r c_i v_i$ with $c_i \in C_{\mathrm{st}}$, $1 \leq i \leq r$, then

$$v_1 \wedge \cdots \wedge v_{i-1} \wedge v \wedge v_{i+1} \wedge v_r = c_i v_0 \in \bigwedge_r V \subset \mathbf{V}_{\mathrm{st}}(\bigwedge_r D') = \mathbb{Q}_p v_0,$$

thus $c_i \in \mathbb{Q}_p$, and thus V is a $\mathbb{Q}_p$-vector space generated by $\{v_1, \cdots, v_r\}$, whose dimension $r = \dim_{K_0} D' \leq \dim_{K_0} D$. Because

$$\mathbf{V}_{\mathrm{st}}(D') = V \quad \text{and} \quad \mathbf{D}_{\mathrm{st}}(V) = D',$$

V is also semi-stable. □

8.7. **Reduction of the reducibility of Theorem A.** In this subsection we prove the following proposition, which reduces the proof of Theorem A to the case D is irreducible.

Proposition 8.14. *If $0 \to D' \to D \to D'' \to 0$ is a short exact sequence of weakly admissible filtered (φ, N)-modules over K, and if*

$$\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D') = \dim_{K_0} D', \quad \dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D'') = \dim_{K_0} D'',$$

then $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) = \dim_{K_0} D$.

Definition 8.15. Let D be a filtered (φ, N)-module, then set

$$\mathbf{V}^0_{\mathrm{st}}(D) := \{b \in B_{\mathrm{st}} \otimes_{K_0} D \mid Nb = 0, \varphi b = b\}$$
$$\mathbf{V}^1_{\mathrm{st}}(D) := B_{\mathrm{dR}} \otimes_K D_K / \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D_K).$$

There is a natural map $\mathbf{V}^0_{\mathrm{st}}(D) \to \mathbf{V}^1_{\mathrm{st}}(D)$ induced by

$$B_{\mathrm{st}} \otimes_{K_0} D \subset B_{\mathrm{dR}} \otimes_K D_K \twoheadrightarrow \mathbf{V}^1_{\mathrm{st}}(D).$$

Then the sequence

$$0 \to \mathbf{V}_{\mathrm{st}}(D) \to \mathbf{V}^0_{\mathrm{st}}(D) \to \mathbf{V}^1_{\mathrm{st}}(D)$$

is exact. We call it the fundamental complex of D.

Proposition 8.16. *If $0 \to D' \to D \to D'' \to 0$ is a short exact sequence of filtered (φ, N)-modules over K, then for $i = 0, 1$, the sequence*

$$0 \to \mathbf{V}^i_{\mathrm{st}}(D') \to \mathbf{V}^i_{\mathrm{st}}(D) \to \mathbf{V}^i_{\mathrm{st}}(D'') \to 0$$

is exact.

Proof. For $i = 1$, by assumption, the exact sequence $0 \to D'_K \to D_K \to D''_K \to 0$ is exact. Tensor with B_{dR}, we get an exact sequence

$$0 \to B_{\mathrm{dR}} \otimes_K D'_K \to B_{\mathrm{dR}} \otimes_K D_K \to B_{\mathrm{dR}} \otimes_K D''_K \to 0,$$

and it follows that the sequence

$$0 \to \mathrm{Fil}^i B_{\mathrm{dR}} \otimes_K \mathrm{Fil}^{-i} D'_K \to \mathrm{Fil}^i B_{\mathrm{dR}} \otimes_K \mathrm{Fil}^{-i} D_K \to \mathrm{Fil}^i B_{\mathrm{dR}} \otimes_K \mathrm{Fil}^{-i} D''_K \to 0$$

is exact. As a quotient of these two exact sequences, we have the exact sequence of $\mathbf{V}^1_{\mathrm{st}}$.

For $i = 0$, notice that there is a $\mathbb{Q}_p$-linear bijection between $\mathbf{V}^0_{\mathrm{cris}}(D) = \{y \in B_{\mathrm{cris}} \otimes_{K_0} D \mid \varphi y = y\}$ and $\mathbf{V}^0_{\mathrm{st}}(D)$ for every filtered (φ, N)-module D. Thus it suffices to show that the sequence

$$0 \to \mathbf{V}^0_{\mathrm{cris}}(D') \to \mathbf{V}^0_{\mathrm{cris}}(D) \to \mathbf{V}^0_{\mathrm{cris}}(D'') \to 0$$

is exact.

In fact, if k is algebraically closed, then the exact sequence

$$0 \to D' \to D \to D'' \to 0$$

splits as a sequence of φ-isocrystals by Dieudonnè-Manin Theorem. Then $D \cong D' \oplus D''$, and hence $\mathbf{V}^0_{\mathrm{cris}}(D) \cong \mathbf{V}^0_{\mathrm{cris}}(D') \oplus \mathbf{V}^0_{\mathrm{cris}}(D'')$. If k is arbitrary, then $\mathbf{V}^0_{\mathrm{cris}}(D) = \{y \in B_{\mathrm{cris}} \otimes_{K_0} D \mid \varphi y = y\} = \{y \in B_{\mathrm{cris}} \otimes_{P_0} (P_0 \otimes \otimes_{K_0} D) \mid \varphi y = y\}$. Since $P_0 \otimes_{K_0} D$ is a φ-isocrystal over P_0 whose residue field is $\bar{k}$, the exact sequence

$$0 \to P_0 \otimes_{K_0} D' \to P_0 \otimes_{K_0} D \to P_0 \otimes_{K_0} D'' \to 0$$

splits and the result follows. □

Proposition 8.17. *If V is semi-stable, let $D = \mathbf{D}_{\mathrm{st}}(V)$, then the sequence*

$$0 \to \mathbf{V}_{\mathrm{st}}(D) \to \mathbf{V}^0_{\mathrm{st}}(D) \to \mathbf{V}^1_{\mathrm{st}}(D) \to 0$$

is exact.

Proof. Notice that

$$\mathbf{V}^0_{\mathrm{st}}(D) = \{x \in B_{\mathrm{st}} \otimes_{K_0} D = B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V \mid Nx = 0, \varphi x = x\} = B_e \otimes_{\mathbb{Q}_p} V$$

and

$$\mathbf{V}^1_{\mathrm{st}}(D) = B_{\mathrm{dR}} \otimes_K D_K / \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D_K) = B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V / \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V) = \tilde{B_{\mathrm{dR}}} \otimes_{\mathbb{Q}_p} V,$$

by the fundamental exact sequence

$$0 \to \mathbb{Q}_p \to B_e \to \tilde{B_{\mathrm{dR}}} \to 0,$$

we have the sequence

$$0 \to V \to B_e \otimes V \to \tilde{B_{\mathrm{dR}}} \otimes V \to 0$$

exact. Since $V = \mathbf{V}_{\mathrm{st}}(D)$, the desired exact sequence follows. □

Now we can prove Proposition 8.14.

Proof of Proposition 8.14 By the reduced Thorem A2, $\mathbf{V}_{\mathrm{st}}(D')$ and $\mathbf{V}_{\mathrm{st}}(D'')$ are semi-stable. The short exact sequence $0 \to D' \to D \to D'' \to 0$ thus induces the following

commutative diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbf{V}_{\mathrm{st}}(D') & \longrightarrow & \mathbf{V}^0_{\mathrm{st}}(D) & \longrightarrow & \mathbf{V}^1_{\mathrm{st}}(D') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbf{V}_{\mathrm{st}}(D) & \longrightarrow & \mathbf{V}^0_{\mathrm{st}}(D) & \longrightarrow & \mathbf{V}^1_{\mathrm{st}}(D) & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbf{V}_{\mathrm{st}}(D'') & \longrightarrow & \mathbf{V}^0_{\mathrm{st}}(D'') & \longrightarrow & \mathbf{V}^1_{\mathrm{st}}(D'') & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

which is exact in rows and columns. Then a diagram chasing shows that $\mathbf{V}_{\mathrm{st}}(D) \to \mathbf{V}_{\mathrm{st}}(D'')$ is onto, and thus we conclude the proof. Precisely the process is as follows: for every $\delta \in \mathbf{V}_{\mathrm{st}}(D'')$, denote $\bar{x}$ the image of δ in $\mathbf{V}^0_{\mathrm{st}}(D'')$, and $x \in \mathbf{V}^0_{\mathrm{st}}(D)$ is a lift of $\bar{x}$. Let y be the image of x in $\mathbf{V}^1_{\mathrm{st}}(D)$, since the image of y in $\mathbf{V}^1_{\mathrm{st}}(D'')$ is 0, y comes from an element in $\mathbf{V}^1_{\mathrm{st}}(D')$, which is denoted by y as well. Such an element $y \in \mathbf{V}^1_{\mathrm{st}}(D')$ is lifted to an element $z \in \mathbf{V}^0_{\mathrm{st}}(D')$, and we denote its image in $\mathbf{V}^0_{\mathrm{st}}(D)$ by z as well. Then $x - z \in \mathbf{V}^0_{\mathrm{st}}(D)$ is an element whose image in $\mathbf{V}^1_{\mathrm{st}}(D)$ is 0 and in $\mathbf{V}^0_{\mathrm{st}}(D'')$ is x. The lift $\xi \in \mathbf{V}_{\mathrm{st}}(D)$ of $x - z$ is a preimage of x, as desired. □

8.8. **$\mathbb{Q}_{p^r}$-representations.** For a $\mathbb{Q}_{p^r}$-representation of G_K, we say it is de-Rham (semi-stable, crystalline, etc.) if it is de-Rham (semi-stable, crystalline, etc.) as a p-adic representation. Now suppose V is a $\mathbb{Q}_{p^r}$ representation of G_K. Then we have the following decomposition

$$B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V = B_{\mathrm{st}} \otimes_{\mathbb{Q}_{p^r}} (\mathbb{Q}_{p^r} \otimes_{\mathbb{Q}_p} V) = \bigoplus_{m=0}^{r-1} B_{\mathrm{st}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V,$$

where ${}_{\sigma^m}\otimes_{\mathbb{Q}_{p^r}}$ is the twisted tensor product by σ^m. Each component of this decomposition is stable by the G_K-action, and

$$\varphi^j : B_{\mathrm{st}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V \to B_{\mathrm{st}\ \sigma^{\overline{m+j}}}\otimes_{\mathbb{Q}_{p^r}}$$

is a bijection, where $\overline{m+j}$ is the remainder of $m+j$ by r. By the same reason, we also have

$$B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V = \bigoplus_{m=0}^{r-1} B_{\mathrm{dR}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V$$

with each component stable by the G_K-action, and the map

$$1 \otimes \varphi^j : B_{\mathrm{dR}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V = B_{\mathrm{dR}} \otimes_{B_{\mathrm{st}}} (B_{\mathrm{st}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} B) \to B_{\mathrm{dR}\ \sigma^{\overline{m+j}}}\otimes_{\mathbb{Q}_{p^r}} V$$

is a bijection.

Definition 8.18. For a $\mathbb{Q}_{p^r}$-representation V, $0 \le m < r$, set

$$\mathbf{D}^{(m)}_{\mathrm{st},r}(V) := (B_{\mathrm{st}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V)^{G_K}, \quad \mathbf{D}^{(m)}_{\mathrm{dR},r}(V) := B_{\mathrm{dR}\ \sigma^m}\otimes_{\mathbb{Q}_{p^r}} V^{G_K}.$$

In particular, write $\mathbf{D}_{\mathrm{st},r}(V) = \mathbf{D}^{(0)}_{\mathrm{st},r}(V)$ and $\mathbf{D}_{\mathrm{dR},r}(V) = \mathbf{D}^{(0)}_{\mathrm{dR},r}(V)$.

On one hand, $\mathbf{D}^{(m)}_{\mathrm{st},r}(V)$ are K_0-vector spaces, stable by the actions of φ^j and N, and we have the decomposition

$$\mathbf{D}_{\mathrm{st}}(V) = \bigoplus_{m=0}^{r-1} \mathbf{D}^{(m)}_{\mathrm{st},r}(V).$$

On the other hand, $\mathbf{D}^{(m)}_{\mathrm{dR},r}(V)$ are filtered K-vector spaces, and

$$\mathbf{D}_{\mathrm{dR}}(V) = \bigoplus_{m=0}^{r-1} \mathbf{D}^{(m)}_{\mathrm{dR},r}(V).$$

Moreover, we have an injection

$$K \otimes_{K_0} \mathbf{D}^{(m)}_{\mathrm{st},r}(V) \hookrightarrow \mathbf{D}^{(m)}_{\mathrm{dR},r}(V).$$

It is obvious that we have the following proposition:

Proposition 8.19. *For $0 \leq m < r$, the maps $\varphi^j : \mathbf{D}^{(m)}_{\mathrm{st},r}(V) \to \mathbf{D}^{(\overline{m+j})}_{\mathrm{st},r}(V)$ and $1 \otimes \varphi^j : \mathbf{D}^{(m)}_{\mathrm{dR},r}(V) \to \mathbf{D}^{(\overline{m+j})}_{\mathrm{dR},r}(V)$ are bijective. Consequently:*

$$\dim_{K_0} \mathbf{D}^{(m)}_{\mathrm{st},r}(V) = \dim_{K_0} \mathbf{D}_{\mathrm{st},r}(V) \leq \dim_{\mathbb{Q}_{p^r}} V$$
$$\dim_{K_0} \mathbf{D}^{(m)}_{\mathrm{dR},r}(V) = \dim_{K_0} \mathbf{D}_{\mathrm{dR},r}(V) \leq \dim_{\mathbb{Q}_{p^r}} V$$

This proposition also shows that the terminology de-Rham (semi-stable, crystalline, etc.) for $\mathbb{Q}_{p^r}$-representations are exactly equivalent to B_{dR}- (B_{st}-, B_{cris}-, etc.) admissible.

Proposition 8.20. *Let V be a $\mathbb{Q}_{p^r}$-representation of G_K, then*
(1) V is semi-stable if and only if $\dim_{K_0} \mathbf{D}_{\mathrm{st},r}(V) = \dim_{\mathbb{Q}_{p^r}} V$, and in this case for every m,

$$\mathbf{D}^{(m)}_{\mathrm{dR},r}(V) = K \otimes_{K_0} \mathbf{D}^{(m)}_{\mathrm{st},r}(V) = K \,{}_{\varphi^m}\!\otimes_{K_0} \mathbf{D}_{\mathrm{st},r}(V).$$

(2) V is de-Rham if and only if $\dim_{K_0} \mathbf{D}_{\mathrm{dR},r}(V) = \dim_{\mathbb{Q}_{p^r}} V$.

Definition 8.21. A filtered (φ^r, N)-module over K is a K_0-vector space Δ equipped with two operators

$$\varphi^r, N : \Delta \to \Delta$$

such that N is K_0-linear, φ^r is σ^r-semi-linear and bijective, and

$$N\varphi^r = p^r \varphi^r N,$$

and there is a structure of filtered K-vector space on

$$\Delta_{K,m} = K \otimes_{K_0} \Delta_m = K \,{}_{\varphi^m}\!\otimes_{K_0} \Delta$$

for each $m = 0, 1, 2, \cdots, r-1$, where $\Delta_m = K_0 \,{}_{\varphi^m}\!\otimes_{K_0} \Delta$.

Definition 8.22. Suppose Δ is a filtered (φ^r, N)-module over K, the associated filtered (φ, N)-module over K is the module

$$D := K_0[\varphi] \otimes_{K_0[\varphi^r]} \Delta = \sum_{m=0}^{r-1} \Delta_m.$$

Δ is called weakly admissible if the associated D is weakly admissible.

Thus for a semi-stable $\mathbb{Q}_{p^r}$-representation V of G_K, $\Delta = \mathbf{D}_{\mathrm{st},r}(V)$ has a natural structure of a filtered (φ^r, N)-module over K, $\Delta_m = \mathbf{D}^{(m)}_{\mathrm{st},r}(V)$ and $\Delta_{K,m} = \mathbf{D}^{(m)}_{\mathrm{dR},r}(V)$, and the associated weakly admissible filtered (φ, N)-module is $D = \mathbf{D}_{\mathrm{st}}(V)$. It follows that:

Proposition 8.23. *Let $\mathbf{Rep}^{st}_{\mathbb{Q}_{p^r}}(G_K)$ denote the category of semi-stable $\mathbb{Q}_{p^r}$-representations of G_K and $\boldsymbol{MF}^{\mathrm{ad}}_K(\varphi^r, N)$ denote the category of weakly admissible filtered (φ^r, N)-modules over K. Then the functor*

$$\mathbf{D}_{\mathrm{st},r} : \mathbf{Rep}^{st}_{\mathbb{Q}_{p^r}}(G_K) \to \boldsymbol{MF}^{\mathrm{ad}}_K(\varphi^r, N)$$

is an exact and fully faithful functor.

For a filtered (φ^r, N)-module Δ, one can define the Galois, φ^r-, N-actions on $B_{\mathrm{st}} \otimes \Delta$, and the filtration on

$$K \otimes_{K_0} (B_{\mathrm{st}} \otimes \Delta) \hookrightarrow B_{\mathrm{dR}} \otimes_K \Delta_K.$$

Definition 8.24. Set

$$\mathbf{V}_{\mathrm{st},r}(\Delta) := (B_{\mathrm{st}} \otimes \Delta)_{\varphi^r=1, N=0} \cap \mathrm{Fil}^0(K \otimes_{K_0} (B_{\mathrm{st}} \otimes \Delta)).$$

Proposition 8.25. *If V is a semi-stable $\mathbb{Q}_{p^r}$-representation, then*

$$\mathbf{V}_{\mathrm{st},r}(\mathbf{D}_{\mathrm{st},r}(V)) = V.$$

The functors $\mathbf{D}_{\mathrm{st},r}$ and $\mathbf{D}_{\mathrm{dR},r}$ also preserve tensor products:

Proposition 8.26. *If V_1 and V_2 are two $\mathbb{Q}_{p^r}$-representations. Suppose they are semi-stable (de-Rham, resp.), then so is $V_1 \otimes_{\mathbb{Q}_{p^r}} V_2$, and in this case, for every $m = 0, \cdots, r-1$,*

$$\mathbf{D}^{(m)}_{\mathrm{st},r}(V_1) \otimes_{K_0} \mathbf{D}^{(m)}_{\mathrm{st},r}(V_2) \to \mathbf{D}^{(m)}_{\mathrm{st},r}(V_1 \otimes_{\mathbb{Q}_{p^r}} V_2)$$

(or

$$\mathbf{D}^{(m)}_{\mathrm{dR},r}(V_1) \otimes_{K} \mathbf{D}^{(m)}_{\mathrm{dR},r}(V_2) \to \mathbf{D}^{(m)}_{\mathrm{dR},r}(V_1 \otimes_{\mathbb{Q}_{p^r}} V_2)$$

resp.) is an isomorphism.

Proposition 8.27. *(1) If V is a de-Rham $\mathbb{Q}_{p^r}$-representation, set $\Delta_{K,m} = \mathbf{D}^{(m)}_{\mathrm{dR},r}(V)$, $t_{H,m}(V) = t_H(\Delta_{K,m})$ and $t_H(V) = t_H(D) = t_H(\mathbf{D}_{\mathrm{dR}}(V))$, then $t_H(V) = \sum_{m=0}^{r-1} t_{H,m}(V)$.*
(2) If V_1 and V_2 are de-Rham $\mathbb{Q}_{p^r}$-representations of $\mathbb{Q}_{p^r}$-dimension h_1 and h_2 respectively, let $V = V_1 \otimes_{\mathbb{Q}_{p^r}} V_2$, then

$$t_H(V) = h_2 t_H(V_1) + h_1 t_H(V_2).$$

In particular, if $s = rb$ is a multiple of r and V is a de-Rham $\mathbb{Q}_{p^r}$-representation, then

$$t_H(\mathbb{Q}_{p^s} \otimes_{\mathbb{Q}_{p^r}} V) = b t_H(V).$$

Proof. (1) follows that $D_K = \bigoplus_{m=0}^{r-1} \Delta_{K,m}$, and (2) is computed by (1). □

Let treat the Lubin-Tate representation $V_r = (B^+_{\mathrm{cris}})^{\varphi^r=p} \cap \mathrm{Fil}^1 B_{\mathrm{dR}}$ as an example. Since $(B_{\mathrm{cris}} \otimes_{\mathbb{Q}_{p^r}} V_r)^{G_K} = (B_{\mathrm{cris}} \otimes_{\mathbb{Q}_{p^r}} \mathbb{Q}_{p^r} t_r)^{G_K} = B^{G_K}_{\mathrm{cris}} = K_0$, for every $r \in \mathbb{N}$, $r \geq 1$, V_r is crystalline. Let $e = t_r^{-1} \otimes t_r \in \mathbf{D}_{\mathrm{st},r}(V_r)$ and $e_m = \varphi^m(e) = \varphi^m(t_r^{-1}) \otimes t_r$, then e_m is a basis of $\mathbf{D}^{(m)}_{\mathrm{st},r}(V_r)$, and $\{e_m \mid 0 \leq m < r\}$ is a basis of $\mathbf{D}_{\mathrm{cris}}(V_r)$. Also we note that $\Delta = \mathbf{D}_{\mathrm{st},r}(V_r) = K_0 e$, and

$$\Delta_{K,m} = K \,{}_{\varphi^m}\!\otimes_{K_0} K_0 e = K e_m, \;\; e_m = 1 \,{}_{\varphi^m}\!\otimes_{K_0} e = \varphi^m(e)$$

for $m = 0, 1, \cdots r-1$.

If $m > 0$, then

$$\mathrm{Fil}^i \Delta_{K,m} = \begin{cases} Ke_m, \text{ if } i \leq 0, \\ 0, \text{ if } i > 0. \end{cases}$$

If $m = 0$, then

$$\mathrm{Fil}^i \Delta_{K,0} = \begin{cases} Ke_0, \text{ if } i < 0, \\ 0, \text{ if } i \geq 0. \end{cases}$$

Thus $t_{H,0} = -1$ and $t_{H,m} = 0$ for $m \neq 0$, and hence $t_H(V_r) = -1$.

Furthermore, for $a \in \mathbb{Z}$, set

$$V_r^a = \begin{cases} \mathrm{Sym}^a_{\mathbb{Q}_{p^r}}, \text{ if } a \geq 0, \\ \mathcal{L}_{\mathbb{Q}_{\underset{\vee}{\ }^{\nabla}}}(V_r^{-a}, \mathbb{Q}_{p^r}), \text{ if } a < 0. \end{cases}$$

Then $\mathbb{Q}_{p^r}$ is a $\mathbb{Q}_{p^r}$-representation of dimension 1 generated by t_r^a, and $\mathbf{D}^{(m)}_{\mathrm{st},r}(V_r^a)$ is generated by $\varphi^m(t_r^{-a} \otimes t_r^a) = \varphi^m(t_r^{-a}) \otimes t_r^a$. By the same computation as for V_r, we have $t_{H,0}(V_r^a) = -a$ and $t_{H,m}(V_r^a) = 0$ for $0 < m < r$, and hence $t_H(V_r^a) = -a$.

8.9. **Reduction of the Hodge number.** If D is a filtered (φ, N)-module over K, let $r \geq 1$ be te largest integer such that we can associate D with a filtered (φ^r, N)-module Δ over K, i.e. $D = \Delta \otimes_{K_0[\varphi^r]} K_0[\varphi]$, then the reduced dimension of D is defined to be the integer $\frac{\dim_{K_0} D}{r} = \dim_{K_0} \Delta$. We have

Proposition 8.28. *For $h \in \mathbb{N}$, $h \geq 1$, the followings are equivalent:*
(1) Any weakly admissible filtered (φ, N)-module D over K of reduced dimension $\leq h$ and with $t_H(D) = 0$ satisfied $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) = \dim_{K_0} D$.
(2) Any weakly admissible (φ, N)-module D over K of reduced dimension $\leq h$ satisfied $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) = \dim_{K_0} D$.

Our main strategy is to reduce the Hodge number of D by tensoring $\mathbf{D}_{\mathrm{st}}(V_r^a)$ for proper integer r and a.

Proof. We need to show (1) $\Rightarrow$ (2). Let D be a weakly admissible filtered (φ, N)-module over K of reduced dimension h and of dimension $d = rh$. Let Δ be the associated (φ^r, N)-module. We need to show $\dim_{\mathbb{Q}_p} \mathbf{V}_{\mathrm{st}}(D) = \dim_{K_0} D = rh$.

By Proposition 8.14, we may assume that D is irreducible.

Claim1:$N = 0$.

Otherwise, $\ker(N : D \to D)$ is a nontrivial weakly admissible sub-object of D.

Claim2: If $D = D_1 \oplus D_2$, the weak admissibility of D implies the weak admissibility of D_1 and D_2.

Since D is weakly admissible, $t_N(D) = t_H(D)$ and $t_N(D_i) \leq t_H(D_i)$ for $i = 1, 2$, and then the equality

$$t_N(D_1) + t_N(D_2) = t_N(D_1 \oplus D_2) = t_N(D) = t_H(D) = t_H(D_1 \oplus D_2) = t_H(D_1) + t_H(D_2)$$

implies that $t_N(D_i) = t_H(D_i)$ for $i = 1, 2$.

Claim3: for any non-zero $x \in D$, D is generated as a K_0-vector space by $\{x, \varphi(x), \cdots, \varphi^{rh-1}(x)\}$, and Δ is generated as a K_0 vector space by $\{x, \varphi^r(x), \cdots, \varphi^{r(h-1)}(x)\}$, and hence D_i are admissible.

Indeed, let $D(x)$ be the subspace of D generated by $\varphi^i(x)$ for $i \in \mathbb{N}$, then $D(x)$ is stable under φ and N since $N = 0$. Then as a vector space, $D = \bigoplus_x D(x)$ for some $x \in D$. By Claim 2 and the irreducibility of D, we must have $D = D(x)$.

Now let $a = t_H(D)$, $b = h$. Let $D_{rh} = \mathbf{D}_{\mathrm{st}}(V^a_{rh})$, and let $\Delta_{(rh)} = \mathbf{D}_{\mathrm{st},rh}(V^a_{rh})$, then $\dim_{K_0} \Delta_{rh} = 1$. In this case, $N = 0$. Let $D' = D \otimes_{K_0[\varphi^r]} D_{rh}$ and $\Delta' = \Delta \otimes_{K_0[\varphi^{rh}]} \Delta_{(rh)}$, then

$$K_0[\varphi] \otimes_{K_0[\varphi^r]} \Delta' = D \otimes_{K_0[\varphi^r]} \Delta_{(rh)} = D \otimes_{K_0[\varphi]} D_{rh},$$

and thus D' is a φ-module associated with the φ^r-module Δ' and is of reduced dimension $\leq h$. Moreover, let $\{e_1, \cdots, e_h\}$ be a K_0-basis of Δ, f be a generator of $\Delta_{(rh)}$, then Δ'_m is generated by $\{\varphi^m(e_1 \otimes f), \cdots, \varphi^m(e_h \otimes f)\}$. We claim that D' is admissible and $t_H(D') = 0$.

The second claim is easy, since by the above construction and the definition of t_H, we have $t_H(D') = h(t_H(D) - a) = 0$.

For the first claim, for $x \neq 0$, $x \in D$, let D_x be the K_0-subspace of D generated by $\varphi^{rhi}(x)$ for $i \in \mathbb{N}$. □

8.10. **Fundamental Lemma of Colmez.** Recall $U = \log(U^+_R) = P^+_{1,1}$ and $B_2 = B^+_{\mathrm{dR}}/\mathrm{Fil}^2 B_{\mathrm{dR}}$. The diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & U & \xrightarrow{\theta} & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle incl} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & C(1) & \longrightarrow & B_2 & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

is commutative, where all rows are exact and the by the Five Lemma all vertical arrows are injective.

Suppose h is an integer ≥ 2, and suppose that $\lambda_1, \cdots, \lambda_h \in C$ are not all zero. Set

$$Y := \{(u_1, u_2, \cdots, u_h) \in U^h \mid \exists\ c \in C, \text{ such that for all } i,\ \theta(u_i) = c\lambda_i\}.$$

If we denote by θ as well the map $U^h \to C^h, (u_1, \cdots, u_h) \mapsto (\theta(u_1), \cdots, \theta(u_h))$, and let $\lambda = (\lambda_1, \cdots, \lambda^h)$, then $Y = \theta^{-1}(\langle\lambda\rangle)$.

If $b = (b_1, \cdots, b_h) \in B_2^h$ is a nonzero vector such that $\lambda \cdot \theta(b) = \sum_{i=1}^h \lambda_i \theta(b_i) = 0$. Then the map

$$\rho : Y \to B_2,\ u = (u_1, \cdots, u_h) \mapsto b \cdot u = \sum_{i=1}^h b_i u_i$$

has image in $C(1)$.

Theorem 8.29. *(Fundamental Lemma of Colmez) Assume the above hypothesis. Then* $\mathrm{im}\rho \subset C(1)$ *and*
(1). either $\mathrm{im}\rho = \rho(\mathbb{Q}_p(1)^{h'})$ *for some* $h' \leq h$ *and hence* $\dim_{\mathbb{Q}_p} \mathrm{im}\rho \leq h$*;*
(2). or $\mathrm{im}\rho = C(1)$ *and* $\dim_{\mathbb{Q}_p} \ker \rho = h$*.*

8.11. **End of the Proof.** Recall that the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & U & \xrightarrow{\theta} & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle incl} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & C(1) & \longrightarrow & B_2 & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

is commutative, all rows are exact and all vertical arrows are injective. Given a finite dimensional $\mathbb{Q}_p$-vector space, tensoring the above diagram with $V(-1)$, we get

$$\begin{array}{ccccccccc} 0 & \longrightarrow & V & \longrightarrow & U(-1)\otimes V & \xrightarrow{\theta} & V_C(-1) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle incl} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & V_C & \longrightarrow & B_2(-1)\otimes V & \xrightarrow{\bar{\theta}} & V_C(-1) & \longrightarrow & 0 \end{array}.$$

Proposition 8.30. *Suppose* $\dim_{\mathbb{Q}_p} V = h \geq 2$*, and suppose that there is a surjective* B_2*-linear map* $\eta : B_2(-1)\otimes_{\mathbb{Q}_p} V \to B_2(-1)$ *which passes to the quotient map* $\bar{\eta} : V_C(-1) \to C(-1)$*. Suppose* $\overline{X}$ *is a sub-*C*-vector space of dimension 1 of* $V_C(-1)$ *and* X *its inverse image in* $U(-1)\otimes_{\mathbb{Q}_p} V$*, i.e. we have a diagram*

$$\begin{array}{ccccccc} X & \longrightarrow & U(-1)\otimes_{\mathbb{Q}_p} V & \longrightarrow & B(-1)\otimes_{\mathbb{Q}_p} V & \xrightarrow{\eta} & B_2(-1) \\ \downarrow{\scriptstyle\theta} & & \downarrow{\scriptstyle\theta} & & \downarrow{\scriptstyle\theta} & & \downarrow{\scriptstyle\theta} \\ \overline{X} & \longrightarrow & V_C(-1) & \longrightarrow & V_C(-1) & \xrightarrow{\bar{\eta}} & C(-1) \end{array}$$

If $\overline{X}\subset \ker\bar{\eta}$*, then the restriction* $\eta_X : X \to B_2(-1)$ *of* η *factors through* $X \to C$*. Moreover, if* $\eta(V)\neq\eta(X)$*, then* η_X *is surjective and its kernel is a* $\mathbb{Q}_p$*-vector space of dimension* h*.*

Proof. Suppose $\{e_1,\cdots,e_h\}$ is a basis of V over $\mathbb{Q}_p$, then $\{e'_n = t^{-1}\otimes e_n\}$ forms a basis of the free B_2-module $B_2(-1)\otimes_{\mathbb{Q}_p} V$. Write $\eta(e'_n) = b_n\otimes t^{-1}$ with $b_n\in B_2$.

The images $\bar{e_n}'$ of e'_n in $V_C(-1)$ forms a basis of its as a C-vector space. Suppose $\lambda = \sum_{n=1}^h \lambda_n \bar{e_n}'$ is a non-zero element of $\overline{X}$, the fact $\overline{X}\subset\ker\bar{\eta}$ implies that $\sum\lambda_n\theta(b_n)=0$, hence $\theta\circ\eta(X)$ and η_X factors through $X\to C$.

Let Y and ρ be given by the above (λ_n) and (b_n). Let $\nu : U^h \to U(-1)\otimes V$, $(u_1,\cdots,u_h)\mapsto \sum(u_n\otimes t^{-1})\otimes e_n$, then ν is a bijection. We can see that the diagram

$$\begin{array}{ccc} Y & \xrightarrow{\rho} & C(1) \\ \downarrow{\scriptstyle\nu_Y} & & \downarrow{\scriptstyle\cdot t^{-1}} \\ X & \xrightarrow{\eta_X} & C \end{array}$$

is commutative, and the proposition is nothing but a reformulation of the Fundamental Lemma of Colmez. □

Proposition 8.31. *Let* V_1 *be a* $\mathbb{Q}_p$*-vector space of finite dimension* $h\geq 2$ *and* $\Lambda_1 = B^+_{\mathrm{dR}}\otimes_{\mathbb{Q}_p} V_1$*. Suppose* Λ_2 *is a sub-*B^+_{dR}*-module of* $\Lambda_1(-1)$ *such that* $(\Lambda_1+\Lambda_2)/\Lambda_1$ *and* $(\Lambda_1+\Lambda_2)/\Lambda_2$ *are simple* B^+_{dR}*-modules. Let* X *be the inverse image of* $\Lambda_1+\Lambda_2$ *in* $U(-1)\otimes_{\mathbb{Q}_p} V_1$ *and*

$$\rho : U(-1)\otimes_{\mathbb{Q}_p} V_1 \to \Lambda_1(-1)/\Lambda_2$$

be the natural projection. Then
(1) either $\dim_{\mathbb{Q}_p}\rho(X)\leq h$ *and* $\ker(\rho)$ *is not finite dimensional over* $\mathbb{Q}_p$*;*
(2) or ρ *is surjective and* $\ker\rho$ *is a sub-*$\mathbb{Q}_p$ *vector space of dimension* h*.*

Proof. Since B^+_{dR} is a DVR whose residue field is C, the hypotheses indicate that $(\Lambda_1+\Lambda_2)/\Lambda_1$ and $(\Lambda_1+\Lambda_2)/\Lambda_2$ are C-vector space of dimension 1. Then we can find elements $\{e_1,\cdots,e_h\}$ in Λ_1 such that

$$\Lambda_1 = B^+_{\mathrm{dR}}e_1\oplus B^+_{\mathrm{dR}}e_2\oplus\Lambda_0,\quad \Lambda_2 = B^+_{\mathrm{dR}}t^{-1}e_1\oplus B^+_{\mathrm{dR}}te_2\oplus\Lambda_0,$$

where $\Lambda_0 = \bigoplus_{i=3}^{h} B_{\mathrm{dR}}^{+} e_i$. One thus has two commutative diagram, which are exact on the rows:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & V_1 & \longrightarrow & X & \longrightarrow & \frac{\Lambda_1+\Lambda_2}{\Lambda_1} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle =} & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & V_1 & \longrightarrow & U(-1)\otimes_{\mathbb{Q}_p} V_1 & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_1} & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & \frac{\Lambda_1(-1)}{\Lambda_1+\Lambda_2} & \xrightarrow{=} & \frac{\Lambda_1(-1)}{\Lambda_1+\Lambda_2} & &
\end{array}$$

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \longrightarrow & U(-1)\otimes_{\mathbb{Q}_p} V_1 & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle \rho} & & & & \\
0 & \longrightarrow & \frac{\Lambda_1+\Lambda_2}{\Lambda_2} & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_2} & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_1+\Lambda_2} & \longrightarrow & 0
\end{array}$$

Let ϵ_i denote the image of $t^{-1}e_i$ in $B_2(-1)\otimes_{\mathbb{Q}_p} V_1 = \Lambda_1(-1)/\Lambda_1(1)$, then $\{\epsilon_i | 1 \le i \le h\}$ is a basis of $\Lambda_1(-1)/\Lambda_1(1)$ as a free B_2-module of rank h.

We denote by $\eta : B_2(-1)\otimes V_1 \to B_2(-1)$ the map which sends $\sum_i a_i\epsilon_i$ to $a_2 t^{-1}$. The image of the restriction η_X of η on X is contained in C and the diagram above induces the commutative diagram with exact rows

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \longrightarrow & U(-1)\otimes V_1 & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_1+\Lambda_2} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \eta_X} & & \downarrow{\scriptstyle \rho} & & \downarrow & & \\
0 & \longrightarrow & C & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_2} & \longrightarrow & \frac{\Lambda_1(-1)}{\Lambda_1+\Lambda_2} & \longrightarrow & 0
\end{array}$$

where $C \to \Lambda_1(-1)/\Lambda_2$ is the map $c \mapsto ct^{-1}\epsilon_2$.

One can see that the image $\overline{X}$ of X in $\Lambda_1(-1)/\Lambda_1 = (C\otimes V_1)(-1)$ is a C-vector space of dimension 1 contained in the kernel of $\bar{\eta}$, and X is the inverse image of $\overline{X}$ in $U(-1)\otimes V_1$. Applying the precedent proposition, if $\eta(V_1) = \eta(X)$, we are in case (1); otherwise, η_X is surjective, so is ρ and $\ker\rho = \ker\eta_X$ is of dimension h over $\mathbb{Q}_p$. □

After the argument we have done, to finish the proof of Theorem A, it suffices to prove the following:

Proposition 8.32. *Let Δ be an admissible filtered (φ^r, N)-module over K_0 of K_0-dimension h, D be the associated filtered (φ, N)-module with $t_H(D) = 0$. Then*

$$\dim_{\mathbb{Q}_{p^r}} \mathbf{V}_{\mathrm{st}}(D) = h.$$

We say V (resp. Δ or $\mathbb{D}$) is of dimension (r, h) if V (resp. Δ) is a $\mathbb{Q}_{p^r}$-representation (resp. a (φ^r, N)-module) of dimension h.

Proof. We prove the theorem by induction on h. Suppose the proposition is known for all V' (resp. Δ') of dimension (r', h') with $h' < h$ and r' arbitrary.

Consider the set of all convex polygon with origin $(0, 0)$ and end point $(hr, 0)$. The Hodge polygon P_H of V (resp. D) is an element of this set. We choose a K-basis $\{\delta_i \mid 1 \le j \le rh\}$ of D_K which is compatible with the filtration $\{\mathrm{Fil}^i D_K\}$ and the decomposition

$D_K = \bigoplus_{m=0}^{r-1} \Delta_{K,m}$. To be precise:
(a). If let

$$i_j := \max\{i \in \mathbb{Z} \mid \delta_j \in \mathrm{Fil}^i D_K\},$$

then the set $\{\delta_j \mid i_j \geq i\}$ is a K-basis of $\mathrm{Fil}^i D_K$ for every $i \in \mathbb{Z}$.
(b). For every $0 \leq m < r$, $\Delta_{K,m}$ has a K-basis $\{\delta_j \mid \delta_j \in \mathrm{Fil}^i D_K\}$.

By this way, then

$$h_i = \dim_K \mathrm{Fil}^i D_K/\mathrm{Fil}^{i+1} D_K = \#\{1 \leq j \leq rh \mid i_j = i\}.$$

$$t_H = \sum_{j=1}^{rh} i_j = 0.$$

As long as P_H is non-trivial, by changing the order of δ_j, we may assume that $i_2 \geq i_1 + 2$. We fix such a basis of D_K.

Now we consider the (φ^r, N)-module Δ' defined as follows:
(1) the underlying (φ^r, N)-module structure is the same as of Δ;
(2) since $D'_K = D_K$, for the basis $\{\delta_j \mid 1 \leq j \leq rh\}$ of D_K, the filtration is given as follows:

$$i'_1 = i_1 + 1,\ i'_2 = i_2 - 1,\ i'_j = i_j \text{ for } j \geq 2.$$

Then Δ' is a filtered (φ^r, N)-module of dimension h. Let D' be the associated (φ, N)-module, then $t_H(D') = t(D) - 1 + 1 = t_H(D) = 0$ and $t_N(D') = t_N(D)$, Moreover, let E' be any sub-object of D' as (φ, N)-module, different from 0 and D. Then one has $t_N(E') = t_N(E)$ and $t_H(E') = t_H(E) + \epsilon$ with $\epsilon \in \{-1, 0, 1\}$. Since D is admissible and irreducible, $t_H(E) < t_N(E)$ and we have $t_H(E') \leq t_N(E')$, which implies that D' is an admissible (φ, N)-module.

Let $V_1 = \mathbf{V}_{\mathrm{st}}(D')$ and $V_2 = \mathbf{V}_{\mathrm{st}}(D)$. We need to show $\dim_{\mathbb{Q}_p} V_2 \geq rh$.

Since the Hodge polygon of D' is strictly above that of D, by induction hypothesis, we have $\dim_{\mathbb{Q}_{p^r}} V_1 = h$, which means that V_1 is semi-stable and $\mathbf{D}_{\mathrm{st}}(V_1) = D'$. Then $B_{\mathrm{st}} \otimes_{K_0} D = B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V_1$ and

$$\mathbf{V}^0_{\mathrm{st}}(D) = \mathbf{V}^0_{\mathrm{st}}(D') = \{x \in B_{\mathrm{st}} \otimes_{K_0} D \mid \varphi(x) = x, Nx = 0\} = B_e \otimes_{\mathbb{Q}_p} V_1.$$

Suppose $W = B_{\mathrm{dR}} \otimes_K D_K = B_{\mathrm{dR}} \otimes_K D'_K$, $\Lambda_1 = \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D'_K) = \sum_{i \in \mathbb{Z}} \mathrm{Fil}^{-i} B_{\mathrm{dR}} \otimes_K \mathrm{Fil}^i D'_K$ and $\Lambda_2 = \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D_K)$. Then we have exact sequences

$$0 \to V_i \to]\mathbf{V}^0_{\mathrm{st}}(D) \to W/\Lambda_i$$

for $i = 1, 2$. Since V_1 is semi-stable,

$$\Lambda_1 = \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_K D'_K) \cong \mathrm{Fil}^0(B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1) = B^+_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V_1.$$

In this case, we have an exact sequence

$$0 \to V_1 \to \mathbf{V}^0_{\mathrm{st}}(D) \to W/\Lambda_1 = \mathbf{V}^1_{\mathrm{st}}(D) \to 0.$$

Note that Λ_2 is a sub-B^+_{dR}-module of $\Lambda_1(-1)$ such that $(\Lambda_1 + \Lambda_2/\Lambda_1)$ and $(\Lambda_1 + \Lambda_2)/\Lambda_2$ are simple B^+_{dR}-modules. We can apply proposition 8.31. By the inclusion $U(-1) \subset B_e$, we have a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \ker\rho & \longrightarrow & U(-1) \otimes_{\mathbb{Q}_p} V_1 & \xrightarrow{\rho} & \Lambda_1(-1)/\Lambda_2 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & V_2 & \longrightarrow & B_e \otimes_{\mathbb{Q}_p} V_1 & \longrightarrow & W/\Lambda_2 & &
\end{array}$$

where $\ker\rho \subset V_2$ implies that ρ must be surjective. Thus $\ker\rho$ must be of finite dimension rh. As a result, $\dim_{\mathbb{Q}_p} V_2 \geq rh$ and the proposition is proved. □

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